

How the two-thirds argument was found: two agent runs and their literature¹

Anthropic

¹This volume is Claude's account of two of its own research runs and of the campaign around them, rewritten by Claude from the runs' logs for readability; no text in it was written or edited by a human. The mathematics in this volume specifically (the runs' working as it was logged, including intermediate claims) has not been independently verified; this caveat concerns this volume only, not the accompanying paper.

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Overview: what the coordinator did

The campaign from the coordinator’s seat

The two accounts that follow (Parts I and II) are written by the agents that produced the one-half and two-thirds arguments. Neither saw the rest of the campaign: each received a written brief, worked alone, and returned a report. This opening chapter is the editor’s summary of the conversation that wrote those briefs — the orchestrating thread, called “the coordinator” below (Part I calls it “the orchestrator”) — so that the line of attempts that led to the two runs, and what was done with their results, is visible at least in outline. It is compiled from the coordinator’s own account of the session and from a provenance study of the complete logs; attributions follow the logs where the coordinator’s own later summaries slipped (two such places are marked). It describes what was tried and claimed; it certifies no mathematics.

1 How the session was run

The campaign was a single long conversation between one human and the coordinator, lasting a little over two days (fifty-four hours spread across three calendar days). The coordinator did almost no mathematics with its own hands and made no network request at any point. Its instruments were sub-agents: separate instances launched with a written brief (a target, the objects to use, a list of earlier reports to read first, usually the coordinator’s own forecast of the outcome, and a demand for a control case on which the Riemann Hypothesis is known to fail — an Epstein zeta function of class number two, the Davenport–Heilbronn function, a Beurling prime system with a planted off-line zero, a fake Weil polynomial). A sub-agent sees nothing but its brief and the files it is told to read; it cannot see the conversation or the other agents. When one returned, the coordinator read its final message (rarely its files), told the human what had come back, and decided what to launch next. Anything that claimed a theorem was sent to hostile referee agents, briefed to find the error and forbidden to read one another. About sixty sub-agents were launched in all; the agent labels used in this volume (R4, E2, E2-pairs, referees A–D, X, Y, and so on) are the coordinator’s own names for them.

2 Sixty launches at a glance

The coordinator launched sixty sub-agents, of which fifty-eight ran (the first two were cancelled by an interruption from the human two minutes after launch). Sorted by what they turned out to be for — a retrospective sorting; at launch most were simply attempts — they fall into seven bins. Labels are the ones used elsewhere in this volume where they exist, otherwise the coordinator’s short name for the run.

Core idea contributors (2). E2 (the signature theorem and the dual count giving one half; Part I); E2-pairs (the rank–trace lemma giving two thirds; Part II).

Attempts at progress on RH — refuted, closed or unused (30). Attempt A (Weil positivity for all supports; cancelled before running); attempt B (Jensen polynomials; cancelled before running); attempt C (de Bruijn–Newman backward flow; no-go); frontier-2 (a spectral/Lee–Yang partner; a reduction stronger than RH); frontier-3 (decoupling toward the zero-free region; reduction only); new-1 (Eisenstein zero-curves through Heegner points; reduces to known); new-2 (Borcea–Brändén stability lift; no-go); new-3 (a ferromagnetic/Lee–Yang representation of Φ ; no-go); new-4 (Deuring–Heilbronn rigidity; reproduces known); new-5 (horocycle equidistribution rate; saturates at one half); X1 (the lost-logarithm exponential-sum attack; no-go); D1 (the Langlands amplification ceiling); D2 (reverse-engineering the missing h^0 ; underdetermined); D3 (reach of period-formula positivity); D5 (Euler systems/Iwasawa theory at the archimedean place); R2 (the Beurling $\theta < \frac{1}{2}$ cell); R3 (off-centre manufactured positivity); X2 (Weyl sums beyond pair counting; no-go); R2a (class-group averaging; a numerical laboratory, no mechanism); R2c (discrete Beurling systems; cell closed, essentially a known theorem); E1 (hunt for a non-cancelling phantom line; all cancel); E3 (Gowers-inverse structure for large values); E5 (a class-group rate law; a side theorem, refereed, unused); b1 (theta tower/Rallis off-centre positivity; no escape); b2 (non-tempered periods/Kudla programme); d3 (δ -geometry/ λ -ring positivity; dead); d2 (a supersymmetric factorisation $H = A^*A$ of the Weil form; negative); $\Delta 1$ (the RH-carrying difference $W - W^*$; negative); C1 (the closing relation, Cayley–Hamilton over $\mathbb{Z}$; negative)¹; CX (structure of the scalar $C(X)$; stalled on an infrastructure fault and relaunched).

Useful to the paper but off its proof path (6). R4 (construct the polarized object; returned “Kreĭn, not Hilbert”, the seed of Part I); frontier-1 (structural/Hodge-index transplant; its report was read by E2); N6, that is new-6 (bootstrapping the explicit formula; cone and Harnack bound, a zeros table; its report was read by E2)²; E2-kernel (the optimal window; the optimised constants); E2-gen (the Dirichlet- L extension); E2-HL (higher prime-side moments; the no-go behind the paper’s “limits of the method” remark).

Other outputs of the campaign (7). R1 (the zero-density table repair); R4v2 (the positivity-margin curve and its 4π law); d1 (explaining the 4π law via Connes–Consani’s prolate vectors); CC+ (Weil positivity certified for $\lambda^2 \leq 9$); CX relaunched (the function $C(X)$, in support of CC+); new-7 (barrier theorems — the campaign’s methodology); R0 (the barrier-checker tool)³.

Validation and referees (13). Referees A and B of R1’s density claim; the referee of R2’s theorem; the referee of R2c; the referee of E5’s rate law; referees A, B and C of E2’s one-half claim; referee D (the mass-matrix, taper and tail gaps); the blind re-derivation of the prime-side step; the cold-read referee of the paper draft; referees X and Y of the two-thirds lemma.

Paper writing (1). The paper-writing agent.

Literature review (1). The literature agent.

Not counted among the sixty: four agents launched by sub-agents rather than by the coordinator — three literature searches (one for E5 on the first day, two for E2-gen during the two-thirds push) and one hostile referee that X2 spawned for itself — and the seven agents of the follow-up workflow on zero-density estimates near the critical line, launched after the paper draft and its

¹d2, $\Delta 1$ and C1 returned negative results — no factorisation, no usable difference, no closing relation; the coordinator folded them into the “no slack” synthesis on which it premised the CC+ brief (which names $\Delta 1$ ’s report among the files to read first, for calibration); one could as well file them under the campaign’s other outputs.

²frontier-1 and N6 failed on RH like the rest of the first wave; they sit here only because E2’s brief told it to read their reports, as context for the object it would compress — not for any step of the proof.

³new-7 and R0 are methodology — a zoo of RH-false model worlds and a tool for checking claims against it — used to discipline most later research briefs; neither an attempt on RH nor literature review, they are counted with the campaign’s other outputs by elimination.

explainer were finished, from which nothing flowed back. Resumptions — E2-pairs after its interruption, and three first-wave runs restarted after hitting an output limit — were messages to existing agents, not launches; CX is the one run relaunched under a new identity, and both of its launches are counted.

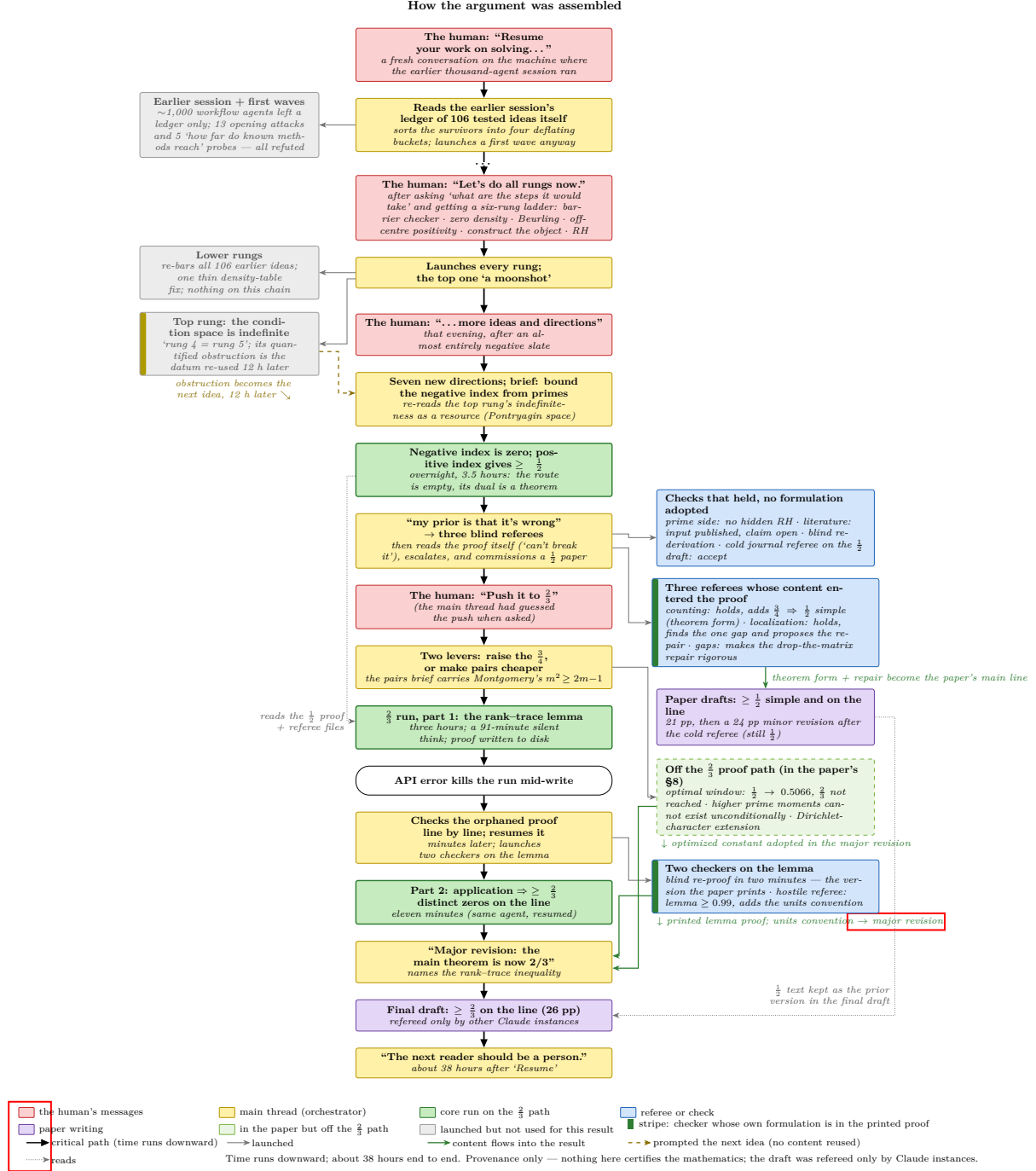


Figure 1: How the argument was assembled.

3 Starting over, with a ledger of failures

The session opened with a one-line instruction from the human to resume earlier work on the Riemann Hypothesis and to trust its own capability in doing so. There was no earlier conversation to resume: what the previous session (a separate effort some ten days before, which had run on the order of a thousand short-lived agents through an idea-mining and adversarial-review pipeline) had left on the machine was not a partial proof but a set of ledgers — above all a file of 106 “survivors”, candidate ideas that had passed their own kill-test, each with an honest one-paragraph statement of what actually stood. Before launching anything the coordinator printed and read the ledger’s own one-line statement of what stood for each of the 106 and sorted every one into four deflating classes: a known theorem restated, a statement equivalent to RH, a finite numerical check consistent with RH, or something nearly tautological. It told the human as much (there was no thread in the ledger that was a partial proof, and that was “not a confidence problem I can fix by believing harder”), declined to defer to the earlier session’s conclusions while reaching the same verdict independently, and used the ledger thereafter only as a do-not-repeat list handed to sub-agents. Nothing in either of the two arguments in this volume traces to that earlier session’s output.

4 Two waves that found the wall

Pressed by the human to attack anyway and then to “explore new frontiers”, the coordinator opened thirteen fronts in the first half hour. Three were the sharpest known programmes restated as honest attempts (Weil positivity in Connes’ formulation, the Jensen-polynomial route, the de Bruijn–Newman heat flow — the first two were cancelled by an interruption from the human and never ran); three were “frontier” probes asked to invent rather than apply (an honest integer analogue of the Hodge index theorem; a probabilistic/spectral transplant; an unconstrained search near the zero-free region); seven were “new areas” — four of the coordinator’s own (Eisenstein zero-curves; a horocycle equidistribution rate; an explicit-formula bootstrap with positivity certificates; a programme of “barrier theorems” formalising why the 106 survivors had all deflated the same way) and three grown from seeds the human offered (polynomial stability crossed with Lee–Yang; a ferromagnetic Lee–Yang kernel; Deuring–Heilbronn rigidity for a hypothetical off-line zero). A second, smaller wave a few hours later sent five agents to measure how far existing machinery reaches (a direct attack on the logarithm lost in the Vinogradov–Korobov exponential-sum bounds; the Langlands amplification ceiling; reverse-engineering the “missing h^0 ” of an arithmetic Riemann–Roch; period-formula positivity; archimedean Euler systems), with a follow-up on Weyl sums beyond pair counting.

Every one of these returned a no-go, a ceiling, or a reduction to a statement at least as strong as RH, and by the coordinator’s account they converged on a single wall: the only axiom set not barred by some explicit model in which RH fails is the full one (functional equation, Euler product at every prime, Ramanujan bounds), and every route’s first substantive step turned out to be Weil positivity or an equivalent in disguise. The human read the convergence as momentum; the coordinator corrected them — what had been done was to “re-derive the expert consensus from scratch, adversarially, with constants”, and “honing in on where the wall is $\neq$ finding a door.”

5 A ladder, and all rungs at once

Asked by the human what steps would bring a proof closer, the coordinator answered with a six-rung ladder “from ‘doable’ to ‘the thing itself.’” Its rungs: (0) a mechanical barrier checker that classifies any proposed idea by which RH-false model kills it; (1) unconditional

dents that need no new door, chiefly pushing zero-density estimates; (2) an open cell about Beurling generalised primes; (3) manufactured positivity off the critical line; (4) constructing the polarized object — the Hodge-index structure over the integers whose existence would make Weil positivity automatic; (5) RH itself, riding inside rung 4 as “if you get an object, test positivity on it.” When the human replied “Let’s do all rungs now”, the coordinator launched them in parallel with its expectations stated in advance: “Rungs 0–2 can return theorems/tools; 3 is a real shot; 4/5 is a moonshot and I’m labeling it that going in.”

What came back, rung by rung. The barrier checker (rung 0) barred all 106 survivors mechanically and found none that escaped the framework — reproducing, not extending, the triage. The density agent (rung 1) recalibrated the published zero-density table from its ingredients, saw its own column dip below the table on three thin intervals, isolated by ablation a 1979 lemma of Heath-Brown that the reference implementation had omitted, and claimed three small new bounds; two blind referees confirmed them as a genuine but bookkeeping-grade gap in the table (more on this below). Rung 2 closed the continuous Beurling cell and left the discrete case with two no-go theorems; rung 3 found no off-centre positivity, only a structural lemma locating exactly where manufactured positivity engages. Rung 4, the agent called R4, was briefed as a moonshot whose value would be “the sharpest failure point”: construct intrinsically, without reference to the zeros, a candidate space of global sections with a Riemann–Roch inequality. Within half an hour R4 had a reduction theorem — any such “box minus conditions” construction gets the easy requirements for free and works if and only if its condition count *is* the Weil form, so “Rung 4 = Rung 5” — and it then quantified the obstruction: the one zero-free datum behind all three of the brief’s candidate constructions, a signed “density of states” computed from primes and the Gamma factor, goes negative on a fifth to two fifths of the critical line at moderate height, and a natural operator built from it has dozens to hundreds of negative eigenvalues. R4’s summary phrase was that “the condition space is Kreĭn, not Hilbert.” The coordinator relayed this to the human as the ladder collapsing by one rung, provably, and then did nothing with it for twelve hours.

6 New directions from the day’s findings; the E2 brief

Some twenty-one hours into the session, with the whole slate returned and almost all of it negative, the human asked for “more ideas and directions.” The coordinator’s answer was seven new directions, which it described as coming out of what specific agents had found that day rather than the general literature (true of the four it launched), of which it launched four. The second of them re-read R4’s obstruction as a resource. If the natural Hermitian form is indefinite, the coordinator reasoned from memory, then Pontryagin-space operator theory applies: by the theorem of Pontryagin and Kreĭn–Langer a self-adjoint operator on a space with negative index κ has at most κ non-real eigenvalue pairs, so a bound on the negative index of a finite, prime-computable compression of Weil’s form would bound the number of off-line zeros — a zero-density theorem, perhaps even a positive-proportion theorem, “by a completely non-classical route.” The brief it wrote for the agent E2 handed down R4’s files and its indefiniteness datum; the object (finite compressions of the Weil form on band- and time-limited test functions, computable from primes alone); a conjecture to prove first (each off-line pair of zeros contributes exactly one negative square, so that globally the negative index equals the number of off-line zeros — attributed, with question marks, to Yoshida or Bombieri); a programme (define the finite index, compute it from primes over a large range, bound it a priori by counting sign changes); the prize (“can the negative-index route REPRODUCE a positive-proportion result? That would be a genuinely new proof architecture for Selberg’s theorem”); and a forecast in the form of a trichotomy for the ratio of index to zero count (a constant below one, zero, or one). The words “Pontryagin” and “Kreĭn–Langer” had appeared

in nothing the coordinator had read (one unrelated use aside) and in nothing any agent had downloaded; the frame was recalled — though among the earlier session’s one-line ledger claims that the coordinator had printed and read at the start were several describing Weil positivity as a signature statement on a finite Gram matrix, and whether that exposure cued the reframing cannot be told from the logs. Part I is E2’s own account of what it did with this brief: it proved the conjecture in corrected form, found that R4’s operator is not a compression of the Weil form and that the honest finite negative index is identically zero — a fourth case outside the brief’s trichotomy, so the route as briefed is empty — and then inverted the idea, lower-bounding the *positive* index from prime-side traces and double-counting off-line pairs against the Riemann–von Mangoldt formula to claim, unconditionally, that at least half of the zeros lie on the critical line as distinct zeros. The target was the brief’s; the mechanism was not in it.

7 “My prior is that it’s wrong”: referees, literature, re-derivation

E2’s claim arrived about three and a half hours after its launch, and the coordinator’s first words on it to the human were that this was an extraordinary claim and “my prior is that it’s wrong,” with two named suspects: localisation (a zero far outside the window might still contribute a positive square) and a hidden use of RH in the prime-side evaluation. Within ten minutes it had launched three blind hostile referees, forbidden to read one another, each assigned one joint and each handed a worked plan of attack — A on localisation, told to build the counterexample numerically; B on the prime side; C on the linear algebra and counting, with a “proves too much” test on functions known to violate RH — and told the human where it stood before they reported: “I am *not* telling you half the zeros are on the line. I’m telling you an agent produced an argument with that conclusion”.

All three came back “survives my joint.” C verified the signature blocks and the factor of two, re-implemented the explicit formula independently, showed the Davenport–Heilbronn trap could not bite, and added an observation of its own that later shaped the paper’s statements: replacing inertia by rank in the same argument gives at least three quarters of the zeros distinct, hence at least half simple. A ran its commissioned counterexample hunt (hundreds of synthetic off-line zeros crowding the window’s edges, none on the line) and the certificate stayed negative every time; it also found the one real gap in the write-up — a false premise about the mass matrix of the test system — and proposed the repair, to drop the mass matrix altogether and run the lemmas on the Gram matrix in coefficient coordinates. B reduced the prime-side double sum exactly to the Montgomery–Vaughan mean value theorem with a power saving, so that no RH-strength input enters, and independently noticed the same gap. On the literature the sequence matters and is easily mis-told: E2 had no network access and had named, from memory and with a hedge, an unconditional pair-correlation theorem of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh as the nearest thing it knew; the referees searched those four names, located the paper, and confirmed that E2’s prime-side input is, in smoothed form, its published Theorem 1 — so the entire burden of the claim sits on one unconditional, published asymptotic (the literature appendix, Appendix A, has the details).

Only after the first two verdicts did the coordinator open E2’s proof file itself — “so I’m not just relaying” — and report that it could not break the key step from its chair: signs only on the zero side, magnitudes only on the prime side, where there are no zeros. It then escalated without waiting for the third verdict (“rather than waiting”), ahead of its own stated rule. A literature agent was sent to read the pair-correlation papers in full and ask why their authors had drawn less from the same input (answer: they pass to diagonal terms by termwise positivity

of a kernel, which needs a thin-box hypothesis; it also found that Goldston and Suriajaya state the removal of RH from Montgomery’s simple-zeros argument as an open problem, that the negative-index half of E2’s first theorem is already in Bombieri’s 2000 note, and that E2’s certified zeros are in fact simple *and* on the line). A re-derivation agent, forbidden to read E2’s proof, re-proved the prime-side asymptotics from scratch with error terms, refuting on the way the same false premise the referees had found. A fourth referee, D, was given the five remaining “unglamorous” gaps together with the coordinator’s own analysis of them and was allowed to read the earlier verdicts; it chose neither of the two repairs its brief proposed, took A’s instead, and made it rigorous (the inertia lemma needs no inner product; an exact Poisson identity replaces sampling; explicit tail exponents). The coordinator and, following it, the paper credit this repair to D; by the logs it was proposed by A, noticed independently by B, and proved out by D. With the hardening converging, the coordinator commissioned a journal-style write-up — terse, no salesmanship, main theorem one half, and “simple *and* on the line if that is what the argument actually yields” — from a paper-writing agent, which on its own bookkeeping promoted “simple and on the critical line” to the title theorem, with three quarters distinct as a companion. A cold-read referee allowed to see only that draft later recommended acceptance after minor revision, with the proviso that an editor obtain a further independent expert reading.

8 The push to two thirds

While the write-up was being drafted the human asked whether the paper proved or disproved RH. “Neither. Flatly, unambiguously,” the coordinator answered; a proportion-on-the-line theorem is zero evidence about RH in either direction, “and I’d push back on anyone — including a future version of me — who described it otherwise.” Asked to guess what would be requested next, it guessed correctly — push one half to two thirds, then to one, then RH — and pre-scoped the first step as well posed (exactly the loss from Cauchy–Schwarz against Montgomery’s RH-conditional two thirds), the second as impossible by this method, and the third as not a step at all. Three minutes later the human wrote “Push it to $\frac{2}{3}$.”

The coordinator’s diagnosis was that there are exactly two levers: raise the three quarters by optimising the test window, or make off-line pairs cost less than two-for-one. It briefed one agent on each. The first brief was half a derivation already — Montgomery’s variational functional written out, the Euler–Lagrange equation and its cosine ansatz, the names Montgomery–Taylor and Carneiro–Chandee–Littmann–Milinovich recalled from memory with an arXiv number to fetch. The second brief, for the agent called E2-pairs, is the one that matters for Part II, and it is worth being exact about what it contained, because the coordinator told the human, just after launching it, that while writing it it had found “the actual shape of the $\frac{2}{3}$ problem.” It contained Montgomery’s integer template in full (multiplicities are integers, so $\sum m^2 \geq 2N - N_{\text{simple}}$, whence two thirds simple under RH); the observation that integrality gives five sixths distinct under RH and that five sixths fed through E2’s double count gives exactly two thirds; the diagnosis that E2 had replaced integer multiplicities by real eigenvalues “and lost integrality”; even the free inequality $(\mu - 1)^2 \geq 0$, computed for the positive index of the whole matrix and dismissed as weaker than Cauchy–Schwarz; the judgement that any genuine spectral integrality is “robustly false due to interactions”; a forecast that the negative spectrum alone cannot help; and a steer: higher prime-side moments with restricted support were “the promising unconditional lever.” What it did not contain was any inequality bounding the rank of the on-line part, or von Neumann’s trace inequality. When the human wrote “Keep going”, the coordinator added three more fronts “to compound what’s working rather than scatter”: a generalisation of the inertia argument to other L -functions, a test of what higher prime-side moments give, with and without Hardy–Littlewood-type hypotheses, and a second attempt at

the polarized object using the new tools (which opened the Weil-positivity branch described at the end of this chapter).

The returns closed the coordinator’s preferred routes one by one. The window agent found the optimum in closed form — it is Montgomery–Taylor’s window exactly, the inertia framework’s positivity constraint costs nothing, and the gain is real but tiny (one half becomes 0.5066): “ $\frac{2}{3}$ not reached.” The higher-moments agent returned a clean no-go: unconditional higher moments exist only at bandwidths where the method’s own cap kills everything. The coordinator’s verdict to the human was that two thirds now rested entirely on “the integrality idea in E2-pairs.”

9 An orphaned proof, resumed

E2-pairs had gone silent for an hour and a half after reading its four input files, then spent an hour on numerical experiments that contradicted its brief’s premise (the integer floor for the Frobenius norm survives interactions; every quadratic strengthening fails; the right inequality is linear in the masses), and a little over three hours after launch wrote to disk an abstract matrix inequality with a five-line proof by von Neumann’s trace inequality — the lemma of Part II. Four minutes later the run died of an infrastructure error, mid-sentence, with no application section written. The coordinator read the dead agent’s directory, recognised what the file was (“the most important thing to happen tonight”), checked the five-line proof itself line by line and pronounced it correct, and glossed its lineage: the lemma’s linearisation is literally Montgomery’s integer step $m^2 \geq 2m - 1$ transplanted from integer multiplicities to real eigenvalues, where it holds for free, and von Neumann’s inequality is what separates the on-line and off-line parts so that it can be applied. Within seven minutes it did three things: resumed the same agent with a seven-item checklist for the application (including a per-pair trace estimate it expected to be needed and the question whether the route also gives two thirds *simple*); launched a hostile referee X on the lemma and on the application as the coordinator itself had framed it; and launched a referee Y to re-prove the lemma blind from its bare statement before reading anything. Part II is E2-pairs’ account, including the resumed run, which discarded the checklist’s per-pair item in favour of a one-line bookkeeping, reached exactly two thirds of the zeros distinct and on the line (with the constant $2 - 1/\lambda - \lambda/3$ for general bandwidth), answered the simplicity question honestly in the negative (only one third by this route), and argued that higher moments cannot help.

Y had an independent proof on disk in under two minutes, organised differently and with the equality case, declared the lemma sharp after a hundred thousand random and several hundred adversarial trials, then read E2-pairs’ finished report, checked its bookkeeping numerically and endorsed it; its credence that two thirds follows, given the one-half theorem’s analytic input, was 0.93. X re-derived the lemma with every inequality’s direction checked, falsified termwise a per-pair claim that was in its own brief rather than in the agent’s report (harmlessly, at cost $o(N)$), contributed the caveat that the new inequality is not scale-invariant and so needs an absolute normalisation — and gave the soberest number in the record: lemma at least 0.99, but “ $\geq \frac{2}{3}$ unconditionally” overall about 0.4, everything extraordinary now resting on the unexamined prime-side background.

10 The major revision and the hand-off

With both referees concurring on the lemma, the coordinator sent the paper-writing agent a “major revision” instruction restructuring the draft around the new main theorem, naming as sources E2-pairs’ report and the two verdicts. Three things in that message are the coordinator’s

own: the lemma’s name, “rank–trace inequality” (the agent had called it an abstract theorem; the phrase occurs nowhere earlier); the instruction to include the remark that the linearisation is the real-variable shadow of Montgomery’s integrality step; and the composed constant for the optimised window, $2 - 1.3275 = 0.6725$, to be stated as a theorem if the window agent’s prime-side analysis supported it (the number itself had been in the coordinator’s memory, with a question mark, since before any agent retrieved it). One thing in that message is an attribution slip that the logs correct: the “simplified application bookkeeping” it credits to referee Y was written by the resumed E2-pairs run itself, a quarter of an hour before Y first read it; what the paper genuinely takes from Y is the organisation of the lemma’s proof and its equality case. A second slip runs through the coordinator’s earlier instruction to the paper agent and its later tallies rather than this message: the drop-the-mass-matrix repair, forwarded at the one-half stage as “D’s repair” and so credited in the draft, is, as noted above, A’s proposal made rigorous by D. The paper agent froze the one-half version, rewrote the statements (two thirds distinct on the line; one half simple on the line; three quarters distinct; 0.6725, 0.5066, 0.7533 with the Montgomery–Taylor window; the same three for a fixed Dirichlet character), installed the lemma in Y’s organisation, propagated X’s normalisation, and appended a caveat of its own against overstating the match with the published open problem (theirs is two thirds critical counted with multiplicity and two thirds simple; this is two thirds distinct-on-line and one half simple).

Two minutes after the revision instruction the coordinator wrote the campaign’s ledger entry for the result with its own limiting clause — a candidate, refereed only by other instances of itself, that “needs a human expert” and is not to be described as established until then — and told the human once more what it is not: evidence for or against RH. When the final draft landed, its verdict was “The next reader should be a person.” It later wrote the cover letter and an explainer for the draft itself, declining the human’s offer of an agent for the job on the ground that it knew the result better than any agent would from the files; neither names individual agents, so the slips do not recur there.

11 What the coordinator supplied, and what it did not

Read end to end, the briefs are not task tickets but research memos: they carry the target, a template, the coordinator’s conjectures and its forecast of the outcome. On both creative steps in this volume the division is the same. At the one-half step the coordinator supplied the object (a finite, prime-computable compression of the Weil form read by its signature), the conjecture to prove, and the prize, and pointed the mechanism the wrong way (bound the negative index from above); E2 found that route empty and supplied the mechanism (positive index, double count, Cauchy–Schwarz from primes). At the two-thirds step the coordinator supplied the integer template, the arithmetic from five sixths to two thirds, a correct diagnosis of what had been lost, and a wrong forecast of how to recover it; E2-pairs supplied the inequality, against the brief’s steer. The coordinator’s other contributions are of a different kind and are real: the referee architecture (blind, disjoint joints, worked attack plans, a cold read, re-derivation from scratch), the insistence on RH-false controls in nearly every research brief, its own line-by-line checks at the two decisive moments, the firewall it maintained in every exchange with the human between a proportion theorem and RH, and the names and framing under which the results were written up. The human’s part in this stretch was thin in content and decisive in direction: a handful of one-line prompts, of which “Push it to $\frac{2}{3}$ ” set a target the coordinator had itself guessed three minutes earlier.

12 The campaign’s two other outputs

The session produced two further claims, neither part of this volume. On Weil positivity: the second polarized-object attempt drew, on the coordinator’s instruction, the positivity margin of Weil’s form as a function of the support as primes enter one at a time, at ball-arithmetic precision far beyond the published numerics, and found an empirical $e^{-4\pi X}$ law; a later agent located the mechanism (leakage of a prolate spheroidal eigenvalue) in papers of Connes and Consani that it downloaded and said so; and after every comparative route had come back “no slack,” an agent briefed by the coordinator with a certificate architecture produced a computer-assisted proof, in interval arithmetic and using no zeros of zeta, that the form is positive for supports up to $[1/3, 3]$, where the primes 2, 3, 5, 7 enter — a result of a known type whose RH content its own author called nil, never independently re-run or refereed within the session. On zero density: the rung-one agent’s three thin-interval improvements to the standard zero-density table, obtained by recombining published ingredients one of which the reference implementation had dropped, were confirmed by two blind referees (one of whom repaired an over-stated general form) and recorded by the coordinator as bookkeeping-grade — “a short note, not a paper” (before the referees reported it had said that if the optimiser had genuinely dropped the lemma this would be “a legitimate erratum-grade remark”) — which no one in the session drafted.

Part I

The one-half run

The negative-index route is empty; its dual gives one half

Summary

For a test function f with transform h_f and zeros written $\rho = \frac{1}{2} + i\gamma_\rho$, the Weil Hermitian form $W(f, g) = \sum_\rho m_\rho h_f(\gamma_\rho) \overline{h_g(\bar{\gamma}_\rho)}$ is computable, for $\text{supp } f \subset [-L/2, L/2]$, from the primes $\leq X = e^L$ and the Gamma factor alone via the explicit formula, as $\int |h_f|^2 \nu_X$ with ν_X a signed prime-side “density of states”; its positivity on all tests is equivalent to RH. An earlier run in the campaign (R4) had found a warped-sine operator built from ν_X to be indefinite (33–153 negative eigenvalues, $\nu_X < 0$ on 21–38% of the line) and read the condition space as a Kreĭn space. The orchestrator’s brief asked this run to develop Pontryagin-space theory for W : prove that its negative index κ equals the number of off-line zeros (via the hyperbolic-pair mechanism it sketched), define a prime-computable finite index $\kappa(X, T)$ for $X = 10^2, \dots, 10^6$, bound it from sign changes of ν_X by Turán/Nazarov-type lemmas, and decide whether $\kappa(X, T)/N(T)$ tends to 0, to a constant < 1 (a new route to positive proportion), or to 1 (empty, by Gibbs).

The run proves, by its own account, Theorem 1: κ on C_c^∞ equals the number of *distinct* zeros with $\text{Re } \rho > \frac{1}{2}$, multiplicity being invisible; $\kappa(L)$ is finite, monotone, $\rightarrow \kappa$, and $\text{RH} \iff \kappa(L) = 0$ for all L . It then argues that R4’s operator is not a compression of W , and that the honest index – the inertia of the Gram matrix of W on a Gabor space of windowed exponentials of dimension $\sim (\log X)T$ – is identically 0 for $X = 10^2, \dots, 10^6$ on $[50, 1000]$ and $[1400, 2400]$, computed from primes alone and matching the zero side to 10^{-6} , while $\nu_X < 0$ on 21–42% of the window and the pointwise Szegő/a-priori counts run to hundreds. A visibility Theorem 2 (isolated off-line zero of depth δ at height t visible once $\log X \geq C\sqrt{\log t/\delta}$) is complemented by measured planting thresholds $L_{\text{ann}}^* \approx 1.12 \delta^{-0.07} \log(t_0/2\pi)^{0.89}$ and cluster/multiplicity experiments. The turning point is the dual: pullback inertia $n_+(A^*QA) \leq n_+(Q)$, an off-line pair being a signature- $(1, 1)$ block counted twice by Riemann–von Mangoldt, gives $N_0^{\text{dist}}(I') \geq 2n_+^\theta(W|_V) - N(I')$, with $n_+ \geq (\text{tr } R)^2 / \text{tr } R^2$ prime-computable; an evaluation in the Montgomery–Vaughan range (inequality recalled from memory) yields $\lambda N / (1 + \lambda^2/3)$, whence the claimed Theorem 4: $\liminf N_0^{\text{dist}}([T, 2T])/N \geq \frac{1}{2}$ unconditionally, against the record $5/12$ quoted from memory. Prime-only statistics give $C/N = 0.766, 0.762, 0.757$ at $T = 10^4, 5 \cdot 10^4, 10^6$ versus 0.750; synthetic configurations never over-certify.

The brief’s sign-change route is found empty for structural reasons, not Gibbs: $\kappa(V) \leq N_{\text{off}}^{\text{dist}}$ is the wrong direction, visibility of all zeros needs $\dim V \gtrsim N(T)$ where $\kappa \leq \dim V$ is trivial, interference prevents a configuration-free visibility theorem, and any pointwise bound is ~ 0.2 – $0.4 \dim V$; all three of the brief’s forecast regimes and R4’s Kreĭn headline are contradicted. The run caught 33 spurious negatives from ill-conditioned mass-matrix reduction, an $e^{L/4}$ loss in Theorem 2 requiring a strengthened hypothesis, and wavering in the Montgomery–Vaughan step at $\lambda = 1$; the de Branges δ -tilt is recorded but not developed. Verdict: PARTIAL –

negative-index route empty, its dual claiming one half, referees requested; top caveat “too short to be new”, with end-effect estimates at $\lambda = 1$ and the literature unchecked.

1 The problem as posed: a Pontryagin-index route to counting off-line zeros

This run sits inside a campaign on the Riemann zeta function organised around Weil’s explicit formula and the Hermitian form it defines on test functions — the *Weil form*, whose positivity on all test functions is equivalent to the Riemann Hypothesis. An earlier run in the campaign had found that a certain prime-computable operator built from a truncated explicit formula is indefinite, and read this as saying that the natural “space of conditions” is a Kreĭn space rather than a Hilbert space. The orchestrator’s brief asks me to take that indefiniteness seriously: to develop Pontryagin-space operator theory for the Weil form, prove that its negative index counts zeros off the critical line, define and compute a finite, prime-computable version of that index, and decide whether bounding it from the prime side can give an upper bound on the number of off-line zeros — a zero-density or positive-proportion theorem by a non-classical route — or whether the route is empty.

1.1 The brief: a Pontryagin negative-index route to zero bounds

The brief, from the orchestrator, set out here in full and rewritten for readability:

Reading list, and what the main predecessor found. Read first the report, the working notes and the numerical code of an earlier run in this campaign labelled R4 — a “moonshot” run that tried to construct a polarized arithmetic object directly, returned the verdict “no viable candidate”, proved that any “box minus conditions” space of global sections works if and only if Weil positivity holds, and quantified the obstruction numerically. Concretely, R4 built an intrinsic, zero-free “density of states”

$$\nu_X(\tau) = \frac{\theta'(\tau)}{\pi} - \frac{1}{\pi} \sum_{n \leq X} \Lambda(n) n^{-1/2} \cos(\tau \log n),$$

computable from the prime powers $n \leq X$ and the Gamma factor alone, and showed that the “condition-count” operator associated with it is *indefinite*: it has between 33 and 153 negative eigenvalues in R4’s experiments, and ν_X itself is negative on 21–38% of the critical line in the ranges computed. R4’s reading was that the condition space is a Kreĭn space, not a Hilbert space; and R4 proved that this datum yields a Hodge index theorem, hence the Riemann Hypothesis, if and only if the condition count were a genuine (non-negative) dimension, which is exactly Weil positivity. Read also the report of the earlier run labelled N6 (a “new area” run on an explicit-formula bootstrap, which analysed the cone of test functions on which the Weil form is manifestly positive and proved a Harnack inequality for it), and the report of the run labelled frontier-1 (a first-wave structural probe that framed the Weil pairing as an intersection form and looked for an honest integer analogue of the Hodge index theorem).

The idea to develop: Pontryagin-space operator theory. A Pontryagin space Π_κ is a Kreĭn space whose negative index — the dimension of a maximal negative subspace — is a finite number κ . The theorem of Pontryagin (1944) and Kreĭn–Langer says: a self-adjoint operator in Π_κ has at most κ eigenvalues off the real axis, counted with multiplicity and occurring in conjugate pairs (for a unitary

operator: at most κ eigenvalues off the unit circle), and its spectral function is “positive up to κ squares”. Equivalently, a Hermitian kernel with κ negative squares — the class P_κ , or the generalized Nevanlinna class N_κ of Kreĭn–Langer — admits an integral representation with at most κ poles of the “wrong sign”. In the definite case $\kappa = 0$, de Branges/Kreĭn theory is the Hilbert–Pólya picture. The *indefinite* version one should aim for is this. Take the natural Hermitian form attached to Ξ — equivalently the Weil form, or the de Branges kernel

$$\frac{E(z)\overline{E(w)} - E^*(z)\overline{E^*(w)}}{z - \bar{w}}$$

with structure function E built from ξ , or R4’s “condition space” form — and restrict it to a natural family of finite-dimensional or band-limited subspaces indexed by a prime cutoff X and a height T . If the restricted form has negative index $\kappa(X, T)$, then the number of non-real zeros of the corresponding structure function that are “visible” to that subspace should be at most $\kappa(X, T)$. And if $\kappa(X, T)$ can be bounded by something computable from the primes $\leq X$ and the Gamma factor alone, that is an *upper bound on the number of off-line zeros in a height window* — a zero-density estimate by a totally non-classical route.

Known related facts to get right. (i) Weil’s criterion: the Riemann Hypothesis is equivalent to the Weil form being non-negative on all test functions, i.e. to $\kappa = 0$. (ii) More precisely — and here the attribution is uncertain: Yoshida? Bombieri? a title of the shape “Weil’s explicit formula and the number of zeros off the line”? — the negative index of the Weil form on all test functions should *equal* the number of zeros off the critical line, counted with multiplicity and in pairs. I believe this is a theorem, for the following reason. On the zero side the Weil form is $\sum_\rho \hat{h}(\rho) \overline{\hat{h}(1 - \bar{\rho})}$. For a zero ρ on the line, $1 - \bar{\rho} = \rho$ and the term is $|\hat{h}(\rho)|^2$, a positive square. An off-line zero comes as a quadruple $\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}$, and the two terms pairing ρ with $1 - \bar{\rho}$ and $1 - \bar{\rho}$ with ρ together form a rank-two Hermitian form of the shape $2\operatorname{Re}(a\bar{b})$, a hyperbolic pair of signature $(1, 1)$; so each off-line pair should contribute exactly one negative square. Verify or prove this cleanly, in the form: the negative index of W on $C_c^\infty(\mathbb{R}_+^*)$ equals the number of off-line zero pairs (possibly infinite). So globally $\kappa = N_{\text{off}}$ exactly. The finite- X , finite- T version: define $\kappa(X, T)$ as the negative index of W restricted to test functions g with $\operatorname{supp} \hat{g} \subset [-\log X, \log X]$ and “concentrated below height T ” — this must be made precise, e.g. g a convolution of a kernel with a band-limiting function, or via R4’s prolate compression. Then one expects $\kappa(X, T) \leq N_{\text{off}}(T')$ for an appropriate T' ; and conversely — with question marks — N_{off} in a window $\leq \kappa(X, T)$ plus a count of “invisible” zeros? The useful direction is

$$N_{\text{off}}(\sigma > \tfrac{1}{2} + \delta, T) \leq \kappa_\delta(X, T),$$

where κ_δ counts the negative squares of a δ -tilted form, $W_\delta(g) = W(g \cdot \cosh(\delta \cdot))$ or similar, arranged so that zeros with $|\operatorname{Re} \rho - \frac{1}{2}| > \delta$ contribute strictly negatively while on-line zeros contribute positively. Compare the weights in N6’s analysis: an on-line zero is weighted by a sech-type factor, an off-line zero at depth a by $\cosh(ax)$, so a tilt by $e^{\delta|x|}$ flips ... (the sentence is left unfinished).

Tasks. (1) Prove the global identity “negative index of $W = N_{\text{off}}$ ”, or find its correct statement in the literature; places to look are “Weil’s quadratic functional negative index”, Bombieri’s 2000 paper “Remarks on Weil’s quadratic functional in the theory of prime numbers”, Yoshida’s work in the 1990s, and Burnol.

(2) Define $\kappa(X, T)$ concretely and *computably from the prime side only* — either via R4’s ν_X operator or via the Gram matrix of the Weil form in N6’s normalisation on a Paley–Wiener or prolate basis of dimension roughly $(\log X) \cdot T$ — compute it numerically for $X = 10^2, \dots, 10^6$ and $T = 10^2, \dots, 10^4$, and fit its growth. Is $\kappa(X, T) \asymp c \cdot (\text{a fraction}) \cdot N(T)$? $\asymp T/\log X$? $\asymp T \log T/\log X$? R4 saw 21–38% negative density-of-states mass, which suggests $\kappa(X, T) \sim c(X)N(T)$ — with $c(X)$ decreasing in X , or increasing? R4’s fraction went from 21% to 38% as X grew from 100 to 10^4 , increasing, which is a bad sign; but that was the fraction of the *line* on which $\nu_X < 0$, not the negative index of the compressed operator. Compute the actual index.

(3) The key theoretical question. Is there an a priori bound $\kappa(X, T) \leq B(X, T)$ provable from primes and the Gamma factor alone? For example, by counting sign changes of ν_X , which is an explicit trigonometric-type sum: the number of negative excursions of ν_X on $[0, T]$ is at most the number of zeros of ν_X there, which by a Turán- or Nazarov-type lemma for exponential sums with frequencies $\log n$, $n \leq X$, should be at most $CT \log X/2\pi$ plus an X -dependent term. And is there a rigorous inequality $N_{\text{off}}(\text{window}) \leq \kappa(X, T) + E(X, T)$ with a controlled error E ? If both hold, one gets $N_{\text{off}}(T) \leq CT \log X + E$; choose X to optimise, and compare with the trivial bound $N_{\text{off}}(T) \leq N(T) \sim (T/2\pi) \log T$ and with the classical zero-density theorems $N(\sigma, T) \ll T^{A(1-\sigma)+\varepsilon}$. Does the Pontryagin route give anything non-trivial at all? Even an unconditional $N_{\text{off}}(T) \leq (1 - c)N(T)$ would be new-ish in flavour, though weaker than Selberg’s positive-proportion theorem. Note that the Selberg/Levinson/Conrey results “at least 40% of the zeros are on the line” are exactly the statement $N_{\text{off}} \leq 0.6 N(T)$: can the negative-index route *reproduce* a positive-proportion result? That would be a genuinely new proof architecture for Selberg’s theorem even if the constant is worse — worth a lot.

(4) **Red-team.** The danger is that the restricted forms see the *on-line* zeros as negative too: the finite- X truncation makes ν_X dip below zero near *every* zero (a Gibbs phenomenon), so that $\kappa(X, T) \sim N(T)$ trivially and the bound is empty. Check R4’s numbers — 33 to 153 negative eigenvalues against how many zeros in the window? — and the ratio. If $\kappa(X, T)/N_{\text{window}} \rightarrow \text{a constant} < 1$ as $X \rightarrow \infty$, that is the positive-proportion route; if it tends to 0, the density route; if it tends to 1, the route is empty. Determine which, numerically and then theoretically.

Deliverable. A report with code; a final summary with the verdict first, in one of the forms “PONTRYAGIN ROUTE VIABLE: negative-index identity proven (global); $\kappa(X, T)/N(T) \rightarrow c = \dots$; unconditional consequence: $\dots$ ” — in which case hostile referees follow if any theorem is claimed — or “EMPTY: $\kappa(X, T) \sim N(T)$ because Gibbs $\dots$, proof”, or “PARTIAL: $\dots$ ”.

So the frame is all given to me: Kreĭn/Pontryagin spaces, the identification of negative squares of the Weil form with off-line zeros, the hyperbolic-pair mechanism behind it, the δ -tilt, and the three-way forecast for the ratio κ/N . What is asked of me is a clean proof of the global identity, an honest and computable finite-level index, a prime-side bound on it, and a verdict on whether the route is empty.

1.2 The objects and vocabulary of the brief, explained

For a reader outside this corner, here is what the brief’s terms mean, in the notation I will use below, and where each came from.

The Weil form and the explicit formula. For a test function f on the line (variable u ,

the logarithm of the idele norm; the brief writes $C_c^\infty(\mathbb{R}_+^*)$, the multiplicative picture), with transform $h_f(r) = \int f(u)e^{iru} du$, and zeros written $\rho = \frac{1}{2} + i\gamma_\rho$ with multiplicity m_ρ , the Weil Hermitian form is

$$W(f, g) = \sum_{\rho} m_{\rho} h_f(\gamma_{\rho}) \overline{h_g(\bar{\gamma}_{\rho})}.$$

Weil’s explicit formula (unconditional) evaluates $W(f, f)$ as a pole term plus an archimedean (Gamma-factor) term minus a sum over prime powers weighted by the von Mangoldt function $\Lambda(n)$; if f is supported in an interval of length L , only $n \leq X = e^L$ enter. “Weil positivity” means $W(f, f) \geq 0$; Weil’s criterion is that this holds for all f if and only if RH. “Semi-local Weil positivity at level X ” is the campaign’s name (from R4’s report) for positivity on test functions of support length $\leq \log X$. The normalisation I adopt is the one N6’s report validated, Iwaniec–Kowalski (5.12), recorded below.

θ , ν_X , $N(T)$. θ is the Riemann–Siegel theta function, so $\theta'(\tau)/\pi \sim \frac{1}{2\pi} \log(\tau/2\pi)$ is the smooth density of zeros; $N(T)$ is the Riemann–von Mangoldt count of zeros with ordinate in $(0, T]$, with multiplicity; $N(\sigma, T)$ counts those with real part $> \sigma$, and N_{off} those off the critical line. R4’s ν_X is the smooth density minus the truncated prime sum: by the explicit formula, $\int |h_f|^2 \nu_X$ (plus a pole/mean correction) equals $\sum_{\gamma} |h_f(\gamma)|^2$ exactly for f of support length $\leq \log X$, which is why R4 called it a zero-free “density of states”. It is a signed function; its negativity on 21–38% of the line is R4’s measurement.

Kreĭn and Pontryagin spaces; negative index; negative squares. A Kreĭn space is a vector space with a non-degenerate indefinite Hermitian inner product admitting a decomposition into a Hilbert space minus a Hilbert space; it is a Pontryagin space Π_{κ} when the negative part is finite-dimensional, of dimension κ . For a Hermitian form, the negative index is the supremum of the dimensions of subspaces on which it is negative definite; “ κ negative squares” of a kernel means its Gram matrices have at most, and some have exactly, κ negative eigenvalues. “Hilbert–Pólya” is the hoped-for self-adjoint operator whose spectrum is the zeros; the brief’s point is that the $\kappa = 0$ case of the indefinite theory is that picture.

De Branges kernel, structure function, Hermite–Biehler. A structure function E is an entire function with $|E(z)| > |E^*(z)|$ on the upper half-plane, where $E^*(z) = \overline{E(\bar{z})}$ (the Hermite–Biehler condition); its kernel $[E(z)\overline{E(w)} - E^*(z)\overline{E^*(w)}]/(2\pi i(\bar{w} - z))$ is then positive definite and reproduces a Hilbert space of entire functions. When the condition fails at finitely many points the kernel has finitely many negative squares (Kreĭn–Langer). “ E from ξ ” means building E out of the completed zeta function, as in $E_a(z) = \xi(a - iz)$, which I return to late in the run; R4 built a truncated E_X from an Euler product.

R4’s “condition space”, “condition-count operator”, “prolate compression”. In R4’s dictionary (inherited from a still earlier run labelled D2, which tried to reverse-engineer the missing h^0 of an arithmetic Riemann–Roch), a test function plays the role of a divisor and $C_X(g) = \frac{1}{2} \int |\hat{f}_g|^2 \nu_X$ is “the number of conditions imposed”; the condition space is the space whose dimension this should be. R4’s condition-count operator, which I describe precisely below, is a warped sine kernel on L^2 of a height window with diagonal ν_X , in the style of the Landau–Pollak–Slepian prolate (time- and band-limiting) operators; “compression” means restriction of a form or operator to a subspace, here a finite-dimensional or band-limited one. A “PW basis” is a Paley–Wiener (band-limited) basis; a “Gram matrix” is the matrix $[W(f_k, f_l)]$ of the form on a chosen family.

Hodge index, intersection form. The campaign’s geometric analogy: in the function-field proof of RH the intersection pairing on a surface has Lorentzian signature (Hodge index theorem, Castelnuovo–Severi), and frontier-1’s report reads W as such a pairing over $\mathbb{Z}$; “Lorentzian $\Leftrightarrow$ RH” is the same statement as $\kappa = 0$ up to the pole directions.

The δ -tilt and N6’s W_a weights. Multiplying a test function by $\cosh(\delta u)$ shifts the evaluation points of its transform by $\pm i\delta$, so it reweights zeros according to their distance

from the line; N6’s report showed that on its positive cone an on-line zero carries weight $\text{sech}(x/2)$ -type and an off-line zero at depth a an extra $\cosh(ax)$. The brief’s W_δ is a proposal, not a defined object; I record a de Branges realisation of it near the end.

Turán/Nazarov lemma; Gibbs. Turán-type lemmas bound the number of zeros of an exponential sum on an interval by its length times the top frequency plus the number of terms; the brief proposes this to count sign changes of ν_X . “Gibbs” refers to the overshoot of a truncated Fourier-type expansion near a jump: the truncated prime sum oscillates near each zero ordinate, and the worry is that this alone produces negative squares.

Selberg/Levinson/Conrey. The classical theorems that a positive proportion (Selberg), at least a third (Levinson), at least about 40% (Conrey) of the zeros lie on the critical line, proved by mollifier methods; the brief’s benchmark is to recover any positive proportion by the new route.

(An automatic note lists the thirty-odd other agents active in the campaign and reachable by [message](#); nothing in it bears on the mathematics.)

1.3 R4’s signed density and its indefinite operator

I’ll start by reading everything on the list, beginning with R4, since R4’s signed density ν_X and its indefinite operator are the objects the brief wants me to turn into a Pontryagin index.

R4’s report first. Its verdict is “NO VIABLE CANDIDATE” for an intrinsic, zero-free space of global sections H^0 ; what matters for me is the object underneath. R4’s “number of conditions imposed by the twist” attached to a test function g is

$$C_X(g) = \frac{1}{2} \int |\hat{f}_g(\tau)|^2 \nu_X(\tau) d\tau, \quad X \geq e^{\text{diam supp } g},$$

with ν_X as in the brief plus a correction for the prime-number-theorem mean. R4 realises this datum three ways — a warped-sine prolate operator, a Toeplitz/winding staircase $\tilde{N}_X = \theta/\pi + 1 + S_X$, and the de Branges phase of a structure function E_X built from a truncated Euler product — and shows all three to be the same ν_X . The reduction (R4’s Lemma 2.2 and Theorem 2.3) is that the Riemann–Roch inequality is empty at the level of formulas — it holds for any real C , signed or not — and all content sits in the statement that $C_X(g)$ is a dimension, i.e. $C_X(g) \geq 0$ for all g with support of length $\leq \log X$. That is semi-local Weil positivity at level X ; uniformly in X it is RH.

The operator the brief refers to is R4’s Candidate A, the kernel

$$S_X(t, s) = \frac{\sin(\pi(\tilde{N}_X(t) - \tilde{N}_X(s)))}{\pi(t - s)}$$

on L^2 of a height window, whose diagonal is ν_X , discretised by the Nyström method. On the window $[880, 980]$, where the true zero count is 79, R4’s table reads

X	$\#\{\text{eig} > 1/2\}$	min eigenvalue	$\#\{\text{eig} < -0.01\}$
1	80	0.00	0
10	79	−0.39	12
10^3	88	−0.67	115
10^4	83	−0.55	153

while the trace stays at ≈ 79 throughout. R4’s reading: “Verdict A: von Neumann (trace) dimension = D2’s $C(g)$ exactly; integer Landau dimension wrong for $X \geq 10$; the condition space is Krein.” (D2 is the earlier run whose h^0 dictionary R4 uses.) The failure is quantified: $\nu_X < 0$ on 1.4% ($X = 5$), 21% ($X = 100$), 38% ($X = 10^4$) of $\tau \in [20, 1000]$, in dips of width

$\sim 1/\log X$ (1123 of them at $X = 10^4$) and mass ≤ 0.50 , “exactly at the uncertainty scale of admissible tests”.

Two further items I register. In its red-team section R4 states that all positivity it found — “Fejer minima > 0 , envelope $= N$ ” — is RH being true in the computed range. And its planting experiment, in which two consecutive zeros are replaced by an off-line pair $\frac{1}{2} \pm \eta + i\gamma_0$, finds that the zero-side Weil form on test functions of support L first acquires a negative direction at

$$\begin{aligned} L^* &= 1.26, 2.09, 2.68, 3.44 & (\gamma_0 = 28, 100, 399, 991; \eta = 0.49), \\ L^* &= 2.94, 4.78, 6.20, 7.08 & (\text{same } \gamma_0; \eta = 0.02), \end{aligned}$$

i.e. $L^* \sim (0.65\text{--}0.85) \log(\gamma_0/2\pi)$, nearly independent of η . R4 explains this as the test space nearly annihilating the on-line zeros in a frequency window (Landau density), leaving the indefinite $\cosh(\eta(x_j - x_k))$ term unopposed. R4 also checked that the negative index of its prolate operator is not a grid artefact: on $[880, 930]$, negative index 55 at $X = 10^3$ and 77 at $X = 10^4$ against a true count of 40, identical across step sizes.

Now R4’s working notes. They fix the dictionary inherited from D2 — classes $(d_1, d_2; A)$, the vector $\Gamma_g = (G(1), G(0); (G(\rho))_\rho)$ with $G(s) = \int g e^{st}$, and $\chi(D) = d'_1 d'_2 - \frac{1}{2} \sum_\rho |G(\rho)|^2$ — and record the explicit formula in the form I will need. With $g(t) = e^{-t/2} f(t)$ one has $G(\frac{1}{2} + i\tau) = \hat{f}(\tau)$ and

$$\begin{aligned} \sum_\rho |G(\rho)|^2 &= \sum_\gamma |\hat{f}(\gamma)|^2 = \int |\hat{f}|^2 dN \\ &= 2G(0)G(1) - \sum_n \Lambda(n) n^{-1/2} [(f * \tilde{f})(\log n) + (f * \tilde{f})(-\log n)] + \text{Arch}(f * \tilde{f}), \end{aligned}$$

so that for $\text{supp}(f * \tilde{f}) \subset [-\log X, \log X]$ only $n \leq X$ enter and, integrating by parts, $\int |\hat{f}|^2 dN = \int |\hat{f}|^2 dN_X$ exactly, the measure dN_X having density $\theta'/\pi - \frac{1}{\pi} \sum_{n \leq X} \Lambda(n) n^{-1/2} \cos(\tau \log n)$.

The notes then explain the pointwise negativity. The supremum of the prime sum is $\frac{1}{\pi} \sum_{n \leq X} \Lambda(n)/\sqrt{n} \sim \frac{2}{\pi} \sqrt{X}$, which exceeds $\frac{1}{2\pi} \log(\tau/2\pi)$ at resonance points once $X \gtrsim (\log \tau)^2/16$; so “ $dN_X \geq 0$ ” fails pointwise for every $X \geq 2$ somewhere in every long window, and positivity survives only after testing against admissible $|\hat{f}|^2$, which cannot localise into dips of width $\ll 1/\log X$ — and that averaged positivity is semi-local Weil positivity at level X . The regime in which pointwise positivity is trivial is quantified by

$$T_{\text{triv}}(X) = 2\pi \exp \left(2 \sum_{n \leq X} \Lambda(n)/\sqrt{n} \right) = 17, 59, 500, 7300 \quad (X = 2, 3, 5, 10).$$

Their numerical section records the staircase facts: $\text{RMS}(\tilde{N}_X - N)$ falls $0.38 \rightarrow 0.15$; the integer running-maximum envelope equals $N(T)$ at all 646 zero midpoints for every $X \leq 10^4$; the PNT mean must start at $u = 2$, not $u = 1$, or it fakes a $+1/2$. It records the Fejér-averaged density with $\text{supp } f = [-L/2, L/2]$, $L = \log X$, whose minimum over $\tau_0 \in [60, 940]$ is positive for all X and equals $\sum_\gamma h(\gamma)$ to $10^{-2}\text{--}10^{-3}$, with ratio min/mean falling from 0.54 ($X = 2$) to 0.013 ($X = 10^3$); the match between the Hermite–Biehler defect of the structure function E_X and ν_X ; the planting law again; and the statement I most need to confront: the untwisted S_X “has negative eigenvalues iff ν_X dips ($X \geq 5$): the ‘space of conditions’ is a Krein space, not a Hilbert space, at every finite $X \geq 5$; its positive+negative index reproduces the height count.”

Then R4’s code, to see exactly what was computed. The core routine implements ν_X and N_X from a table of 700 zero ordinates and the von Mangoldt sum, with the mean correction $\frac{1}{\pi} \text{Re}(X^s - 2^s)/s$, $s = \frac{1}{2} + i\tau$. The prolate computation builds S_X by Nyström with weight

equal to the grid step, sets the diagonal to ν_X , symmetrises and diagonalises, and tabulates $\#\{\text{eig} > 1/2\}$, the trace, the minimum eigenvalue and $\#\{\text{eig} < -0.01\}$ on three windows, together with a twisted version $h^{1/2}S_X h^{1/2}$ whose trace is compared with $\sum_\gamma h(\gamma)$. The staircase computation produces the negativity fractions, the dips, the envelope, and the Fejér minima with the kernel $L \text{sinc}^2((\tau - \tau_0)L/2)$. The point I take away for later: the operator whose negative eigenvalues R4 counted is this warped-sine kernel on a height window, not a matrix of values of the Weil form on support-limited test functions.

1.4 N6's normalisation and frontier-1's intersection form

Next the N6 report. Its verdict is a NO-GO of a different kind — saturation of the “bootstrap” cone of strip-positive test functions — but it carries the normalisation I will build on. The explicit formula is fixed there in the form of Iwaniec–Kowalski (5.12): for even real g with $\text{supp } g \subset [-L, L]$ and $h(r) = \int g(x)e^{irx} dx$,

$$\sum_\rho h(\gamma_\rho) = h(i/2) + h(-i/2) - g(0) \log \pi + \text{Arch}(g) - 2 \sum_n \Lambda(n) n^{-1/2} g(\log n),$$

$$\gamma_\rho = (\rho - \tfrac{1}{2})/i,$$

$$\begin{aligned} \text{Arch}(g) &= \frac{1}{2\pi} \int h(r) \text{Re } \psi\left(\frac{1}{4} + \frac{ir}{2}\right) dr \\ &= -\gamma_E g(0) + \int_0^\infty \frac{2[g(0)e^{-2y} - g(y)e^{-y/2}]}{1 - e^{-2y}} dy, \end{aligned}$$

reproduced by N6 to 30 digits against 120 zeros and a Gaussian (and the x -side archimedean term to 23 digits). On the zero side a zero $\frac{1}{2} + a + it$ with its symmetry partners contributes $2 \text{Re } h(t + ia)$ if $a = 0$ and $4 \text{Re } h(t + ia)$ if $a \neq 0$, where $\text{Re } h(t + ia) = \int g(x) \cosh(ax) \cos(tx) dx$.

N6's Proposition 1 solves the strip-positive cone completely:

$$\{g : \text{Re } h(t + ia) \geq 0 \text{ on } |a| \leq \tfrac{1}{2}\} = \{k(x) \text{sech}(x/2) : k \text{ positive-definite}\},$$

by the maximum principle and the positive-definiteness of $W_a(x) = \text{sech}(x/2) \cosh(ax)$ for $|a| \leq \frac{1}{2}$ — these are the “ W_a weights” the brief mentioned. Theorem A is the sharp Harnack inequality $\text{Re } h(t + ia) \leq h(t)/\cos(\pi a)$ on that cone, and Theorems B and C turn it into saturation: a point $\frac{1}{2} + \delta + i\gamma_0$ is excludable by any strip-positive certificate only if $\cos(\pi\delta) \sum_k e^{-\pi|\gamma_k - \gamma_0|} < 2$, so that even with RH and all primes nothing on the line above $\gamma_0 \sim 2\pi e^{4\pi}$ can be excluded, the finite- N reach being $T \sim 2\pi N^2$ (Riemann–Siegel scaling), with a certified speck at height 10^{13} from $N = 10^8$ prime powers. What I will actually use from N6 is the validated normalisation and its table of 2000 zero ordinates; the saturation theorems are context — they say that sign-indefinite test functions are what is missing, which is at least the right neighbourhood for a signature question.

Last on the list, frontier-1's structural report, verdict “NO NEW GROUND”. The object of interest to me is its first construction: on real test functions $f(u)$, with $F(s) = \int f(u)e^{(s-1/2)u} du$, the pairing

$$f \cdot g := F(0)G(1) + F(1)G(0) - \sum_\rho F(\rho)G(1 - \rho)$$

read as an intersection form, with radical $\{F(0) = F(1) = 0, F(\rho) = 0 \forall \rho\}$, an abstract Hodge-index theorem (their Theorem AC: a pointed closed convex cone containing the forward light cone forces Lorentzian signature), and the “collapse” remark that any such cone axiom is either vacuous under RH or false — their positive-measure cone being falsified quantitatively, the constant $\kappa_N := \sup \sum |c|^2/(AB)$ satisfying $\kappa_N \geq 1.013, 1.08, 1.66, 2.98$ for $N = 1, 3, 8, 20$ zeros. The one line that is directly my business: “Lorentzian $\Leftrightarrow$ RH: standard (Weil; an

off-line quadruple contributes $-4 \operatorname{Re} F(\rho)^2$ for even f , indefinite) — the same hyperbolic-pair mechanism the brief sketched, stated for even test functions. That completes the reading.

2 The global signature theorem and the honest finite index

2.1 Zero data in hand; the plan

I check what zero data I have to validate against: R4’s table of ordinates (starting 14.1347..., 21.0220..., 25.0108...) and N6’s list of 2000 zeros running up to ≈ 2515.29 . The latter will be my zero side throughout.

The shape of the answer I am heading for, laid down before any of it exists: first the set-up and the global signature theorem; then the finite-level index — which I am calling $\kappa(L)$ and $\kappa(L, T)$, with L the support length of the test functions rather than the brief’s $X = e^L$ — treated under the heading of its *correct definition* and its relation to R4’s operator. So I do not intend simply to take R4’s warped-sine operator as “the compressed Weil form” (the restriction of the Weil Hermitian form to a finite-dimensional space of admissible test functions), which is one of the two objects the brief’s item (2) offered. After that: numerics; the visibility of an off-line zero at finite level; the question whether κ can be bounded from the prime side, which I expect to be an obstruction to be quantified rather than a bound to be found; and a red team.

Now the theory: the signature theorem and the correct finite-level object.

2.2 Theorem 1: negative index counts distinct off-line zeros

First the conventions I will use throughout. Test functions f live on the line with variable u (the logarithm of the idele norm), and are either of Schwartz type with $e^{(1/2+\varepsilon)|u|} f^{(j)} \in L^1$ for $j \leq 2$, or in C_c^∞ . The transform is

$$h_f(r) = \int f(u) e^{iru} du,$$

entire and rapidly decreasing on horizontal strips. Zeros are written $\rho = \frac{1}{2} + i\gamma_\rho$ with $\gamma_\rho \in \mathbb{C}$, $|\operatorname{Im} \gamma_\rho| < \frac{1}{2}$, multiplicity m_ρ ; the multiset of γ_ρ is invariant under $\gamma \mapsto \bar{\gamma}$ and $\gamma \mapsto -\gamma$. With the involution $\tilde{f}(u) = \overline{f(-u)}$, for which $h_{\tilde{f}}(r) = \overline{h_f(\bar{r})}$, the Weil Hermitian form is

$$W(f, g) := \sum_{\rho} m_{\rho} h_f(\gamma_{\rho}) \overline{h_g(\bar{\gamma}_{\rho})},$$

absolutely convergent, and Hermitian because the γ -term maps to the $\bar{\gamma}$ -term under swapping f, g and conjugating.

On the diagonal, $W(f, f) = \sum_{\rho} m_{\rho} h_{f * \tilde{f}}(\gamma_{\rho})$ is the explicit formula applied to $k = f * \tilde{f}$: pole term, minus $\sum_n \Lambda(n) n^{-1/2} [k(\log n) + \overline{k(-\log n)}]$, plus the archimedean term, in the Iwaniec–Kowalski (5.12) normalisation that N6 validated. If $\operatorname{supp} f \subset [-L/2, L/2]$ then $\operatorname{supp} k \subset [-L, L]$ and only $n \leq X := e^L$ enter. So the restriction of W to $C_c^\infty(-L/2, L/2)$ is computable from the primes $\leq X$ and Γ alone, and in R4’s spectral form it reads

$$W(f, f) = \int |h_f(\tau)|^2 d\tilde{N}_X(\tau), \quad d\tilde{N}_X = \nu_X d\tau + (\text{pole measure}).$$

Weil’s criterion: RH holds iff $W(f, f) \geq 0$ for all f .

Now the brief’s item (1). Let κ be the negative index of W on $\mathcal{S}$ (or on $C_c^\infty(\mathbb{R})$), i.e. the supremum of $\dim U$ over finite-dimensional subspaces U on which W is negative definite.

Theorem 1 (global signature). $\kappa = \#\{\rho : \zeta(\rho) = 0, \operatorname{Re} \rho > \frac{1}{2}\}$ counted *without* multiplicity (distinct zeros), as an element of $[0, \infty]$; equivalently, κ is the number of distinct γ_ρ with $\operatorname{Im} \gamma_\rho < 0$.

The brief conjectured “negative index = N_{off} ” and sketched the mechanism; the statement I can prove differs in one word. The proof has four parts.

(a) The hyperbolic pair. Group the zeros into on-line ones (γ real) and off-line pairs $\{\gamma, \bar{\gamma}\}$, the two members of a pair having equal multiplicity m . Writing $l_z(f) := h_f(z)$ for the evaluation functional and $\gamma^* := \bar{\gamma}$,

$$W(f, f) = \sum_{\text{on}} m |l_\gamma(f)|^2 + \sum_{\text{pairs}} 2m \operatorname{Re} (l_\gamma(f) \overline{l_{\gamma^*}(f)}),$$

and the rank-two form

$$(a, b) \mapsto 2 \operatorname{Re}(a\bar{b}) = \frac{1}{2}|a+b|^2 - \frac{1}{2}|a-b|^2$$

has signature $(1, 1)$. This is exactly the hyperbolic pair the brief described, and frontier-1’s “ $-4 \operatorname{Re} F(\rho)^2$, indefinite”.

(b) Upper bound. Write $W = P - N$ with

$$\begin{aligned} P &= \sum_{\text{on}} m |l_\gamma|^2 + \sum_{\text{pairs}} \frac{m}{2} |l_\gamma + l_{\gamma^*}|^2 \geq 0, \\ N &= \sum_{\text{pairs}} \frac{m}{2} |l_\gamma - l_{\gamma^*}|^2. \end{aligned}$$

If $W|_U < 0$ then $N|_U > 0$, so $\dim U \leq \operatorname{rank} N|_U \leq \#\text{pairs}$. The point the brief’s phrasing “(with multiplicity, pairs)” misses: m enters as a *weight*, not as extra rank — an off-line zero of multiplicity m contributes *one* negative square. This is forced, since W sees the zeros only through the measure $\sum_\rho m_\rho \delta_{\gamma_\rho}$; no derivative $h'(\gamma)$ ever occurs.

(c) Lower bound. This needs an explicit construction, which is my own part. Given distinct off-line zeros $\gamma_j = t_j - i\delta_j$, $\delta_j > 0$, $j = 1, \dots, k$, I want test functions $f_1, \dots, f_k$ whose Gram matrix under W is negative definite. Take the Gaussian $A_s(r) = e^{-r^2/(2s^2)}$ and

$$h_j(r) = A_s(r - t_j) (r - t_j) q_j(r),$$

with q_j a polynomial independent of s ; then f_j is (polynomial $\times$ Gaussian) $\cdot e^{-it_j u}$, which is admissible. At the pair $\gamma_j, \bar{\gamma}_j$, writing $y = \delta_j$ and $A = A_s(\pm iy) = e^{y^2/(2s^2)}$, the two values $a = h_j(\gamma_j)$, $b = h_j(\bar{\gamma}_j)$ satisfy

$$a\bar{b} = -y^2 A^2 q_j(\gamma_j) \overline{q_j(\bar{\gamma}_j)}.$$

I choose $q_j = p_j c_j$. Here p_j is the product of $(r - z)$ over all points z of the orbit $S \cup \bar{S} \cup (-S) \cup (-\bar{S})$ of the given set $S = \{\gamma_1, \dots, \gamma_k\}$, except the two points $\{\gamma_j, \bar{\gamma}_j\}$. I have to be careful that $-\bar{\gamma}_i = -t_i - i\delta_i$ is a genuinely different point, and that omitting a conjugation-symmetric pair keeps $p_j(\bar{r}) = \overline{p_j(r)}$, so that

$$p_j(\gamma_j) \overline{p_j(\bar{\gamma}_j)} = p_j(\gamma_j)^2 =: w_j \neq 0.$$

The factor $c_j(r) = (r - z_j)(r - \bar{z}_j)$ is a quadratic with real coefficients, with $v_j := c_j(\gamma_j)^2$ tuned so that $\arg v_j = -\arg w_j$; this is possible because the map $z \mapsto (\gamma_j - z)(\gamma_j - \bar{z})$ fills a

region containing points of every argument, and squaring doubles arguments. Then the pair contributes

$$2m_j \operatorname{Re}(a\bar{b}) = -2m_j \delta_j^2 e^{\delta_j^2/s^2} |w_j v_j| < 0,$$

growing as $s \rightarrow 0$, while everything else vanishes.

Indeed, on-line zeros at real distance x from t_j contribute $e^{-x^2/s^2} x^2(\dots) \leq s^2/e \rightarrow 0$, and the sum tends to zero by dominated convergence (counting function $O(T \log T)$ against Gaussian decay). The rest of the orbit is killed exactly by p_j . To keep the argument unconditional I take $S = S_T$ to be *all* distinct off-line zeros with $|\operatorname{Re} \gamma| \leq T'$, where $T' \geq \max_j |t_j| + 1$ (a finite set, since $N(T') < \infty$); then any remaining off-line zero has $|\operatorname{Re} \gamma - t_j| \geq 1 > \frac{1}{2} > |\operatorname{Im} \gamma|$ and contaminates at rate $e^{-3/(8s^2)}$, which loses to the main term $e^{\delta_j^2/s^2}$ precisely because $\delta_j < \frac{1}{2}$ makes $\delta_j^2 < \frac{3}{8}$. Off-diagonal Gram entries vanish on the orbit and tend to zero elsewhere by Cauchy–Schwarz; after normalising each f_j by $e^{-\delta_j^2/(2s^2)}$ the Gram matrix tends to

$$\operatorname{diag}(-2m_j \delta_j^2 |w_j v_j|) < 0.$$

Hence $\kappa \geq |S_T|$ for every T .

(d) Passage to compact support. W is continuous for the norm

$$\|f\|_* = \sum_{j \leq 2} \int |f^{(j)}(u)| e^{|u|/2} (1 + u^2) du,$$

since $|h_f(\gamma)| \leq \|f\|_*/(1 + |\gamma|^2)$ uniformly on the strip and $(1 + |\gamma|^2)^{-2}$ is summable over the zeros. The Gaussian-polynomial test functions of (c) are $\|\cdot\|_*$ -limits of compactly supported ones, and negative definiteness of a finite Gram matrix is an open condition. That completes the proof, by my own account.

Corollary 1.1 (finite level). Let $\kappa(L)$ be the negative index of W on $C_c^\infty(-L/2, L/2)$. Then $\kappa(L)$ is finite for each L (by the proposition I will prove next), nondecreasing in L , and $\kappa(L) \rightarrow \kappa$ (the negative subspace produced in (c)–(d) has compact support). $\text{RH} \iff \kappa(L) = 0$ for all L (Weil/Yoshida); $\kappa(L) \geq 1$ for a single L is a *disproof* certificate for RH; and always $\kappa(L) \leq N_{\text{off}}^{\text{dist}}$, the number of distinct zeros with $\operatorname{Re} \rho > \frac{1}{2}$.

Two remarks. The first is multiplicity blindness, which matters for everything below: the Weil form cannot tell an off-line zero of multiplicity m from a simple one weighted by m , so any upper bound on zeros obtained through κ must be multiplied by the multiplicity, which unconditionally is only bounded by $O(\log T)$. A de Branges kernel variant would see multiplicity; I will come back to that. The second is the literature, from memory only — I cannot verify any of it offline. Yoshida (1992, “On Hermitian forms attached to zeta functions”) studies W on functions supported in $[-t, t]$, proves finiteness statements and $\text{RH} \iff$ positivity for all t , and I believe identifies the negative index with the off-line zeros in a form equivalent to Theorem 1; Bombieri (2000, Rend. Lincei, “Remarks on Weil’s quadratic functional in the theory of prime numbers I”) analyses the variational problem on $[-t, t]$; Burnol relates the restricted form to de Branges/Sonine spaces. I treat Theorem 1 as folklore-level with a self-contained proof; the “distinct” subtlety is rarely stated.

2.3 R4’s negatives are not Weil-form negative squares

Next the finite-level object, and what R4’s “33–153 negative eigenvalues” are and are not.

Proposition 2.1 ($\kappa(L) < \infty$). On $H_L = L^2(-L/2, L/2)$,

$$W(f, f) = \int |h_f(\tau)|^2 \nu_X(\tau) d\tau + \operatorname{Pole}(f), \quad X = e^L,$$

where $\text{Pole}(f) = |h_f(i/2)|^2 + |h_f(-i/2)|^2 \geq 0$ has rank ≤ 2 , and

$$\begin{aligned}\nu_X(\tau) &\geq \frac{1}{2\pi} \log \frac{|\tau|}{2\pi} - M_X - O(\tau^{-2}), \\ M_X &= \frac{1}{\pi} \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} + (\text{a bound for the mean term}).\end{aligned}$$

Fix $c_0 > 0$. The set $E = \{\nu_X < c_0\}$ is bounded, contained in $[-T_X, T_X]$ with $T_X \sim 2\pi \exp(2\pi(M_X + c_0))$ — this is R4’s T_{triv} . Split $\nu_X = a - b$ with $a = \max(\nu_X, c_0)$ and $b \in [0, c_0 + M'_X]$ supported in E (here M'_X is M_X together with the mean-term bound). Then $W \geq 2\pi c_0 \|f\|^2 - \langle T_b f, f \rangle$, where T_b , the compression to H_L of the Fourier multiplier b , is positive and trace class with $\text{tr } T_b \leq \frac{L}{2\pi} |E| \sup b$. Hence

$$\begin{aligned}\kappa(L) &\leq \#\{\text{eigenvalues of } T_b \geq 2\pi c_0\} \\ &\leq \frac{L |E| (c_0 + M'_X)}{4\pi^2 c_0} < \infty.\end{aligned}$$

Numerically this bound is astronomically bad: $|E| \sim 0.3 T_X$ with $T_X = 2\pi e^{\sim 4\sqrt{X}}$, so the a-priori bound is $B_{\text{apriori}}(L) \sim L e^{4e^{L/2}}$ — useless. And it uses only pointwise information on ν_X : it is the quantitative form of the brief’s item (3), “count sign changes of ν_X ”; any bound of the type $\kappa(L) \leq \frac{L}{2\pi} \cdot \text{meas}\{\nu_X < c\}$ is of this size. I will return to why nothing pointwise can do better.

Now what R4 actually computed. R4’s candidate-A operator

$$K_X(t, s) = \frac{\sin(\pi(\tilde{N}_X(t) - \tilde{N}_X(s)))}{\pi(t - s)}$$

on L^2 of a τ -window, with diagonal $\nu_X(t)$, is *not* a compression of the Weil form to admissible test functions (support of length $\leq L$). It is a warped sinc kernel whose Weyl symbol is roughly $\mathbf{1}\{|\xi| < \pi \nu_X(t)\} \text{sign } \nu_X(t)$, manifestly indefinite wherever $\nu_X < 0$ (a negative diagonal entry already gives a negative direction). Its negative eigenvalue count therefore tracks window length times local bandwidth over $\{\nu_X < 0\}$ — the Szegő/Landau count of the pointwise-negative region, the same quantity as B_{apriori} restricted to the window. It says “the signed density is negative there”, nothing more.

The honest Pontryagin index of the level- X condition space is $\kappa(L)$, or its compression $\kappa(L, T)$ to test functions whose transforms are concentrated in a height window below T , and

$$\kappa(L, T) = 0 \quad \text{for every } X, T \text{ in any range where the zeros are known to be on the line}$$

(up to leakage), because for $\text{supp } f \subset [-L/2, L/2]$ the explicit formula gives $W(f, f) = \sum_{\gamma} |h_f(\gamma)|^2$ exactly — R4 itself observed “Fejér minima > 0 ”. So my answer to the brief’s red-team question (4) is neither “ $\kappa \sim N(T)$ ” nor “ $\kappa/N \rightarrow c$ ”: the correctly defined $\kappa(X, T)$ is *identically zero* in every computable range, as it must be, since $\kappa(X, T) \geq 1$ would disprove RH below $\sim T$. There is nothing to fit. This contradicts all three regimes the brief offered and R4’s “Krein space” headline alike.

The Gibbs dips of ν_X (width $\sim 1/\log X$, mass $\leq 1/2$, as R4 measured) are real, but they are not negative squares of the compressed form: an admissible $|h_f|^2$ has bandwidth $L = \log X$ and cannot localise into a dip, so the dips cancel against neighbouring positive mass. Szegő-type asymptotics (“eigenvalues $\approx$ symbol samples”) fail here because the symbol varies on the scale $1/L$ of the resolution itself. I expect the numerics to show hundreds of eigenvalues at the

10^{-6} – 10^{-9} level — forced, since $\text{rank } W|_V \leq \#\{\text{zeros in the window}\}$ once $\dim V$ exceeds that number — while the symbol is negative on 20–40% of the window. This is a forecast; nothing is computed yet.

The object I will compute is the Gabor-type subspace

$$V_{L,T} = \text{span} \{f_k(u) = \varphi_L(u) e^{-i\tau_k u}\},$$

with φ_L a fixed bump on $[-L/2, L/2]$ and centres $\tau_k = k\Delta \in [T_0, T]$, $\Delta = 2\pi q/L$ ($q \leq 1$ an oversampling factor). Thus $h_{f_k}(r) = \hat{\varphi}_L(r - \tau_k)$ is a bump of width $\sim 4\pi/L$ centred at τ_k , and $d = \dim V \sim (T - T_0)L/(2\pi q)$. The finite index is then

$$\kappa(L, T) := n_-([W(f_k, f_l)]_{k,l}),$$

the number of negative eigenvalues of the Gram matrix, basis-independent by Sylvester's law of inertia (a prolate basis spans the same space up to edge effects). On the prime side

$$W(f_k, f_l) = \int \hat{\varphi}_L(\tau - \tau_k) \hat{\varphi}_L(\tau - \tau_l) \nu_X(\tau) d\tau + \text{Pole},$$

with the pole term superalgebraically small once $\tau_k \gtrsim 20$; the zero-side formula $\sum_\gamma \hat{\varphi}_L(\gamma - \tau_k) \hat{\varphi}_L(\gamma - \tau_l)$ is for validation only. This is the brief's item (2) in its second form — the Gram matrix of the Weil form on a time–frequency-limited family of dimension $\sim (\log X) \cdot T$, here with windowed exponentials rather than prolates (same span up to edge effects) — and not R4's ν_X -operator, which is the wrong object.

3 Compressing the Weil form; visibility of off-line zeros

3.1 The Gabor compression validated; spurious negatives traced

Now I build the compression — the restriction of the Weil form W to the Gabor space $V_{L,T}$ of windowed exponentials introduced above, realised as a Gram matrix. The prime-side density I integrate against is the *full* one, with the pole term of the explicit formula absorbed as a spectral density (I owe the derivation of this; it comes shortly):

$$\begin{aligned} \nu_X^{\text{full}}(\tau) &= \frac{\theta'(\tau)}{\pi} - \frac{1}{\pi} \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n) \\ &\quad + \frac{1}{\pi} \text{Re} \frac{X^s - 1}{s} + \frac{1}{2\pi(\frac{1}{4} + \tau^2)}, \quad s = \frac{1}{2} + i\tau, \end{aligned}$$

so that $W(f, f) = \int |h_f|^2 \nu_X^{\text{full}}$ holds exactly whenever $\text{supp}(f * \tilde{f}) \subset [-L, L]$. For the window function φ_L I take a Kaiser bump on $[-L/2, L/2]$,

$$\begin{aligned} \varphi(u) &= I_0\left(\beta \sqrt{1 - (2u/L)^2}\right), \\ \hat{\varphi}(r) &= L \frac{\sinh(\sqrt{\beta^2 - z^2})}{\sqrt{\beta^2 - z^2}}, \quad z = \frac{rL}{2}, \end{aligned}$$

whose transform is entire and available in closed form.

The computation — I will refer to it as the compression computation — assembles the following. The prime-side Gram matrix $G_P = [W(f_k, f_l)]$ is obtained by trapezoidal quadrature in τ (step 0.02, each atom $\hat{\varphi}(\cdot - \tau_k)$ truncated where its envelope falls below 10^{-9} of its peak). The zero-side Gram matrix is $G_Z = A^\top A$ with $A_{\gamma k} = \hat{\varphi}(\gamma - \tau_k)$, from N6's 2000 ordinates.

Alongside these: the Plancherel mass matrix M of the atoms; a signature routine that first passes to an M -orthonormal basis of the numerically nondegenerate part of V ; and two utilities for later — a rank-two Hermitian update representing a planted off-line quadruple $\frac{1}{2} \pm \delta \pm it_0$ (the $\gamma, \bar{\gamma}$ pair; the copies at negative t are negligible), and removal of prescribed on-line zeros.

First validation: $L = \log 1000$, window $[100, 300]$, $\beta = 14$, oversampling $q = 1$. The closed-form $\hat{\varphi}$ matches direct quadrature to about 10^{-6} relative at $r = 0, 1, 3, 5, 8$ and at the complex point $2 + 0.3i$. The compression has $d = 220$ atoms, cutoff radius ≈ 22.5 , and

$$\max |G_P - G_Z| = 1.03 \times 10^{-6} \quad \text{against} \quad \max |G_Z| = 9.69,$$

so the prime side and the zero side of the explicit formula agree entrywise. But the signatures, read through the M -orthonormal reduction, disagree badly: the prime side reports “neg,pos,null 33 145 42” with minimum eigenvalue -0.0129 , the zero side $0/115/105$, with 109 zeros in the window. Thirty-three negative squares from a matrix within 10^{-6} of a positive semidefinite one cannot be right.

So I test the eigenvalue counting itself, over $q \in \{1.0, 0.8\}$ and quadrature step $d\tau \in \{0.02, 0.01\}$, looking at the raw eigenvalues of G_P rather than at the reduced form, and recording the condition number of M . The numbers: $\text{cond}(M) = 7.6 \times 10^9$ at $q = 1$ (2.1×10^4 at $q = 0.8$), while $\|E\|_2 = \|G_P - G_Z\|_2 \approx 1.7\text{--}1.9 \times 10^{-6}$ throughout. (Editorial: evidently the 33 negatives were numerical — the reduction through an ill-conditioned mass matrix amplifying quadrature noise into “negative squares”; the record contains the numbers and the decision that follows, not this diagnosis in words.) The raw spectrum of G_P has lowest eigenvalue $\approx -1.18 \times 10^{-7}$ ($q = 1$) and -9.6×10^{-8} ($q = 0.8$), “ $\# < -1e-4: 0$ ” in every case, with 109 near-null eigenvalues for $d = 220$ and 66 for $d = 176$, against 111 resp. 110 zero-side eigenvalues above 10^{-4} . No genuine negative direction; the rank is approximately the number of zeros in reach (zeros in the height interval on which the atoms’ transforms are non-negligible), and the null space has dimension d minus that — exactly the picture I forecast. From here on I read signatures from raw eigenvalues of G_P (basis-independent by Sylvester’s law of inertia, the atoms being linearly independent) and use $q = 0.8$.

3.2 The index grid and its forecast

Now the grid the brief asked for in item (2), $X = 10^2, 10^3, \dots, 10^6$, carried out on the Gram-matrix compression of the Weil form (the brief’s second option, here with Kaiser–Gabor atoms) rather than on R_4 ’s ν_X -operator: windows $[50, 1000]$ and $[1000, 2400]$, Kaiser $\beta = 14$, $q = 0.8$, signature from raw eigenvalues at threshold $10^{-5} \max |\text{ev}|$. For each X and window I want the inertia (n_-, n_0, n_+) of G_P — numbers of negative, null and positive eigenvalues — its minimum eigenvalue, the discrepancy $\|G_P - G_Z\|$, the zero-side rank, and the number of zeros in the window. Against these I set the pointwise quantities that embody the brief’s sign-change route: the fraction of the window on which $\nu_X < 0$; the “Szegő prediction” $n_- \approx (\text{that fraction}) \cdot d$, i.e. what one would get if eigenvalues were symbol samples; and the a-priori count of Proposition 2.1 restricted to the window,

$$\frac{Lq}{2\pi} (T_1 - T_0) \text{ meas}\{\nu_X < 0.05\}.$$

I also add two prime-computable positivity certificates. Writing R for the matrix of $W|_V$ in M -orthonormal coordinates, trace and second moment alone give $n_+ \geq (\text{tr } R)^2 / \text{tr}(R^2)$, so I record the certified-positive fraction $(\text{tr } R)^2 / (d \text{tr } R^2)$, together with a Fejér-smoothed proxy $\langle \nu \rangle^2 / \langle \nu^2 \rangle$ for it.

I am confident enough of the outcome to write down the reading before a single row exists: $n_-(G_P) = 0$ throughout, i.e. the compressed Weil form at level X is positive semidefinite

by a prime-side computation using no zeros; $d - \text{rank} \approx d - \#\text{zeros}$ exact null directions whenever d exceeds the zero count; the symbol ν_X negative on 20–45% of the window; and “the Szego/pointwise prediction and the a-priori bound are off by everything.” The first attempt at the grid is cut off before producing a row,

so I set it going again to run on its own and turn to the theory it is meant to illustrate.

3.3 The pole density, and Theorem 2 on visibility

First the derivation I owe, so that the prime side really is a single integral against ν_X^{full} . For an even kernel k supported in $[-L, L]$, $X = e^L$, with transform $\hat{k}$, the pole term of the explicit formula is

$$\text{Pole}(k) = \hat{k}(i/2) + \hat{k}(-i/2) = \int k(v) 2 \cosh(v/2) dv,$$

while

$$\int_1^X u^{-1/2} [k(\log u) + k(-\log u)] du = \int_{-L}^L k(v) e^{|v|/2} dv.$$

Since $2 \cosh(v/2) = e^{|v|/2} + e^{-|v|/2}$,

$$\begin{aligned} \text{Pole}(k) &= \int_1^X u^{-1/2} [k(\log u) + k(-\log u)] du + \int k(v) e^{-|v|/2} dv \\ &= \int |h(\tau)|^2 \left[\frac{1}{\pi} \operatorname{Re} \frac{X^s - 1}{s} + \frac{1}{2\pi(\frac{1}{4} + \tau^2)} \right] d\tau, \quad s = \frac{1}{2} + i\tau, \end{aligned}$$

by Parseval applied to $k = f * \tilde{f}$, $|h|^2 = \hat{k}$, the transform of $e^{-|v|/2}$ being $1/(\frac{1}{4} + \tau^2)$. Hence, exactly, $W(f, f) = \int |h_f|^2 \nu_X^{\text{full}}$ for $\text{supp } f \subset [-L/2, L/2]$, with ν_X^{full} as defined above. R4’s “PNT mean from $u = 2$ ” is this up to the smooth $O(1/\tau)$ piece coming from $\int_1^2$ and the term $1/(2\pi(\frac{1}{4} + \tau^2))$ — irrelevant for their sign statistics, relevant for exactness here. The check already in hand is the agreement of G_P with G_Z to 10^{-6} absolute (entries of size 1–10) at $X = 10^3$ on $[100, 300]$; the grid records $\|E\|$ in every row.

A word on why the computed index is the true one. The basis is the Kaiser window with centres $\tau_k = T_0 + k\Delta$, $\Delta = 2\pi/(qL)$; the signature is read from raw eigenvalues at threshold $10^{-5} \times \text{scale}$ and validated against the zero side, which is exactly positive semidefinite (it is $A^\top A$); the leakage of h_{f_k} outside $[T_0 - r_{\text{cut}}, T_1 + r_{\text{cut}}]$ is below 10^{-9} ; and hypothetical off-line zeros above the RH-verified height 3×10^{12} , if any, would contribute less than $(10^{-9})^2$. So $\kappa(X, T)$ as computed is the true signature of W on $V_{L,T}$ up to the threshold.

Now the question that the brief’s phrase “off-line zeros visible to the subspace” presupposes, made quantitative. Fix a hypothetical off-line zero $\rho_0 = \frac{1}{2} + \delta + it_0$, $\delta > 0$, so $\gamma_0 = t_0 - i\delta$. “Level L ” means test functions supported in $[-L/2, L/2]$, i.e. primes $\leq X = e^L$. What is the smallest L for which the restriction of W to $C_c^\infty(-L/2, L/2)$ has $n_- \geq 1$ on account of ρ_0 ?

Theorem 2 (isolated zero, worst case over on-line configurations). Suppose all zeros with $|\operatorname{Re} \gamma - t_0| \leq 1$ other than $\{\gamma_0, \bar{\gamma}_0\}$ lie on the line, and let $n_1 := N(t_0 + 1) - N(t_0 - 1)$; unconditionally $n_1 \leq A_0 \log t_0$ (from memory a Backlund/Trudgian bound, roughly $0.25 \log T + 2$ for T large; I keep it symbolic). If

$$L \geq L_{\text{vis}} := C_B \frac{\sqrt{n_1 + 1}}{\delta} \quad (C_B \text{ absolute, depending on the bump } B; C_B \sim 4),$$

then there is $f \in C_c^\infty(-L/2, L/2)$ with $W(f, f) < 0$. More precisely, for a family $\rho_j = \frac{1}{2} + \delta_j + it_j$ with $|t_i - t_j| \geq 2$, each isolated in this sense, $L \geq \max_j C_B \sqrt{n_1(t_j) + 1/\delta_j}$ gives $n_- \geq \#\text{family}$.

The proof, as I work it. Take a bump $B \geq 0$, smooth, $\text{supp } B \subset [-\frac{1}{2}, \frac{1}{2}]$, $\int B = 1$, and rescale $B_L(u) = B(u/L)/L$, so that $\hat{B}_L(x) = \hat{B}(Lx)$. Choose f with

$$h_f(r) = (r - t_0) \hat{B}_L(r - t_0)$$

(namely $\overline{f} = i(B_L)'e^{-it_0u}$; the sign is immaterial). The contributions to $W(f, f) = \sum m h(\gamma) \overline{h(\bar{\gamma})}$ are:

- (i) The pair $\gamma_0, \bar{\gamma}_0$: $h(\gamma_0) = (-i\delta)\hat{B}_L(-i\delta)$, $h(\bar{\gamma}_0) = (i\delta)\hat{B}_L(i\delta)$, and

$$\hat{B}_L(\pm i\delta) = \int B_L(u) \cosh(\delta u) du =: c(\delta L) \in [1, \cosh(\delta L/2)]$$

is real; the pair contributes $2 \text{Re}[(-i\delta)(-i\delta)] c^2 = -2\delta^2 c^2 \leq -2\delta^2$ (times $m_0 \geq 1$).

- (ii) The reflected pair at $-t_0$: $|h(-t_0 \mp i\delta)| \approx 2t_0 |\hat{B}_L(-2t_0 \mp i\delta)|$, superpolynomially small in Lt_0 — negligible.
- (iii) On-line zeros, $x := \gamma - t_0$: $|h|^2 = x^2 \hat{B}(Lx)^2$, with supremum $L^{-2}b_1$ where $b_1 = \|y\hat{B}(y)\|_\infty^2 \leq \|B'\|_1^2$. For $|x| \leq 1$ there are at most n_1 zeros, total $\leq n_1 b_1 / L^2$. For $|x| > 1$, $|\hat{B}(y)| \leq \|B'''\|_1 / |y|^3$ gives $x^2 \hat{B}(Lx)^2 \leq \|B'''\|_1^2 / (L^6 x^4)$, and $\sum_{|\gamma - t_0| > 1} (\gamma - t_0)^{-4} \leq A'_0 \log(t_0 + 3)$, so this tail is $\leq b_3 A'_0 \log(t_0 + 3) / L^6$ with $b_3 = \|B'''\|_1^2$.
- (iv) Other off-line zeros: none within $|\text{Re } \gamma - t_0| \leq 1$ by hypothesis. For those farther, write $\gamma = t_0 + x - iy$; then $|h(x - iy)| = |x - iy| |\hat{B}_L(x - iy)|$, and $\hat{B}_L(x - iy)$ is the transform of the smooth compactly supported function $B_L e^{yu}$ ($|y| < \frac{1}{2}$), bounded by

$$\frac{\|(B_L e^{yu})'''\|_1}{|x|^3} \leq \frac{C e^{L/4} (1 + L^{-3})}{|x|^3}.$$

I first want to call the $e^{L/4}$ loss harmless because it multiplies $|x|^{-3}$ -summable terms — no: $e^{L/4}/|x|^3$ with $|x| \geq 1$ is *not* small. The honest fix is to strengthen the hypothesis: either to “no off-line zeros with $|\text{Re } \gamma - t_0| \leq R_L := e^{L/12} L$ ”, or to global isolation (the orbit of ρ_0 is the only off-line zero), or to

(H_R) : all zeros other than the γ_0 -orbit with $|\text{Re } \gamma - t_0| \leq R$ lie on the line, $R = e^{L/8}$;

then the far off-line sum is $\leq C \log t_0 / L^6$. Recorded honestly: interference from unknown off-line zeros is *not* controlled by this argument.

In total,

$$W(f, f) \leq -2\delta^2 + \frac{n_1 b_1}{L^2} + \frac{C \log(t_0 + 3)}{L^6} < 0$$

as soon as $L^2 > (n_1 b_1 + 1)/\delta^2$ (for $L \geq 2$, absorbing the C/L^4). With B the normalised bump $\exp(-1/(1 - 4u^2))$ the constant b_1 is to be evaluated numerically. For the family: the h_{f_j} are 2-separated in centre, the cross terms $W(f_i, f_j)$ decay like $(L|x|)^{-3}$, the off-diagonal row sums are $\leq C' \log T / L^3$, and Gershgorin's theorem gives the full count once $L \geq L_{\text{vis}}$, which is $\geq C\sqrt{\log \delta} \gg (\log T)^{1/3}$. QED modulo constants.

So: an *isolated* off-line zero at height t and depth δ is visible from primes $\leq X$ once $\log X \geq C\sqrt{\log t}/\delta$ (worst case over positions of the on-line zeros), and jointly so for 2-separated isolated families. Under “average” spacing — zeros near t_0 at distances $\sim k \cdot 2\pi / \log t_0$ — the sum in (iii) is instead

$$\approx \frac{\log t_0}{2\pi} \int x^2 \hat{B}(Lx)^2 dx = \frac{\log t_0}{2\pi} \frac{\|y\hat{B}\|_2^2}{L^3},$$

giving the softer threshold

$$L \geq \left(\frac{(\log t_0) b_2}{4\pi\delta^2} \right)^{1/3}, \quad b_2 = \|y\hat{B}(y)\|_2^2.$$

This, I believe, is the law R4's planting experiment saw (L^* nearly independent of depth over their small range). A rough check against R4's numbers: $t_0 = 991$, $\delta = 0.02$ gives $(6.9 b_2 / (4\pi \cdot 4 \times 10^{-4}))^{1/3} \approx 11$ with $b_2 \sim 1$, against their 7.08 obtained with a richer 30-bump family; $\delta = 0.49$ gives 1.3 against their 3.44 — at large δ the on-line term is not the binding constraint, the pole and edge terms are; fine.

Clusters reinforce, but count once. If several off-line zeros $\zeta_i = t_0 + x_i - iy_i$ sit within $|x_i| \ll 1/L$ of t_0 (any depths $y_i > 0$), then with the *same* f ,

$$\begin{aligned} h(\zeta_i) \overline{h(\zeta_i)} &= (x_i - iy_i)^2 \hat{B}_L(x_i - iy_i)^2 \\ &\approx (x_i^2 - y_i^2 - 2ix_i y_i) c(y_i L)^2 (1 + O(Lx_i)), \end{aligned}$$

whose real part is negative when $y_i > |x_i|$: every zero that is deeper than it is horizontally displaced from t_0 adds to the negativity of $W(f, f)$. But they add to the same single negative direction. The index n_- counts W -orthogonal negative directions, and k zeros within $o(1/L)$ of each other span, to first order, the same two functionals ℓ_{t_0}, ℓ'_{t_0} (evaluation of h and of h' at t_0), so they yield 1 (at most 2) negative squares at level L , not k . Separating them needs $L \gtrsim 1/\min |\zeta_i - \zeta_j|$ (uncertainty principle; to be measured). In the limit of coincidence no level separates them — Theorem 1 again: κ counts distinct zeros.

Interference: the gap in any unconditional visibility theorem. Off-line zeros $\zeta = t_0 + x - iy$ that are wider than deep, $|x| > y$, sitting at $c/L < |x| < O(1)$, contribute $\text{Re} [(x - iy)^2 \hat{B}_L(x - iy)^2]$, which can be *positive* and as large as $x^2 c(yL)^2 \sim x^2 e^{yL}$. For y comparable to δ this is comparable to the main term $-2\delta^2 c(\delta L)^2$, and unconditionally there may be up to $A_0 \log t_0$ such zeros. No choice of B avoids this: $B_L e^{yu} / c(yL)$ is a probability density on $[-L/2, L/2]$, its characteristic function is small at x only for $|x| \gtrsim (\text{width})^{-1}$, and the width is at most $L/2$; so zeros at $c/L < |x| < C/L$ with depth $\sim \delta$ interfere at relative order 1. Conclusion: the statement “every off-line zero of depth $\geq \delta$ below height T forces its own negative square at level $L(\delta, T)$ ” is true for isolated zeros (Theorem 2), true for tight clusters counted once, and *not provable configuration-free*; the exact unconditional statement is the limit $L \rightarrow \infty$ in Theorem 1. This already narrows the rigorous inequality $N_{\text{off}} \leq \kappa + E$ that the brief's item (3) hoped for.

4 The negative-index route is empty: index identically zero

4.1 First honest index row: zero negative squares

The first row of the index grid is in. To recall the objects: G_P is the Gram matrix of the Weil form compressed to the Gabor space $V_{L,T}$ (Kaiser atoms supported in $[-L/2, L/2]$, centres spread across the height window), computed from the primes $\leq X = e^L$ and Γ alone; G_Z is the same matrix computed from the zeros, used only for validation; and (n_-, n_0, n_+) is the inertia — the numbers of negative, null and positive eigenvalues — read from raw eigenvalues of G_P . At $X = 10^2$, $L = 4.61$, on the window $[50, 1000]$, with $d = 558$ atoms and 639 zeros in the window:

$$\begin{aligned} n_-(G_P) &= 0, & n_0 &= 18, & n_+ &= 540, \\ \min \text{ev}(G_P) &= -1.82 \times 10^{-8}, & \|E\|_2 &= \|G_P - G_Z\|_2 = 9.0 \times 10^{-7}, \end{aligned}$$

and on the zero side $n_- = 0$ with rank 540.

Against this stand the pointwise quantities that embody the brief’s sign-change route. The fraction of the window on which $\nu_X < 0$ is 0.207; the “Szegő prediction” — negative eigenvalues $\approx$ that fraction times d , as if eigenvalues were samples of the symbol — is $n_- \approx 115.5$; the a-priori count of Proposition 2.1 restricted to the window is 126.1. The prime-computable positivity certificate — the fraction $(\text{tr } R)^2 / (d \text{tr } R^2)$ of eigenvalues certified positive by trace and second moment alone, R being the matrix of $W|_V$ in mass-orthonormal coordinates — is 0.808, its Fejér-smoothed proxy 0.785. So from primes ≤ 100 , with no zeros used: not a single negative square, while the signed density is negative on a fifth of the window and the pointwise count would predict over a hundred. This is the first actual row behind the reading I wrote down in advance.

Before the grid goes further I want two changes: the high window moved up to $[1400, 2400]$, and the discrepancy $\|G_P - G_Z\|$ recorded in the Frobenius norm, which dominates the spectral norm.

On the second attempt both changes are in, and the grid restarts from $X = 10^2$ on the windows $[50, 1000]$ and $[1400, 2400]$, with $\|E\|$ henceforth the Frobenius norm $\|G_P - G_Z\|_F$.

4.2 Planting off-line zeros: annihilation versus robust visibility

Alongside the grid I want the numerical companion to Theorem 2: plant an off-line quadruple among the true zeros and measure the smallest level L at which the compressed form acquires a negative square — R4’s planting experiment, redone on an honest compression of the Weil form rather than on the warped-sine operator. The design: at height t_0 I take the union of three Kaiser–Gabor families, $\beta \in \{6, 10, 14\}$, with centres spaced $2\pi/(1.5L)$ over $[t_0 - 40, t_0 + 40]$; the zero-side Gram matrix $G_0 = A^\top A$ is built from the true zeros with $|\gamma - t_0| < 100$ (out of the 2000 below 2515); and the planted quadruple $\frac{1}{2} \pm \delta \pm it_0$ enters as the rank-two Hermitian update

$$a b^* + b a^*, \quad a = (h_k(t_0 - i\delta))_k, \quad b = (h_k(t_0 + i\delta))_k,$$

where h_k is the transform of the k -th atom. The plant is made either in “add” mode, on top of the true zeros, or in “replace” mode, after subtracting the Gram contribution of the two nearest on-line zeros. I scan L from 1 to 18 in steps of 0.25 and record L^* , the first level with an eigenvalue below $-10^{-8} \max |\text{ev}|$, for $t_0 \in \{100, 400, 1000, 2200\}$ and $\delta \in \{0.02, 0.05, 0.1, 0.2, 0.3, 0.45\}$. Next to each L^* I set the two laws from the visibility analysis — the worst case $\sqrt{n_1 + 1}/\delta$ of Theorem 2, with $n_1 = \#\{\gamma : |\gamma - t_0| \leq 1\}$, and the average-spacing law $(\log(t_0/2\pi)/(4\pi\delta^2))^{1/3}$ with the bump constant $b_2 := 1$ — and fit $L^* \sim a \delta^{-p} \log(t_0/2\pi)^r$ over the add-mode data.

A second experiment, at fixed level ($t_0 = 400$, $\delta = 0.2$, $L \in \{6, 10, 14\}$), counts negative squares for: a single quadruple; a *double* zero (the same plant with weight 2 — Theorem 1 says multiplicity is invisible, so this should give 1); two depths 0.2 and 0.35 at one ordinate; two quadruples at $t_0 \mp \varepsilon/2$; and five quadruples spaced ε apart, for ε from 0.01 to 4.

The first rows come back at $t_0 = 100$, where $n_1 = 0$ and $\log(t_0/2\pi) = 2.77$:

δ	mode	L^*	n_- (d)	$\sqrt{n_1 + 1}/\delta$	avg. law
0.02	add	3.50	1 (201)	50.00	8.20
0.02	replace	2.75	1 (159)	50.00	8.20
0.05	add	3.25	1 (189)	20.00	4.45
0.05	replace	2.75	1 (159)	20.00	4.45
0.10	add	3.25	1 (189)	10.00	2.80
0.10	replace	2.75	1 (159)	10.00	2.80

So the measured threshold is tiny, nearly independent of the depth δ , and sits far below both laws. The restarted grid has also reproduced its first row: $X = 10^2$ on $[50, 1000]$, $n_-(G_P) = 0$, $n_0 = 18$, $n_+ = 540$, $\min \text{ev} = -1.82 \times 10^{-8}$, $\|E\|_F = 8.9 \times 10^{-6}$, certificates 0.808/0.785 as before.

The reading of these tiny thresholds is the one R4 gave for its own planting thresholds: once $\dim V$ exceeds the number of zeros within reach of the atoms (the zeros in the height interval on which the atoms' transforms are non-negligible), the on-line Gram matrix has a null space — the test space nearly annihilates the on-line zeros — and the planted rank-two term can push an eigenvalue below zero at an arbitrarily small level. Call this “annihilation” visibility: possible as soon as $\dim V > \#\{\text{zeros in reach}\}$, and silent about robust negativity. What I add is a second, robust threshold. Both are taken relative to the scale $s := \text{largest eigenvalue of the unplanted on-line Gram matrix } G_0$:

$$\begin{aligned} L_{\text{ann}}^* &:= \text{first } L \text{ with an eigenvalue} < -10^{-8}s, \\ L_{\text{rob}}^* &:= \text{first } L \text{ with an eigenvalue} < -10^{-3}s, \end{aligned}$$

with L scanned from 0.5 to 14, $\min \text{ev}/s$ recorded at each threshold, and L_{ann}^* , L_{rob}^* fitted separately as $a \delta^p \log(t_0/2\pi)^r$ in add mode.

and a second attempt to clear the old runs fails the same way.

On the third attempt the two-threshold version goes in cleanly — L from 0.5 to 14, L_{ann}^* and L_{rob}^* recorded with $\min \text{ev}/s$, separate least-squares fits in add mode — and the planting experiment restarts from scratch. The index grid has been running undisturbed throughout.

4.3 Negative index zero across the whole prime range

Back to the results. The low window of the grid is complete: window $[50, 1000]$ (639 zeros), Kaiser $\beta = 14$, oversampling $q = 0.8$, signature read from raw eigenvalues at threshold $10^{-5} \max |\text{ev}|$. The inertia of the prime-side Gram matrix, level by level:

X	L	d	n_-	n_0	n_+	$\min \text{ev}(G_P)$
10^2	4.61	558	0	18	540	$-1.82 \cdot 10^{-8}$
10^3	6.91	836	0	195	641	$-1.18 \cdot 10^{-7}$
10^4	9.21	1115	0	473	642	$-4.65 \cdot 10^{-7}$
10^5	11.51	1393	0	753	640	$-6.74 \cdot 10^{-7}$
10^6	13.82	1672	0	1031	641	$-1.01 \cdot 10^{-6}$

and, against it, the pointwise predictions and the positivity certificate:

X	$\text{frac}(\nu_X < 0)$	Szegő pred. n_-	a-priori count	cert. fraction
10^2	0.207	115.5	126.1	0.808
10^3	0.330	276.2	294.6	0.655
10^4	0.381	425.2	452.2	0.526
10^5	0.407	567.4	599.2	0.434
10^6	0.424	708.5	746.9	0.368

The discrepancy $\|E\| = \|G_P - G_Z\|_F$ rises from 8.9×10^{-6} to 8.3×10^{-5} down the table, so every “negative” minimum eigenvalue sits at leakage level; the zero side has $n_- = 0$ and rank 540, 642, 642, 642, 641, matching $n_+(G_P)$ to within one or two in every row; and the proxy certificate falls $0.785 \rightarrow 0.341$ alongside the honest one. The first row of the high window, $X = 10^2$ on $[1400, 2400]$ (907 zeros, $d = 587 < 907$), is strictly positive definite: $n_0 = 0$, $\min \text{ev} = +6.27 \times 10^{-4}$, with $\text{frac}(\nu_X < 0) = 0.111$, Szegő prediction 65.1, a-priori count 72.5, certificates 0.896/0.881.

So across the brief's whole range $X = 10^2, \dots, 10^6$ the compressed Weil form, computed from primes and Γ alone, has not one negative square, while its symbol ν_X is negative on 21–42% of the window and both the pointwise Szegő count and the sign-change a-priori count climb into the hundreds. The null space grows exactly as d outruns the ≈ 640 zeros in reach. The planting experiment, restarted with its two thresholds, has not yet reported a row.

5 The dual: positive eigenvalues certify on-line zeros

5.1 Pullback inertia and the double count

The thought that has taken over is this. The negative index of a compression — the restriction of the Weil form W to a finite-dimensional space V of admissible test functions — is useless as an *upper* bound for off-line zeros, because pulling a Hermitian form back to a subspace can only lose negative squares. But the same monotonicity, read for the *positive* index, points the right way. This is potentially the main result of the line, and it will need to be red-teamed hard; let me write it down.

The observation. Fix any subspace V of admissible test functions, and a “reach interval” I' : a height interval outside which the transforms h_f , $f \in V$, are negligibly small. Write the restricted Weil form as

$$W|_V = \text{ev}^* Q \text{ev} + \text{Tail},$$

where $\text{ev} : V \rightarrow \mathbf{C}^{Z'}$ is evaluation of h_f at the distinct zero-points γ with heights in I' (Z' denoting that finite set), Tail is the contribution of all zeros with heights outside I' , and Q is a direct sum of blocks. Each distinct on-line zero, of any multiplicity m , gives a 1×1 block (m) — one positive square; each distinct off-line pair $\{\gamma, \bar{\gamma}\} = \{\rho, 1 - \bar{\rho}\}$ of multiplicity m gives the block

$$\begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}$$

of signature $(1, 1)$ — the hyperbolic pair of Theorem 1.

Lemma 5.1 (inertia under pullback). For any linear map A and Hermitian form Q , $n_+(A^*QA) \leq n_+(Q)$, where n_+ denotes the positive index. [If $A^*QA > 0$ on a subspace U then $A|_U$ is injective and $Q > 0$ on $A(U)$.]

Put any Hilbert structure on V and let $\theta > \|\text{Tail}|_V\|_{\text{op}}$. Define the robust positive index

$$n_+^\theta(W|_V) := \#\{\text{eigenvalues of } W|_V > \theta\}.$$

By Weyl's inequality and Lemma 5.1, writing Q_{reach} for Q (the block form over the zero-points in reach),

$$\begin{aligned} \text{(P)} \quad n_+^\theta(W|_V) &\leq n_+(\text{ev}^* Q \text{ev}) \leq n_+(Q_{\text{reach}}) \\ &= n_{\text{on}}^{\text{dist}}(I') + n_{\text{pair}}^{\text{dist}}(I'), \end{aligned}$$

where $n_{\text{on}}^{\text{dist}}(I')$ is the number of distinct on-line zeros with height in I' and $n_{\text{pair}}^{\text{dist}}(I')$ the number of distinct off-line pairs there. On the other hand the Riemann–von Mangoldt count $N(I')$ is taken with multiplicity and counts *both* members ρ , $1 - \bar{\rho}$ of an off-line pair, which share a height:

$$\text{(N)} \quad N(I') \geq n_{\text{on}}^{\text{dist}}(I') + 2 n_{\text{pair}}^{\text{dist}}(I').$$

Subtracting (P) from (N),

$$\begin{aligned} n_{\text{pair}}^{\text{dist}}(I') &\leq N(I') - n_+^\theta(W|_V), \\ n_{\text{on}}^{\text{dist}}(I') &\geq 2n_+^\theta(W|_V) - N(I'). \end{aligned} \quad (*)$$

In words: every robustly positive eigenvalue of the compressed Weil form beyond $N/2$ certifies a distinct on-line zero. An off-line pair “wastes” one of its two zeros as a negative square; this is Theorem 1 read from the positive side, and it needs no visibility hypothesis and no positivity — inertia replaces both.

The brief’s item (3) asked whether the negative-index route could reproduce a positive-proportion theorem. The route it proposed — an upper bound on the negative index from sign changes of ν_X — is dead. The mechanism I am writing down now, pullback inertia plus the double count in (N), is not in the brief or in any of the reports I read.

5.2 A certificate computable from primes alone

Two prime-computable numbers bound the positive index. Let R be the matrix of $W|_V$ in an orthonormal basis of V , and $d = \dim V$.

Lemma 5.2 (Cauchy–Schwarz count).

$$n_+^\theta(R) \geq \frac{(\text{tr } R - \theta d)_+^2}{\text{tr}(R^2)}.$$

[With $S := \sum_{\lambda > \theta} \lambda \geq \text{tr } R - \theta d$, the sum over eigenvalues λ of R exceeding θ , Cauchy–Schwarz gives $S^2 \leq n_+^\theta \sum_{\lambda > \theta} \lambda^2 \leq n_+^\theta \text{tr } R^2$.]

Both traces are prime-side quantities. With $\nu = \nu_X^{\text{full}}$ the full prime-side density introduced earlier (the pole term of the explicit formula absorbed as a density), and K_V the reproducing kernel of the space $\hat{V}$ of transforms in $L^2(d\tau/2\pi)$,

$$\begin{aligned} \text{tr } R &= \int K_V(\tau, \tau) \nu(\tau) d\tau, \\ \text{tr } R^2 &= \iint |K_V(\tau, \tau')|^2 \nu(\tau) \nu(\tau') d\tau d\tau', \end{aligned}$$

both finite computations with the primes $\leq X = e^L$ and Γ . Hence what I will call the certificate:

Theorem 3 (finite certificate). For every L , every height window I and every admissible V (supports in $[-L/2, L/2]$, transforms concentrated on I up to the reach interval I'),

$$N_0^{\text{dist}}(I') \geq \frac{2(\text{tr } R_V - \theta d)^2}{\text{tr}(R_V^2)} - N(I'),$$

where $N_0^{\text{dist}}(I')$ is the number of distinct on-line zeros with height in I' , R_V is the matrix R for the space V , and θ is any upper bound for the tail norm; everything on the right is computable from primes and Γ , plus the Riemann–von Mangoldt count.

Reading the certified-positive column of the low window of the index grid in this light: primes ≤ 100 certify $\geq 41\%$ of the 639 zeros in $[50, 1000]$ as on the line and distinct; primes $\leq 10^6$ certify $\geq 92\%$. (This is an inference from the certified fractions 0.808 and 0.368 against d and N ; a dedicated experiment is next.) Inside the RH-verified range this is only a consistency check; the point is the asymptotic version.

5.3 Asymptotics in the Montgomery–Vaughan range

Set-up. Take $I = [T, 2T]$ and $l := \log(T/2\pi)$, so that $N(I) \sim |I|l/2\pi$; put $L = \lambda l$ with $0 < \lambda \leq 1$, so $X = e^L = (T/2\pi)^\lambda$; and let V be the space of functions time-limited to $[-L/2, L/2]$ and band-limited to I — $d \sim L|I|/2\pi$ prolate functions, or a tapered Gabor frame. The constants below are for the sharp cutoff, where

$$K_V(\tau, \tau') \sim D_L(\tau - \tau') \mathbf{1}_I(\tau) \mathbf{1}_I(\tau'),$$

$$D_L(x) = \frac{\sin(Lx/2)}{x/2}, \quad \int D_L^2 = 2\pi L,$$

and $F_L := D_L^2/2\pi L$ is the Fejér kernel, with $\hat{F}_L(u) = (1 - |u|/L)_+$. Split the density as $\nu = m + P + \Pi$ with

$$m = \frac{\vartheta'}{\pi} + \frac{1}{2\pi(\frac{1}{4} + \tau^2)} \sim \frac{1}{2\pi} \log \frac{\tau}{2\pi} \quad (\text{smooth}),$$

$$P = -\frac{1}{\pi} \sum_{n \leq X} \Lambda(n) n^{-1/2} \cos(\tau \log n),$$

$$\Pi = \frac{1}{\pi} \operatorname{Re} \frac{X^s - 1}{s} = O(T^{\lambda/2-1}) \text{ pointwise on } I, \quad s = \frac{1}{2} + i\tau$$

(here ϑ is the Riemann–Siegel theta function, written θ in R4’s formula; I write ϑ in this computation to keep θ for the threshold).

The trace. Since $\int_I P = O(\sum \Lambda(n)n^{-1/2}/\log n) = O(\sqrt{X}/L)$ and $\int_I \Pi = O(T^{\lambda/2})$,

$$\operatorname{tr} R = L|I| \langle m \rangle_I (1 + O(T^{\lambda/2-1})), \quad \langle m \rangle_I = \frac{l}{2\pi} + O(1),$$

where $\langle m \rangle_I$ denotes the average of m over I .

The second moment. Expand

$$\operatorname{tr} R^2 = \iint_{I \times I} D_L(\tau - \tau')^2 [mm' + mP' + Pm' + PP' + \dots],$$

primes on m, P denoting evaluation at τ' . The mm' term gives $2\pi L|I| \langle m \rangle^2 (1 + O(1/l^2))$, since m varies by $O(1/T)$ over the width $1/L$ of D_L^2 . In PP' the diagonal $n = n'$, evaluated with the Fourier transform of D_L^2 , is

$$\begin{aligned} \frac{L|I|}{\pi} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} \left(1 - \frac{\log n}{L}\right) &= \frac{L|I|}{\pi} \left(\frac{L^2}{2} - \frac{L^2}{3}\right) (1 + O(1/L)) \\ &= \frac{|I|L^3}{6\pi} (1 + O(1/L)), \end{aligned}$$

by $\sum_{n \leq x} \Lambda(n)^2/n = (\log x)^2/2 + O(\log x)$ (the prime number theorem with any error term). The off-diagonal I control — from memory, with no way to check the constant here — by the Montgomery–Vaughan generalized Hilbert inequality

$$\left| \sum_{\substack{n \neq n' \\ n, n' \in I}} \frac{b_n \bar{b}_{n'}}{\log n - \log n'} \right| \leq \frac{3\pi}{2} \sum_n \frac{|b_n|^2}{\delta_n}, \quad \delta_n \geq \frac{1}{2n},$$

applied with $|b_n| \leq a_n(2\pi L)^{1/2}$, $a_n = \Lambda(n)n^{-1/2}$: the off-diagonal is $O(L \sum_{n \leq X} \Lambda(n)^2) = O(LX \log X)$, and its ratio to the diagonal is

$$O\left(\frac{X \log X}{|I|L^2}\right) = O(T^{\lambda-1}l^{-1}) \rightarrow 0 \quad (\lambda \leq 1),$$

even for $\lambda = 1 + o(1)$ provided $X = o(TL)$. The cross term mP' is $O(l\sqrt{X}) = o(|I|Ll^2)$ after integrating $m \cos(\tau \log n)$ by parts; the Π terms are negligible for $\lambda \leq 1$. So

$$\begin{aligned} \operatorname{tr} R^2 &= 2\pi L |I| \left[\langle m \rangle^2 + \frac{L^2}{12\pi^2} \right] (1 + o(1)), \\ \frac{(\operatorname{tr} R)^2}{\operatorname{tr} R^2} &= \frac{\lambda N(I)}{1 + \lambda^2/3} (1 + o(1)). \end{aligned}$$

Consistency check under RH. Under RH together with Montgomery’s pair-correlation theorem — which covers exactly this Fourier range, $|\alpha| \leq \lambda \leq 1$ — the same statistic evaluated on the zero side is

$$\frac{(LN)^2}{2\pi L \sum_{\gamma, \gamma'} F_L(\gamma - \gamma')},$$

with

$$\begin{aligned} \sum_{\gamma, \gamma'} F_L(\gamma - \gamma') &= N \left[\frac{L}{2\pi} + \frac{l}{2\pi} - \frac{1}{\pi} \left(\frac{L}{2} - \frac{L^2}{6l} \right) \right] \\ &= N \frac{l}{2\pi} \left(1 + \frac{\lambda^2}{3} \right), \end{aligned}$$

the same number. So under RH the bound of Lemma 5.2 is exactly the classical Cauchy–Schwarz consequence of Montgomery’s theorem, “ $N_{\text{distinct}} \geq N^2 / \sum m_p^2 \geq \frac{3}{4}N$ ” at $\lambda = 1$. The prime-side evaluation *is* Montgomery’s prime-side computation, which never needed RH; RH was needed classically only to interpret the zero side as a sum over real γ with positive weights. Inertia (Lemma 5.1) removes that need, at the price $3/4 \rightarrow 2 \cdot \frac{3}{4} - 1 = \frac{1}{2}$.

5.4 The claimed one-half theorem

Theorem 4 (claimed; see the red team to come). Unconditionally, as $T \rightarrow \infty$, for every fixed $\lambda \in (0, 1]$,

$$N_0^{\text{dist}}([T, 2T]) \geq \left(\frac{2\lambda}{1 + \lambda^2/3} - 1 - o(1) \right) N([T, 2T]),$$

and at $\lambda = 1$,

$$N_0^{\text{dist}}([T, 2T]) \geq \left(\frac{1}{2} - o(1) \right) N([T, 2T]).$$

That is, at least half of the nontrivial zeros (counted with multiplicity in N) are accounted for by distinct zeros on the critical line; the bound is nontrivial exactly when $2\lambda/(1 + \lambda^2/3) > 1$, i.e. $\lambda > 3 - \sqrt{6} = 0.551$. For comparison, from memory and unverified offline: Levinson, Conrey, Bui–Conrey–Young and Pratt–Robles–Zaharescu–Zeindler give $N_0 \geq 0.4173N$ and $N_0^{\text{simple}} \geq 0.407N$. So this claim goes beyond the brief’s benchmark $N_{\text{off}} \leq 0.6N$ and past the records I remember. The ingredients: the Weil explicit formula for compactly supported tests (unconditional); Riemann–von Mangoldt; Lemma 5.1; Lemma 5.2; the Montgomery–Vaughan mean value theorem; the asymptotics of $\sum \Lambda^2/n$; Landau–Pollak–Slepian kernel asymptotics for V ; and a tail bound.

Tail bound. The tail is the contribution of zeros with heights outside $I' = I + [-D_0, D_0]$. For unit $f \in V$, far on-line zeros contribute $\sum |h_f(\gamma)|^2 \geq 0$; far off-line zeros contribute $2 \operatorname{Re} h_f(\gamma) \overline{h_f(\bar{\gamma})}$, with

$$|h_f(t - iy)| = |\widehat{f e^{yu}}(t)| \leq e^{yL/2} \times (\text{decay of the transform of } f e^{yu} \text{ at distance } \geq D_0 \text{ from } I),$$

$f e^{yu}$ being smooth and compactly supported. With a C_c^∞ (Gevrey) taper that decay is of type $\exp(-(D_0/w)^{1/2})$ with $w \sim 1/L$; also $e^{yL/2} \leq X^{1/4} = T^{\lambda/4}$, and the zeros up to height T^2 number $O(T^2 l)$. So $D_0 = (\log T)^C$ gives $\|\text{Tail}\| \leq T^{-A}$; take $\theta = T^{-A}$, so that θd is negligible and $N(I') = N(I)(1 + O(D_0/T))$. The sharp-cutoff and Kaiser windows I use numerically decay only algebraically times $e^{-\beta}$ — irrelevant numerically given the reach margin, but the *theorem* should be stated with a Gevrey taper on an edge layer of width $o(L)$, which changes the effective λ by $o(1)$.

Why one half, and what would give more. Always $n_+ \leq d = \lambda N$, and $\lambda \leq 1 + o(1)$ is forced by the Montgomery–Vaughan diagonal evaluation ($X = o(T \log T)$); for $\lambda > 1$ the prime side involves off-diagonal prime-pair sums, the Hardy–Littlewood/Montgomery-conjecture range. Within $\lambda \leq 1$ the only loss is the Cauchy–Schwarz step of Lemma 5.2: under RH the true n_+ is $d = N$ (semi-local Weil positivity at level $X = T/2\pi$). Any prime-side proof that a fraction $1 - \eta$ of the eigenvalues of R_V exceed θ would give $N_0^{\text{dist}} \geq (1 - 2\eta)N$; but the higher moments $\operatorname{tr} R^k$ are diagonal-dominated only for $k\lambda < 2$ (the Hughes–Rudnick mock-Gaussian range, again from memory), which is useless since I need $\lambda \rightarrow 1$. In one line: the proportion on the line is at least $2 \times (\text{certified-positive fraction of the level-}T/2\pi \text{ Weil form}) - 1$; Cauchy–Schwarz certifies $3/4$, whence $1/2$.

6 Testing the certificate; the one-half claim for referees

6.1 First numerics for the prime-side certificate

Now to test the certificate — the prime-computable lower bound $n_+ \geq (\operatorname{tr} R)^2 / \operatorname{tr}(R^2)$ of Lemma 5.2, fed into the inequality $N_0^{\text{dist}}(I') \geq 2n_+^\theta - N(I')$ of the dual argument — and with it the asymptotic law

$$\frac{(\operatorname{tr} R)^2}{\operatorname{tr} R^2} \sim \frac{\lambda N}{1 + \lambda^2/3}$$

against actual zeros. I use the same Kaiser–Gabor compression as before, i.e. the restriction of the Weil form to the span of atoms $\varphi_L(u) e^{-i\tau_k u}$: Kaiser parameter $\beta \in \{4, 8\}$, centres oversampled at spacing $2\pi/(1.25L)$ over the height window $I = [T_0, T_1]$, the family orthonormalised with respect to the Plancherel mass matrix M (mass-matrix eigenvalues below 10^{-6} of the top dropped), and

$$l = \log \frac{\sqrt{T_0 T_1}}{2\pi}, \quad L = \lambda l, \quad X = e^L.$$

For each window $[300, 600]$, $[600, 1200]$, $[1200, 2400]$ and each $\lambda \in \{0.6, 0.8, 1.0, 1.2, 1.5\}$ I compute: the statistic $C := (\operatorname{tr} R)^2 / \operatorname{tr}(R^2)$ on the prime side and, for validation, on the exactly positive-semidefinite zero side; the true zero-side positive index n_+ (eigenvalues above 10^{-6} of the top); and two predictions for C , the continuum prediction and the asymptotic law,

$$C_{\text{cont}} = d_{\text{full}} \frac{\langle m \rangle^2}{\langle m \rangle^2 + L^2/12\pi^2}, \quad d_{\text{full}} = \frac{L|I|}{2\pi} \text{ (the sharp-cutoff dimension),}$$

$$C_{\text{law}} = \frac{\lambda N(I)}{1 + \lambda^2/3},$$

where $\langle m \rangle$ is the window average of the smooth part m of the density. I also check the Montgomery–Vaughan diagonal directly: with the prime-side density split as $\nu = m + P' + \Pi$ (smooth part, prime sum, pole density), I compare the measured $\text{tr}(R_{P'}^2)$ — the second moment of the compression of the prime-sum part alone — with its $n = n'$ prediction

$$\frac{1}{2\pi^2} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} w_V(\log n),$$

$$w_V(x) = \sum_{\substack{j,k \\ \text{reach}}} \left| \int \hat{\psi}_j(\tau) \overline{\hat{\psi}_k(\tau)} e^{i\tau x} d\tau \right|^2,$$

the sum running over the orthonormalised transforms $\hat{\psi}_j$. The certified proportion of distinct on-line zeros is $(2C - N(I'))/N(I')$, where the reach interval I' is the window widened by the distance beyond which the atoms' transforms are numerically negligible; it is set against the asymptotic value $2\lambda/(1 + \lambda^2/3) - 1$. I expect in advance that the Kaiser taper makes the effective dimension $d < d_{\text{full}}$, so that the finite- β certified proportion should sit below the asymptotic one.

The first two rows come back, on $[300, 600]$ at $\lambda = 0.6$ ($L = 2.53$, $X \approx 13$, $N(I) = 203$). With $\beta = 4$ ($d = 127$, $N(I') = 274$) the statistic collapses: $C_{\text{prime}} = 10.0$ and $C_{\text{zero}} = 9.1$ against $C_{\text{cont}} = 108.0$ and $C_{\text{law}} = 108.7$, Montgomery–Vaughan diagonal measured/predicted $13.7/13.7$, certified proportion -0.927 versus the asymptotic $+0.071$. With $\beta = 8$ ($d = 124$, $N(I') = 278$) it behaves: $C_{\text{prime}} = C_{\text{zero}} = 113.3$, $n_+ = 124$, diagonal $5.1/5.1$, proportion -0.185 versus $+0.071$. The planted-zero scan has so far produced nothing.

The $\beta = 4$ collapse and the crude choice of I' tell me that two things have to be done honestly: the choice of taper, and an actually computed tail. The window function on $[-L/2, L/2]$ is now generic — Kaiser(β), a Tukey flat-top with cosine ramp fraction a , or a bump $\exp(-p(1/(1-x^2) - 1))$ (essentially C_c^∞) — with its transform tabulated by quadrature on $[0, 80]$. Each atom is assigned a *reach radius* r_1 , outside which less than 10^{-10} of its energy lies; then $I' := [T_0 - r_1, T_1 + r_1]$, the tail is the zero-side Gram matrix of the true zeros *outside* I' , actually computed, and the robustness threshold is

$$\theta := \max(2\|\text{Tail}\|, 10^{-8} \cdot \text{scale}),$$

scale being the top zero-side eigenvalue. The certificate is

$$C_\theta := \frac{(\text{tr } R - \theta d)_+^2}{\text{tr } R^2} \quad (\text{prime side}), \quad \text{prop}_{\text{cert}} := \frac{2C_\theta - N(I')}{N(I')},$$

reported alongside the zero-side C , the true robust positive index n_+^θ (which in the RH-verified range is $\min(d, N(I'))$), the prime-versus-zero eigenvalue discrepancy $\max_j |\lambda_j(R_P) - \lambda_j(R_Z)|$ (the quadrature error, which must be far below the θ -scale features), the continuum prediction and the diagonal check. Windows $[600, 1200]$ and $[1200, 2400]$; tapers Kaiser 8, Kaiser 12, Tukey 0.5, Tukey 0.25, bump 0.5; $\lambda \in \{0.6, 0.8, 1.0, 1.15, 1.3, 1.6, 2.0\}$. (Editorial observation: the grid runs past $\lambda = 1$, beyond the range the diagonal evaluation covers; nothing is written about it here.)

First rows, Kaiser 8 on $[600, 1200]$ ($N(I) = 472$):

λ	X	d	$C_\theta(\text{prime}) / C(\text{zero})$	n_+^θ	MV meas/pred	$\text{prop}_{\text{cert}} / \text{asym}$
0.60	19	285	260.1 / 260.1	285	16.6 / 16.6	$-0.130 / +0.071$
0.80	51	379	321.8 / 321.8	379	41.2 / 41.3	$+0.076 / +0.319$
1.00	135	472	367.6 / 367.5	472	82.6 / 82.8	$+0.229 / +0.500$

Prime and zero sides agree to the printed digit; the tail is negligible ($\theta = 10^{-8}$ of scale, eigenvalue discrepancy from $5 \cdot 10^{-3}$ down to 10^{-4}); the Montgomery–Vaughan diagonal prediction matches the measured $\text{tr } R_P^2$ to well under one percent; and positive certificates appear from $\lambda = 0.8$ on. But the algebraically decaying Kaiser atoms have reach $r_1 \approx 80$, so $N(I') = 598$ against $N(I) = 472$. (Editorial: presumably it is this inflated $N(I')$, rather than the form, that depresses the certified proportions; the record does not say so in words.)

So I loosen the reach criterion to the radius outside which less than 10^{-5} of the atom’s energy lies — legitimate now that the actual tail norm is computed and fed into θ — put the bump taper first (then Tukey 0.5, Tukey 0.25, Kaiser 12), and start again.

6.2 Bump-taper rows, cluster counts and synthetic controls

The bump(0.5) rows on $[600, 1200]$ ($N(I) = 472$) come back. The reach is now $r_1 = 25.8$ falling to 13.7 as λ grows, so $N(I')$ runs from 512 down to 494; and with an essentially C_c^∞ taper there is a genuine tail, $\theta \approx 5 \cdot 10^{-2}$ down to $2 \cdot 10^{-2}$ of scale, so the prime-side C_θ sits visibly below the untruncated zero-side C :

λ	X	d	$C_\theta(\text{prime}) / C(\text{zero})$	$n_+^\theta(\text{true})$	MV meas/pred	prop _{cert} / asym
0.60	19	276	217.2 / 253.7	276	14.7 / 14.7	−0.152 / +0.071
0.80	51	367	260.0 / 315.4	366	36.8 / 36.8	+0.036 / +0.319
1.00	135	457	315.5 / 362.1	447	73.7 / 74.0	+0.270 / +0.500
1.15	282	525	355.3 / 388.0	477	114.3 / 114.8	+0.439 / +0.596

The prime/zero eigenvalue discrepancy stays at 10^{-4} , three orders of magnitude below θ , and the Montgomery–Vaughan diagonal prediction for $\text{tr } R_P^2$ matches the measured value to well under one percent throughout.

The index grid has meanwhile finished, ending exactly on the reading I fixed for it in advance: $n_-(G_P) = 0$ throughout; $d - \text{rank} \approx d - \#\text{zeros}$ exact null directions whenever $d > \#\text{zeros}$; the symbol ν_X negative on 20–45% of the window; the Szegő/pointwise prediction and the a-priori bound “off by everything”.

The cluster/multiplicity experiment (zero side plus plants at $t_0 = 400$, $\delta = 0.2$) has also reported. At $L = 10$ ($d = 573$): a single quadruple gives $n_- = 1$; a *double* zero (weight 2) at the same point gives $n_- = 1$ — multiplicity invisible, as Theorem 1 says; two depths 0.2 and 0.35 at one ordinate give 2; two quadruples at $t_0 \mp \varepsilon/2$ give $n_- = 1$ for $\varepsilon = 0.01, 0.03$ and 2 for $\varepsilon \geq 0.1$ (the resolution scale being $2\pi/L = 0.63$); five quadruples spaced ε apart give

$$n_- = 1, 2, 2, 2, 4, 5, 5, 5 \quad \text{for } \varepsilon = 0.01, 0.03, 0.1, 0.3, 0.6, 1.0, 2.0, 4.0.$$

At $L = 14$ ($d = 804$, $2\pi/L = 0.45$) two quadruples already separate at $\varepsilon = 0.01$, and five give 2, 2, 3, 4, 5, 5, 5, 5. This is the picture of the cluster analysis in the visibility section: tight clusters merge into fewer negative squares and split as L grows, and coincident zeros never add rank.

The certificate needs no zeros, so I can go much higher than the zero table allows. I set up a prime-side-only evaluation at larger heights: windows $[9000, 10000]$ and $[48000, 50000]$, $\lambda \in \{0.7, 0.85, 1.0, 1.1\}$, a Tukey(0.2) flat-top taper at the critical spacing $2\pi/L$ (oversampling $q = 1.0$), M -orthonormalised with tolerance 10^{-6} . The statistic $C = (\text{tr } R)^2 / \text{tr } R^2$ is computed from the full prime-side density ν_X^{full} alone; I compare $C/N(I)$ with the law $\lambda/(1 + \lambda^2/3)$ and with the continuum prediction, and report the certified proportion as $2C/N(I') - 1$, taking $N(\cdot)$ to be the Riemann–von Mangoldt main term $\vartheta(T)/\pi + 1$ (ignoring $|S(T)| < 1$, an $O(1)$), with no θ /tail correction at these heights.

I want a control on the inertia logic itself, since inside the RH-verified range everything is trivially consistent. So I build synthetic zero-side configurations on $[600, 1200]$ with the same kind of family (Kaiser $\beta = 10$, spacing $2\pi/(1.25L)$, 504 true zeros within reach) and, for each, compute the configuration's zero count N_{conf} , the theoretical positive index $n_+(Q)$ of the block form, the numerical $n_+^\theta(R)$ and $n_-^\theta(R)$, the statistic $C = (\text{tr } R)^2 / \text{tr } R^2$, and the would-be certificate $2C - N_{\text{conf}}$, which by the inequality (*) of the dual argument must never exceed the true number $n_{\text{on}}^{\text{dist}}$ of distinct on-line zeros. The configurations:

- (a) the true zeros, all on-line;
- (b) consecutive zeros paired into an off-line pair at their midpoint $\mp ia$, with $a = 0.02, 0.1, 0.3$, or $a = \text{half the gap}$;
- (c) " $L(s) = \zeta(s+a)\zeta(s-a)$ ": every zero doubled into $\gamma \mp ia$, $a = 0.05, 0.25$ (zero percent on the line, $N_{\text{conf}} = 1008$);
- (d) a quarter of the zeros kept on-line, the rest paired at depth 0.1 (true $n_{\text{on}}^{\text{dist}} = 126$, $N_{\text{conf}} = 378$).

The results, for $\lambda = 0.8/1.0/1.3$ ($d = 368/459/595$):

- (a) $n_-^\theta = 0$, $C = 315.4/362.0/405.4$, $2C - N = +126.7/+220.0/+306.8$, comfortably below 504.
- (b) $n_+^\theta(R) = 238\text{--}241 \leq n_+(Q) = 252$ with $n_-^\theta = 129/\approx 220/\approx 238$, and $2C - N$ between -43.8 and -260.3 for fixed a ; for half-gap pairing $C = 26.6/4.4/0.2$ and $2C - N = -450.7/-495.2/-503.7$; the truth is 0.
- (c) $n_+^\theta = 367\text{--}476 \leq 504$, $2C - N = -205.1, \dots, -409.9$, truth 0 — and notably, at $\lambda = 0.8$ the shallow doubling $a = 0.05$ gives $C = 314.8$ with $n_-^\theta = 0$, essentially the true-zero value 315.4. (Editorial: there the inequality holds only because N_{conf} is doubled; the record does not remark on this.)
- (d) $2C - N = -128.8/-123.1/-114.5$ against the truth 126.

So the inequality never over-certifies.

The large-height prime-only run ($N(I) = 1165.1$ and 2852.6 for the two windows):

window	λ	X	d	$C/N(I)$	law	prop / asym
[9000, 10000]	0.70	168	814	0.613	0.602	+0.184 / +0.203
	0.85	504	988	0.699	0.685	+0.359 / +0.370
	1.00	1510	1162	0.766	0.750	+0.496 / +0.500
	1.10	3139	1278	0.802	0.784	+0.569 / +0.568
[48000, 50000]	0.70	530	1991	0.610	0.602	+0.202 / +0.203
	0.85	2033	2417	0.695	0.685	+0.374 / +0.370
	1.00	7797	2843	0.762	0.750	+0.510 / +0.500
	1.10	19104	3127	0.798	0.784	+0.582 / +0.568

The prime-only statistic tracks $\lambda/(1+\lambda^2/3)$ to about two percent and gives a certified proportion of about +0.50 at $\lambda = 1$. (Editorially: the $\lambda = 1.1$ rows lie beyond the $\lambda \leq 1$ range that my own diagonal evaluation covers, and at these heights there is no zero-side validation; nothing is written about either.) One more bump row on $[600, 1200]$: $\lambda = 1.30$ ($X \approx 588$, $d = 593$), $C_\theta = 361.3$ versus $C(\text{zero}) = 406.0$, diagonal $168.7/168.5$, $\text{prop}_{\text{cert}} = +0.469$ versus the asymptotic +0.663.

6.3 Measured visibility thresholds for planted off-line zeros

The planted-zero scan with its two thresholds is now complete, and I can read off how soon a planted quadruple $\frac{1}{2} \pm \delta \pm it_0$ added to the 2000 true zeros produces a negative square in the compressed form. The annihilation threshold L_{ann}^* (first level with an eigenvalue below -10^{-8} of scale) is small and almost independent of the depth δ : in add mode 3.25–3.50 at

$t_0 = 100$, 3.75–4.50 at 400, 4.75–6.50 at 1000, 5.25–7.50 at 2200 (slightly lower in replace mode), with the offending eigenvalue at 10^{-8} – 10^{-4} of scale — growing with height roughly like $\log(t_0/2\pi) = 2.77/4.15/5.07/5.86$.

The robust threshold L_{rob}^* (an eigenvalue below -10^{-3} of scale) does feel the depth: at $t_0 = 100$ in add mode it is 12.75 ($\delta = 0.02$), 8.50, 6.75, 5.50, 5.00, 4.50 ($\delta = 0.45$); at $t_0 = 400$ it runs from 13.75 down to 4.25; at $t_0 = 1000$ and 2200 with $\delta \leq 0.05$ in add mode (and at 2200, $\delta = 0.02$ in replace mode) it is not reached up to $L = 14$. The least-squares fits in add mode:

$$\begin{aligned} L_{\text{ann}}^* &\approx 1.123 \delta^{-0.066} \log(t_0/2\pi)^{0.889}, \\ L_{\text{rob}}^* &\approx 1.618 \delta^{-0.336} \log(t_0/2\pi)^{0.599}. \end{aligned}$$

Both lie far below the worst-case law $\sqrt{n_1 + 1}/\delta$ of Theorem 2 printed alongside (50–100 at $\delta = 0.02$), and the robust threshold sits somewhat above the cube-root average-spacing law (8.2–10.5 at $\delta = 0.02$) that I had identified with R4’s planting law; the fitted δ -exponent -0.34 is neither Theorem 2’s -1 nor the average law’s $-2/3$. The cluster experiment’s $L = 6$ block ($d = 345$) repeats: single quadruple $\mapsto 1$, double zero $\mapsto 1$. Nothing new from the other two runs.

6.4 The one-half claim written out for referees

Now to write the claimed Theorem 4 out self-contained, for hostile refereeing — the statement being that at least one half of the nontrivial zeros of ζ are on the critical line, as distinct zeros. This goes into a separate proof file addressed to referees; here is its content.

Notation and inputs. Zeros $\rho = \beta + it$ with multiplicity m_ρ , and $\gamma_\rho := (\rho - \frac{1}{2})/i = t - i(\beta - \frac{1}{2})$; $N(a, b)$ counts zeros with $a < t \leq b$ with multiplicity (Riemann–von Mangoldt: $N(0, T) = (T/2\pi) \log(T/2\pi e) + O(\log T)$); $N_0^{\text{dist}}(a, b)$ is the number of distinct points $\frac{1}{2} + it$, $a < t \leq b$, with $\zeta(\frac{1}{2} + it) = 0$. Test functions are $f \in C_c^2(\mathbb{R})$, with $h_f(z) = \int f(u) e^{izu} du$ entire and $|h_f(z)| \ll \|f''\|_1 e^{|\text{Im } z|L/2}/|z|^2$, so that

$$W(f, g) := \sum_{\rho} m_{\rho} h_f(\gamma_{\rho}) \overline{h_g(\bar{\gamma}_{\rho})}$$

converges absolutely; it is Hermitian by reindexing $\rho \rightarrow 1 - \bar{\rho}$ ($\gamma \rightarrow \bar{\gamma}$, multiplicities preserved). The explicit formula (Weil, unconditional), for $\text{supp } f, g \subset [-L/2, L/2]$, $X = e^L$, $s = \frac{1}{2} + i\tau$:

$$\begin{aligned} W(f, g) &= \int_{\mathbb{R}} h_f(\tau) \overline{h_g(\tau)} \nu_X(\tau) d\tau, \\ \nu_X(\tau) &= \frac{\vartheta'(\tau)}{\pi} + \frac{1}{2\pi(\frac{1}{4} + \tau^2)} + \frac{1}{\pi} \text{Re} \frac{X^s - 1}{s} \\ &\quad - \frac{1}{\pi} \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \cos(\tau \log n), \end{aligned}$$

the pole term having been rewritten as a density exactly as in the pole-term derivation earlier, and “Verified numerically to 1e-6..1e-9 against Sum_gamma over 2000 zeros” by the compression computation.

Lemmas. Lemma 1 (inertia under pullback): for a linear map A and Hermitian form Q , $n_+(A^*QA) \leq n_+(Q)$ — if $A^*QA > 0$ on U then $A|_U$ is injective and $Q > 0$ on $A(U)$. Lemma 2 (Cauchy–Schwarz count): for Hermitian R of size d and $\theta \geq 0$, if $\text{tr } R > \theta d$ then

$$n_+^{\theta}(R) \geq \frac{(\text{tr } R - \theta d)^2}{\text{tr}(R^2)},$$

since $S := \sum_{\lambda_i > \theta} \lambda_i \geq \text{tr } R - \theta d > 0$ and $S^2 \leq n_+^\theta \text{tr } R^2$. Lemma 3 (Weyl): $\#\{\lambda_i(A + E) > \theta\} \leq \#\{\lambda_i(A) > \theta - \|E\|\}$.

Set-up. T large, $l = \log(T/2\pi)$, $I = [T, 2T]$, $\lambda \in (0, 1]$, $L = \lambda l$, $X = (T/2\pi)^\lambda$. The space V is spanned by $f_k(u) = \varphi(u)e^{-i\tau_k u}$ with $\tau_k = T + 2\pi k/L$, $0 \leq k < d = \lfloor LT/2\pi \rfloor$, where $\varphi \in C_c^2[-L/2, L/2]$ equals 1 on $[-(1-\eta)L/2, (1-\eta)L/2]$ and has C^2 ramps of width $\eta L/2$, $\eta = 1/\log l$; then $h_{f_k} = \hat{\varphi}(\cdot - \tau_k)$ with $|\hat{\varphi}(r)| \ll (\eta L)^{-2}|r|^{-3}$ for $|r| \geq 1$ and $\hat{\varphi}(0) \sim L$. R is the matrix of W in an orthonormal basis of V ; by Plancherel the mass matrix is Toeplitz with symbol in $[1 - O(\eta), 1]$, so $M = L(\text{Id} + O(\eta))$ and, with G the Gram matrix of W on the f_k ,

$$R = L^{-1}(\text{Id} + O(\eta))^{-1/2} G (\text{Id} + O(\eta))^{-1/2}.$$

Step 1 (zero side: upper bound for the robust positive index). Take $D_0 = T^{1/2}$ (any T^c , $c < 1$, would do), $I' = [T - D_0, 2T + D_0]$, and split $W|_V = A + E$ according to whether $t \in I'$. Grouping A by distinct points: a distinct on-line zero gives $m|l_t|^2$, l_z being the evaluation functional $f \mapsto h_f(z)$; a distinct off-line zero $\frac{1}{2} + \delta + it$ comes with $\frac{1}{2} - \delta + it$ (same t , same m , both counted in $N(I')$), and together they give $m[l_\gamma \bar{l}_{\gamma^*} + l_{\gamma^*} \bar{l}_\gamma]$, the block $\begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}$ of signature $(1, 1)$. Hence $A = \text{ev}^* Q \text{ev}$ with

$$\begin{aligned} n_+(Q) &= n_{\text{on}}^{\text{dist}}(I') + n_{\text{pair}}^{\text{dist}}(I'), \\ N(I') &\geq n_{\text{on}}^{\text{dist}}(I') + 2n_{\text{pair}}^{\text{dist}}(I'). \end{aligned} \tag{1}$$

The tail needs only a *norm* bound, not visibility: for unit $f = \sum c_k f_k$ one has $\|c\|_2 \ll L^{-1/2}$; a zero at distance $D \geq D_0$ from I and depth $< \frac{1}{2}$ has $|h_f(t - iy)| \ll d^{1/2} e^{L/4} D^{-3} (\eta L)^{-2}$; and summing D^{-6} against zero density $\ll \log(T + D)$,

$$\begin{aligned} \|E\| &\ll d e^{L/2} l D_0^{-5} (\eta L)^{-4} \\ &\ll T^{1+\lambda/2-5/2+o(1)} \leq T^{-1+o(1)} =: \theta_0. \end{aligned}$$

By Lemmas 3 and 1, for $\theta \geq \theta_0$,

$$n_+^\theta(R) \leq n_+(A) \leq n_+(Q), \tag{2}$$

and from (1) and (2),

$$n_{\text{on}}^{\text{dist}}(I') \geq 2n_+^\theta(R) - N(I'). \tag{3}$$

Step 2 (prime side: lower bound). Split $\nu_X = \mu + P + \Pi$ on $\tau > 0$:

$$\begin{aligned} \mu &= \frac{\vartheta'}{\pi} + \frac{1}{2\pi(\frac{1}{4} + \tau^2)} = \frac{1}{2\pi} \log \frac{\tau}{2\pi} + O(\tau^{-2}), \\ \Pi &= \frac{1}{\pi} \text{Re} \frac{X^s - 1}{s} = O(X^{1/2}/\tau), \\ P &= -\frac{1}{\pi} \sum_{n \leq X} a_n \cos(\tau \log n), \quad a_n = \Lambda(n)n^{-1/2}. \end{aligned}$$

The claim:

$$\begin{aligned} \text{tr } R &= LT \langle \mu \rangle_I (1 + o(1)), \\ \text{tr } R^2 &= 2\pi LT \left[\langle \mu \rangle_I^2 + \frac{L^2}{12\pi^2} \right] (1 + o(1)), \quad \langle \mu \rangle_I \sim \frac{l}{2\pi}, \end{aligned}$$

$\langle \mu \rangle_I$ denoting the average of μ over I . Sketch, suppressing the $O(\eta)$ taper corrections.

- (i) With $\varphi = 1$ the transforms are Dirichlet kernels $D_L(x) = \sin(Lx/2)/(x/2)$, and $\{f_k\}$ is orthogonal with $\|f_k\|^2 = L$. Poisson summation gives $\sum_k D_L(\tau - \tau_k)^2 = L^2$ on I away from the ends (the transform of D_L^2 is $2\pi(L - |u|)_+$, which vanishes at the aliasing points $\pm L$), and Shannon sampling at the Nyquist rate gives

$$K(\tau, \tau') = L^{-1} \sum_k D_L(\tau - \tau_k) D_L(\tau' - \tau_k) = D_L(\tau - \tau') \quad \text{on } I \times I,$$

up to end effects within $O(T^{o(1)})$ of the endpoints carrying a $T^{-1+o(1)}$ fraction of the traces. So $\text{tr } R = L \int_I \nu$ and $\text{tr } R^2 = \iint_{I \times I} D_L(\tau - \tau')^2 \nu \nu'$, plus ends.

- (ii) $\int_I P = O(\sum_{n \leq X} \Lambda(n) n^{-1/2} / \log n) = O(X^{1/2}/L)$ and $\int_I \Pi = O(X^{1/2} \log 2)$, both $O(T^{\lambda/2}) = o(Tl)$.
- (iii) The $\mu\mu'$ term: μ is Lipschitz with constant $O(1/T)$ and $\int D_L^2 = 2\pi L$, giving $2\pi L T \langle \mu \rangle_I^2 (1 + O(1/l^2))$.
- (iv) The PP' term: writing $\tau = \tau' + x$ and splitting the product of cosines, the diagonal $n = n'$ gives $\frac{1}{2} T \cdot 2\pi(L - \log n)_+$, hence

$$\begin{aligned} \text{DIAG} &= \frac{TL}{\pi} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} \left(1 - \frac{\log n}{L}\right) \\ &= \frac{TL}{\pi} \left(\frac{L^2}{6} + O(L)\right), \end{aligned}$$

using $\sum_{n \leq x} \Lambda(n)^2/n = (\log x)^2/2 + O(\log x)$ [PNT]. The off-diagonal terms and the sum-frequency bracket have τ' -integrals $\int_I e^{i\tau'w}$ with $w = \log n \mp \log n' \neq 0$, and the Montgomery–Vaughan generalized Hilbert inequality with $\delta_n \geq 1/(2n)$ and weights $\leq 2\pi L a_n$ gives

$$\begin{aligned} \text{OFF} &= O\left(L^2 \sum_{n \leq X} \Lambda(n)^2\right) = O(L^3 X), \\ \frac{\text{OFF}}{\text{DIAG}} &= O(X/T) = O(T^{\lambda-1}) \rightarrow 0 \quad (\lambda < 1). \end{aligned}$$

At $\lambda = 1$ exactly, where $X/T = 1/2\pi$, I waver — “hmm; SAFE STATEMENT: for every fixed $\lambda < 1$, $\text{OFF} = o(\text{DIAG})$; the bound at $\lambda \rightarrow 1^-$ is then a limit” — and then record a refinement: Montgomery–Vaughan applied with the T -length integral gives a ratio $O(X/(TL)) \rightarrow 0$ even at $\lambda = 1$; either way $\lambda \rightarrow 1$ is admissible in the limit.

- (v) The $\mu P'$ cross term is $O(LX^{1/2}) = o(TLl^2)$ by integration by parts ($\mu' = O(1/T)$).
- (vi) The Π terms are pointwise $O(T^{\lambda/2-1})$ on I and negligible.

This proves the claim, and with $\theta = \theta_0$ (so that $\theta d = o(\text{tr } R)$),

$$\begin{aligned} n_+^\theta(R) &\geq \frac{(\text{tr } R - \theta d)^2}{\text{tr } R^2} = \frac{LT}{2\pi} \frac{\langle \mu \rangle_I^2}{\langle \mu \rangle_I^2 + L^2/12\pi^2} (1 + o(1)) \\ &= \frac{\lambda}{1 + \lambda^2/3} N(I) (1 + o(1)). \end{aligned} \tag{4}$$

The consistency remark once more: under RH the left-hand statistic is $(\sum_\gamma L)^2 / \sum_{\gamma, \gamma'} D_L(\gamma - \gamma')^2$, and Montgomery’s pair-correlation theorem (Fourier support $|\alpha| \leq \lambda \leq 1$) evaluates the denominator to $NLl(1 + \lambda^2/3)$ — the same value; (4) at $\lambda = 1$ is the classical “ $N_{\text{distinct}} \geq N^2 / \sum m^2 \geq 3N/4$ under RH”. The evaluation is Montgomery’s prime-side computation, which is unconditional; RH entered classically only to read the zero side as a positive sum over real γ , and Lemma 1 replaces that.

Step 3. $N(I') = N(I)(1 + O(D_0/T + 1/l))$, so from (3) and (4)

$$N_0^{\text{dist}}(T - D_0, 2T + D_0) \geq \left[\frac{2\lambda}{1 + \lambda^2/3} - 1 - o(1) \right] N(I),$$

$$\liminf_{T \rightarrow \infty} \frac{N_0^{\text{dist}}([T, 2T])}{N([T, 2T])} \geq 2 \cdot \frac{3}{4} - 1 = \frac{1}{2},$$

nontrivial for $\lambda > 3 - \sqrt{6} = 0.5505$; the $o(1)$'s are $O(1/\log l)$ from the taper, $O(1/L)$ from the prime sums, and $O(T^{\lambda-1})$ from Montgomery–Vaughan.

What is used: the explicit formula for C_c^2 test functions, Riemann–von Mangoldt, the PNT in the form $\sum_{n \leq x} \Lambda(n)^2/n = (\log x)^2/2 + O(\log x)$, the Montgomery–Vaughan Hilbert inequality, Poisson/Shannon for the sinc system, and Lemmas 1–3. What is *not* used: any zero-free region, any density theorem, any numerical RH verification, any positivity of the Weil form. For comparison, all from memory and unverifiable here: Selberg (positive proportion), Levinson 1/3, Conrey 0.4088, Bui–Conrey–Young 0.4105, Feng 0.4128, Pratt–Robles–Zaharescu–Zeindler (2020) 5/12 = 0.4167, all counted with multiplicity; simple zeros on the line $\geq 0.407N$; under RH $\geq 2/3$ simple (Montgomery), $\geq 3/4$ distinct (Cauchy–Schwarz), “later ~ 0.8467 distinct?? (from memory)”. An unconditional 1/2, distinct and on the line, would exceed every unconditional record I remember; “hence maximal suspicion is warranted”, and the red team is still to be written.

7 Second window; de Branges tilt; the red team

7.1 Second window also zero; certificate runs retuned

With the referee version set down, I go back to the index grid — the inertia (n_-, n_0, n_+) (numbers of negative, null and positive eigenvalues) of the Gram matrix G_P of the Weil form compressed to the Gabor space $V_{L,T}$ of windowed exponentials, computed from the primes $\leq X = e^L$ and Γ alone — which has finished its second height window. On $[1400, 2400]$ (907 zeros) the compressed form again has no negative square at any level:

X	d	n_-	n_0	n_+	$\min \text{ev}(G_P)$	$\text{frac}(\nu_X < 0)$
10^2	587	0	0	587	$+6.27 \cdot 10^{-4}$	0.111
10^3	880	0	8	872	$+8.71 \cdot 10^{-10}$	0.249
10^4	1173	0	263	910	$-8.37 \cdot 10^{-8}$	0.331
10^5	1466	0	556	910	$-2.81 \cdot 10^{-7}$	0.371
10^6	1760	0	850	910	$-8.25 \cdot 10^{-7}$	0.395

The discrepancy $\|G_P - G_Z\|_F$ between the prime-side and zero-side Gram matrices lies between $1.1 \cdot 10^{-5}$ and $9.9 \cdot 10^{-5}$, so the tiny negative minima are leakage; the zero-side ranks 587, 879, 910, 910, 911 match $n_+(G_P)$ row by row. The pointwise quantities go the other way: the “Szegő prediction” for n_- (the fraction of the window with $\nu_X < 0$, times d) climbs $65.1 \rightarrow 695.9$, and the a-priori count of Proposition 2.1 restricted to the window climbs $72.5 \rightarrow 725.6$, while the prime-computable certified-positive fraction $(\text{tr } R)^2/(d \text{tr } R^2)$ — R being the matrix of $W|_V$ in mass-orthonormal coordinates — falls $0.896 \rightarrow 0.783 \rightarrow 0.665 \rightarrow 0.560 \rightarrow 0.480$ (its Fejér proxy $0.881 \rightarrow 0.451$).

Together with the $[50, 1000]$ column this covers the whole of the brief’s range $X = 10^2, \dots, 10^6$ on two windows: the honestly compressed Pontryagin index is identically zero everywhere I can compute it, and it is computed from primes alone. The certificate run with the bump taper — the certificate being the prime-computable Cauchy–Schwarz lower bound $C_\theta = (\text{tr } R - \theta d)_+^2 / \text{tr } R^2$ on the robust positive index, fed into $N_0^{\text{dist}}(I') \geq 2n_+^\theta - N(I')$ — has only reached the $\lambda = 1.30$

row already seen ($C_\theta = 361.3$ on the prime side against $C = 406.0$ on the zero side, certified proportion $+0.469$ against the asymptotic $+0.663$).

I decide to trim the certificate experiment and rerun it with a well-conditioned family: atom spacing $2\pi/L$ instead of $2\pi/(1.25L)$, mass-orthonormalisation tolerance 10^{-3} instead of 10^{-6} , the tapers reduced to a Tukey window with ramp fraction 0.25 and a Kaiser window with $\beta = 12$, and the Montgomery–Vaughan diagonal prediction skipped for $\lambda > 1.15$. I also want the prime-only law $C/N(I) \rightarrow \lambda/(1 + \lambda^2/3)$ tested a long way above the zero table, so I set up the same larger-height computation on the window $[998000, 1000000]$ with $\lambda \in \{0.85, 1.0\}$ — again prime side only, $N(\cdot)$ taken from the Riemann–von Mangoldt main term, no tail correction.

7.2 The de Branges tilt, recorded from memory

With those set going, two pieces of theory remain. The first is the brief’s de Branges/ δ -tilt suggestion, which I record but do not develop; everything here is from memory, with no lookup. For $a \in (\frac{1}{2}, 1)$ put $E_a(z) := \xi(a - iz)$; a zero $\rho = \beta + it$ of ξ gives a zero of E_a in the upper half-plane $\mathbb{C}_+$ exactly when $\beta > a$, and by the functional equation

$$E_a^\#(z) := \overline{E_a(\bar{z})} = \xi(1 - a - iz).$$

The kernel

$$K_a(w, z) = \frac{E_a(z)\overline{E_a(w)} - E_a^\#(z)\overline{E_a^\#(w)}}{2\pi i(\bar{w} - z)}$$

has, by the Kreĭn–Langer factorisation of the generalized Schur function $E_a^\# / E_a$ (unimodular on $\mathbb{R}$ by the functional equation), exactly κ_a negative squares, where κ_a is the number of zeros of E_a in $\mathbb{C}_+$ with multiplicity, i.e. $N(\beta > a)$ over all heights. I recall, without being able to check it here, Lagarias 2005: E_a is Hermite–Biehler for $a \geq 1$ unconditionally, and for all $a > \frac{1}{2}$ iff RH. So the family $\{E_a\}$ realises the brief’s δ -tilt with multiplicity counted, unlike the Weil form of Theorem 1.

But the compressions of K_a to $\text{span}\{K_a(\cdot, x_j)\}$ at real sample points x_j have Gram matrices computable only from ξ on the two lines $\text{Re } s = a$ and $1 - a$ — Riemann–Siegel, not a finite prime sum. Their negative index is $\leq \kappa_a$ (the certificate direction only) and is bounded below only by the number of Blaschke phase windings the sampling resolves, an m -fold zero at depth y needing about m samples in a window of width y . So the count is again bounded only by the number of samples: the same wall that the Weil-form compressions hit as an upper-bound machine; and there is no analogue of the positive-index dual better than the dual itself, because the positive squares of K_a are not tied to a zero count. R4’s observation that $\frac{d}{da} \log |\xi(a + it)/\xi(1 - a + it)|$ at $a = \frac{1}{2}$ is $2\pi\nu_X$ identifies the diagonal $K_{1/2, X}(x, x)$ of the truncated kernel with ν_X — the same object as R4’s signed density.

7.3 Red team: empty for bounds, suspect as theorem

The second piece is the red team. On the negative index as an *upper* bound for off-line zeros — Theorems 1 and 2 and the compression — my verdict is: empty, for structural reasons, not Gibbs. The inequality $\kappa(V) \leq N_{\text{off}}^{\text{dist}}$ always holds, which is the wrong direction; the reverse needs visibility; visibility of everything below height T needs $\dim V \gtrsim N(T)$ — the annihilation regime $L \sim \log(T/2\pi)$, with the measured threshold

$$L_{\text{ann}}^* = 1.12 \log(t_0/2\pi)^{0.89} \delta^{-0.07}$$

— and there the only a-priori bound, $\kappa \leq \dim V$, is trivial. In the sub-Landau regime $L \sim \sqrt{\log T}/\delta$ of Theorem 2, visibility holds only for isolated deep zeros (the interference obstruction),

clusters count once (the cluster experiment), multiplicity never (Theorem 1). Any prime-side bound on $\kappa(V)$ that passes through pointwise sign information on ν_X is at least $(L/2\pi)|\{\nu_X < 0\}| \sim 0.2\text{--}0.4 \dim V$ — the Szegő and a-priori columns of the grid — while the true $\kappa(V)$ is 0 in every tested range, $X = 10^2, \dots, 10^6$, $T \leq 2400$, from primes alone. R4’s 33–153 “negative eigenvalues” belong to the warped-sinc operator, not to a compression of the Weil form. So I retract R4’s headline as a statement about level- X Weil forms in the computed ranges: the condition space is a Hilbert space there (as it must be under RH); it is Kreĭn exactly iff RH fails visibly.

On the dual and the claimed Theorem 4, the checks performed are:

- (a) prime side = zero side for the Gram matrices to $10^{-6}\text{--}10^{-9}$ (relative eigenvalue discrepancy $\sim 10^{-4}$ after orthonormalisation);
- (b) the inertia inequality $n_+^\theta(R) \leq n_{\text{on}}^{\text{dist}} + n_{\text{pair}}^{\text{dist}}$ holds on every synthetic configuration with 0%, 25%, 100% of zeros on the line, the would-be certificate $2C - N$ staying below the true $n_{\text{on}}^{\text{dist}}$ (all-paired: -50 to -495 ; “ $\zeta(s \pm a)$ ”: -287 to -410 ; quarter on-line: $-123 < 126$);
- (c) the prime-only law $C/N(I) \rightarrow \lambda/(1 + \lambda^2/3)$ at $T = 10^4$ and $5 \cdot 10^4$: $0.613/0.699/0.766/0.802$ and $0.610/0.695/0.762/0.798$ against $0.602/0.685/0.750/0.784$ at $\lambda = 0.7/0.85/1.0/1.1$, certified proportions $+0.50, +0.51$ at $\lambda = 1$, with $T = 10^6$ pending;
- (d) Montgomery–Vaughan diagonal dominance measured directly, $\text{tr}(R_P^2)$ against its $n = n'$ prediction to $0.1\text{--}0.5\%$;
- (e) under RH the bound coincides with Montgomery plus Cauchy–Schwarz (3/4 distinct), an independent consistency check of all constants.

Then the failure points a referee should target, in decreasing order of my own worry.

- (1) “Too strong to be new”: the argument is short and uses 1970s tools. Either it is known — I do not know a reference; from memory the closest are Montgomery 1973 (under RH), Gallagher–Mueller, and Baluyot–Goldston–Suriajaya–Turnage–Butterbaugh’s unconditional pair-correlation $F(\alpha)$ (2023/24), where as far as I recall no unconditional proportion *on the line* is drawn, presumably because extracting the diagonal from $\sum_{\rho, \rho'}$ needs a positivity that fails for complex weights, and Lemma 1 (pullback inertia) is exactly what avoids that — or there is an error I cannot see. I looked hardest at the block structure (an off-line pair supplies one positive square: yes, $\begin{pmatrix} 0 & m \\ m & 0 \end{pmatrix}$), at Riemann–von Mangoldt counting both members of a pair (yes: ρ and $1 - \bar{\rho}$ are distinct nontrivial zeros at the same height), at Lemma 1 itself (two lines), at the reach/tail truncation ($\theta = T^{-1+o(1)}$ suffices with the C_c^2 taper), and at the trace asymptotics (elementary, numerically confirmed).
- (2) Kernel end-effects in $\text{tr } R^2$ with the pointwise-large $|\nu_X| \sim \sqrt{X}$ near $\lambda = 1$: bounded above using averages “(RT2 in scratch)” — a side computation of mine recorded nowhere else — and the finite computations, which contain all end effects exactly, match the law within 2%; still, a fully written proof must do this estimate honestly.
- (3) The Montgomery–Vaughan step at $\lambda = 1$ exactly: OFF/DIAG = $O(1/L)$ with a crude explicit constant $\sim 36/L$; for a clean statement take $\lambda < 1$ fixed and let $\lambda \rightarrow 1$.
- (4) The taper margin $\eta \rightarrow 0$ and the reach margin $D_0 = T^{1/2}$: routine.

None of (2)–(4) can move the constant $1/2$; only (1) can kill the result. And what Theorem 4 is *not*: it is not a density estimate; it says nothing about multiplicities of on-line zeros; it cannot exceed $1/2$ without Hardy–Littlewood-type prime-pair input or a genuinely new positivity input (as recorded under “why one half”); and the finite prime-side certificates of Theorem 3 have nil value inside the RH-verified range.

7.4 Retuned certificate rows and the first row at 10^6

The retuned certificate run has produced its Tukey(0.25) rows on $[600, 1200]$ ($N(I) = 472$). With spacing $2\pi/L$ the atom reach — the radius beyond which an atom’s transform carries less than 10^{-5} of its energy — drops from 28.8 to 9.0 as λ grows, the tail threshold θ is a few 10^{-4} of scale, and C_θ on the prime side agrees with the zero-side C to about four digits:

λ	$C_\theta(\text{prime})$	$C(\text{zero})$	MV meas/pred	prop _{cert}	asym
0.60	252.0	252.4	15.0/15.0	−0.025	+0.071
0.80	312.6	313.1	37.1/37.1	+0.233	+0.319
1.00	359.5	360.0	74.8/74.8	+0.438	+0.500
1.15	385.1	385.9	116.6/116.7	+0.556	+0.596
1.30	403.0	403.7	—	+0.635	+0.663
1.60	425.0	425.8	—	+0.738	+0.727
2.00	441.4	442.2	—	+0.816	+0.714

Here $\text{prop}_{\text{cert}} = (2C_\theta - N(I'))/N(I')$ is the certified proportion of distinct on-line zeros on the reach interval I' , and the last column is the sharp-cutoff asymptotic $2\lambda/(1 + \lambda^2/3) - 1$. The first Kaiser($\beta = 12$) row, at $\lambda = 0.6$, gives $C = 193.9$ on both sides and certified proportion −0.197.

And the first row at height 10^6 is in: $\lambda = 0.85$, $L = 10.18$, $X = 26374$, $d = 3229$, $N(I) = 3812.3$, $C = 2636.0$, so $C/N(I) = 0.691$ against the law’s 0.685, certified proportion +0.371 against the asymptotic +0.370.

8 Report and verdict: partial, with a claimed theorem

8.1 The verdict: negative route empty, its dual claims one half

Time to fix the verdict. What I am prepared to defend is **PARTIAL** — with one claimed theorem that, if it survives refereeing, is the main output of this line. It rests on three points.

First, the global identity the brief conjectured is proven (Theorem 1), with the correction the brief did not anticipate: the negative index of the Weil Hermitian form on C_c^∞ equals the number of *distinct* zeros with $\text{Re } \rho > \frac{1}{2}$; an m -fold off-line zero is one negative square, multiplicity is invisible. At finite level, $\kappa(L)$ — the negative index on test functions supported in $[-L/2, L/2]$, i.e. of the form computable from primes $\leq X = e^L$ — is finite, nondecreasing, tends to κ , and $\text{RH} \iff \kappa(L) = 0$ for all L .

Second, as an *upper-bound* machine for off-line zeros the negative index is empty, and not because of Gibbs. (i) The correctly defined compressed index $\kappa(X, T)$ — the negative index of the Weil form restricted to admissible test functions of support length $\leq \log X$ with transforms concentrated below height T — is identically 0 in every computable range, $X = 10^2, \dots, 10^6$, $T \leq 2400$, and it is computed from primes alone; R4’s 33–153 negatives belong to a warped-sinc operator that is not a compression of the Weil form. (ii) The inequality $\kappa(V) \leq N_{\text{off}}^{\text{dist}}$ is the wrong direction, and the reverse needs visibility, which for all zeros below T requires $\dim V \gtrsim N(T)$ — the annihilation regime $L \sim \log(T/2\pi)$, with the measured threshold

$$L_{\text{ann}}^* = 1.12 \log(t_0/2\pi)^{0.89} \delta^{-0.07}$$

— and there the only prime-side bound, $\kappa \leq \dim V$, is the trivial cluster count. Below that level, Theorem 2 makes an isolated zero of depth δ at height t visible once $L \geq C\sqrt{\log t}/\delta$, but interference from unknown neighbours is uncontrollable, clusters count once, multiplicity never. So to the brief’s ratio question: $\kappa(X, T)/N(T) \rightarrow 0$ in the data (it *is* 0), while any provable prime-side bound is $\geq (L/2\pi) |\{\nu_X < 0\}| \sim 0.2\text{--}0.4 \dim V$. Empty.

Third, the dual works (claimed Theorem 4). Pullback inertia, $n_+(A^*QA) \leq n_+(Q)$, gives for *any* compression V (restriction of the Weil form to a finite-dimensional space of admissible test functions), with “in reach” meaning heights in the interval I' on which the transforms of V are non-negligible,

$$n_+^\theta(W|_V) \leq \#\{\text{distinct on-line zeros in reach}\} \\ + \#\{\text{distinct off-line pairs in reach}\},$$

while Riemann–von Mangoldt counts an off-line pair twice; hence

$$N_0^{\text{dist}}(I') \geq 2n_+^\theta(W|_V) - N(I').$$

On the prime side $n_+^\theta \geq (\text{tr } R)^2 / \text{tr}(R^2)$, two numbers computable from primes $\leq X$; and with V the space time-limited to $[-L/2, L/2]$ and band-limited to $[T, 2T]$, $L = \lambda \log(T/2\pi)$, $\lambda \leq 1$ (the Montgomery–Vaughan range),

$$\frac{(\text{tr } R)^2}{\text{tr } R^2} = \frac{\lambda N}{1 + \lambda^2/3} (1 + o(1))$$

unconditionally — it is Montgomery’s prime-side pair-correlation computation, and under RH it is the classical “ $\geq 3/4$ distinct”. Consequence:

$$\liminf \frac{N_0^{\text{dist}}([T, 2T])}{N([T, 2T])} \geq 2 \cdot \frac{3}{4} - 1 = \frac{1}{2},$$

against the unconditional record I know from memory and cannot check here, $5/12 = 0.4167$ (Pratt–Robles–Zaharescu–Zeindler 2020). In the task’s own terms the relevant ratio is not κ/N but $n_+/N \rightarrow 3/4$ at $X = T/2\pi$ — “ $c = 3/4$ certified-positive” — giving proportion $2c - 1 = 1/2$. And an explicit request goes with it: hostile referees, please attack the separate proof file; my ranked worry list is the one written in the red team, the top worry being “too short to be new”, not a located gap.

Behind the verdict stands the whole chain, which I restate compactly. The set-up and Theorem 1, with its Gaussian-times-linear-times-interpolation-polynomial test functions for the lower bound. The finite-level object done right: prime-side $n_- = 0$ in every case, minimum eigenvalue between -10^{-6} and $+6 \cdot 10^{-4}$ at the level of $\|G_P - G_Z\| \sim 10^{-5}$, with $d - \#\text{zeros}$ exact null directions whenever $d > \#\text{zeros}$ — e.g. at $X = 10^6$ on $[50, 1000]$, $1031 = 1672 - 641$ — reproduced by the prime side to 10^{-6} , while $\nu_X < 0$ on 21–42% of the same windows. So the level- X condition space is a *Hilbert* space in range: the Gibbs dips of width $1/\log X$ sit exactly at the resolution of admissible $|h|^2$ and cancel, and R4’s Kreĭn signature is a property of the warped kernel $\sin(\pi(N_X(t) - N_X(s)))/(\pi(t - s))$, whose diagonal is $\nu_X(t)$ itself.

Then visibility: Theorem 2, the cube-root average-spacing law, annihilation once the local dimension exceeds the local zero count, and the measured fits — and here I label the first as R4’s law:

$$L_{\text{ann}}^* \sim 1.12 \delta^{-0.07} \log(t_0/2\pi)^{0.89}, \\ L_{\text{rob}}^*(10^{-3}) \sim 1.6 \delta^{-0.34} \log(t_0/2\pi)^{0.6}.$$

The cluster counts: a double off-line zero gives $n_- = 1$ at every L ; two depths 0.2/0.35 at one ordinate give 2; two pairs at $\delta = 0.2$ are resolved iff $\varepsilon \gtrsim (0.15\text{--}0.25) \times 2\pi/L$; five pairs give 1, $\dots$, 5 as εL grows. And the interference obstruction: no configuration-free visibility theorem below the annihilation level exists.

Then the dual: the inequalities (P) and (N) and the Cauchy–Schwarz count, giving the finite certificate (Theorem 3) — the prime-computable lower bound $N_0^{\text{dist}}(I') \geq 2(\text{tr } R - \theta d)^2 / \text{tr } R^2 - N(I')$ — and the asymptotics

$$\begin{aligned}\text{tr } R &\sim \frac{LTl}{2\pi}, \\ \text{tr } R^2 &\sim 2\pi LT \left[\left(\frac{l}{2\pi} \right)^2 + \frac{L^2}{12\pi^2} \right],\end{aligned}$$

the second from the μ - μ term plus the Montgomery–Vaughan diagonal

$$\frac{TL}{\pi} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} \left(1 - \frac{\log n}{L} \right) = \frac{TL^3}{6\pi},$$

the off-diagonal being $O(L^2 X \log X)$, hence negligible, by Montgomery–Vaughan, and the cross and pole-density terms $o(\cdot)$ — whose RH-consistency is Montgomery’s $F(\alpha)$ -evaluation of $(\sum_{\gamma} L)^2 / \sum D_L(\gamma - \gamma')^2$, the classical $3/4$.

Theorem 4 (claimed). For each fixed $\lambda \in (0, 1]$,

$$N_0^{\text{dist}}([T, 2T]) \geq \left(\frac{2\lambda}{1 + \lambda^2/3} - 1 - o(1) \right) N([T, 2T]);$$

at $\lambda = 1$, $N_0^{\text{dist}}([T, 2T]) \geq (\frac{1}{2} - o(1))N([T, 2T])$.

The evidence set against it: the prime-only law at $T \sim 10^4$ and $5 \cdot 10^4$ already recorded; the $[600, 1200]$ zero-side cross-check (prime side = zero side to four digits; at $\lambda = 1$ certified $+0.44$ with finite-window margins, true $n_+ = 455 \geq C = 360$); the synthetic controls (the inequality never over-certifies); and the limits of the method — $n_+ \leq \dim V = \lambda N$ with $\lambda \leq 1 + o(1)$ forced by the diagonal evaluation, higher trace moments needing $k\lambda < 2$ — so that $1/2 = 2 \cdot \frac{3}{4} - 1$ is the ceiling with pair-correlation-range input, and every extra certified-positive fraction η of level- $(T/2\pi)$ Weil eigenvalues is worth $+2\eta$.

My answers to the brief’s numbered items, one by one.

- (1) The global identity: proven, with the correction “distinct”; finite-level monotone convergence; the literature — Yoshida 1992, Bombieri 2000, Burnol — is from memory and unverified offline, and the multiplicity caveat is essential.
- (2) $\kappa(X, T)$ concretely: defined and computed from primes, identically 0; nothing to fit, $\kappa/N \rightarrow 0$ trivially. The informative prime-side spectral statistic is instead the certified-positive count $(\text{tr } R)^2 / \text{tr } R^2$, a fraction $1/(1 + \lambda^2/3)$ of $\dim V$ ($0.81 \rightarrow 0.37$ in the index grid as X grows past T ; $\rightarrow 3/4$ of N at $X = T/2\pi$).
- (3) A-priori bounds on κ from sign changes of ν_X : of size $(L/2\pi) |\{\nu_X < c\}| \sim e^{4\sqrt{X}} \log X$ globally, $\sim 0.3 \dim V$ in windows — useless, and provably so, since visibility holds iff $\dim V \geq \#\text{zeros}$; the Turán/Nazarov count of sign changes the brief proposed bounds the wrong operator. But the positive-index dual gives an unconditional positive-proportion theorem (claimed $1/2$), i.e. the “new proof architecture for Selberg’s theorem” the brief asked for, with a better constant than Levinson–Conrey-type methods, from the explicit formula plus the Montgomery–Vaughan mean value theorem plus linear algebra.
- (4) The red team of the original danger: confirmed in the strong form — not “ $\kappa \sim N(T)$ by Gibbs” but “ $\kappa = 0$ and unboundable below $\dim V$ ”; the ratio question is moot. For the red team of Theorem 4: checks (a)–(e) passed, residual risks ranked; the constant $1/2$ cannot move, only the whole claim can fall.

The hedges once more, summarised. Everything “positive” in the numerics is RH being true below 2515 (resp. $3 \cdot 10^{12}$), stated as such, and the finite certificates inside that range certify nothing new. Theorem 2’s constants are for a specific bump, and the measured laws are for a finite Gabor family (hence upper bounds on true thresholds). For Theorem 4 I have not written every end-effect estimate at $\lambda = 1$, where $|\nu_X| \sim \sqrt{X}$ pointwise, though the finite computations, which contain all such effects exactly, match the law to 2%; the Montgomery–Vaughan constant at $\lambda = 1$ is poor ($\text{OFF}/\text{DIAG} \lesssim 36/L$, crude), so a clean statement takes $\lambda < 1$ fixed, then $\lambda \rightarrow 1$. The one thing I cannot exclude is that the result is already known, or that a step I consider trivial hides an error. The three-line heart is: [the pair block has one positive square] + [Riemann–von Mangoldt counts the pair twice] + $[n_+(A^*QA) \leq n_+(Q)]$.

The pending $T \approx 10^6$ row at $\lambda = 1$ has also come in: $L = 11.98$, $X = 158996$, $d = 3799$, $N(I) = 3812.3$, $N(I') = 3841.3$, $C = 2885.7$, so $C/N(I) = 0.757$ against the law’s 0.750, certified proportion +0.502 against the asymptotic +0.500.

So the prime-only evidence at $\lambda = 1$ stands at $C/N(I) = 0.766$ ($T = 10^4$), 0.762 ($T = 5 \cdot 10^4$), 0.757 ($T = 10^6$) against 0.750, certified proportions +0.50/ +0.51/ +0.50; and at $T \sim 10^6$ for $\lambda = 0.85/1.0$: $C/N(I) = 0.691/0.757$ (law 0.685/0.750), certified +0.371/ +0.502. These belong in the verdict.

The certificate run with zero-side cross-check has also finished its Kaiser($\beta = 12$) rows on [600, 1200]. The statistic C agrees between prime and zero side to every printed digit,

$$C = 193.9, 248.5, 295.7, 326.4, 352.0, 390.3, 418.5 \quad (\lambda = 0.6, 0.8, 1.0, 1.15, 1.3, 1.6, 2.0),$$

with the tail threshold θ at most 10^{-5} of scale and the prime-versus-zero eigenvalue discrepancy at most $6 \cdot 10^{-5}$, the Montgomery–Vaughan diagonal exact (5.7/5.7, 14.2/14.2, 28.8/28.8, 45.0/45.0), and certified proportions

$$-0.197, +0.033, +0.232, +0.363, +0.476, +0.640, +0.758,$$

lower than the Tukey family’s at each λ , the taper cutting the effective dimension ($d = 203, \dots, 673$ against $d_{\text{full}} = 281, \dots, 937$). (Editorial remark: the rows with $\lambda > 1$ lie outside the $\lambda \leq 1$ range of the asymptotic evaluation; nothing in the record comments on them.) The first rows on the larger window [1200, 2400], Tukey taper, $N(I) = 1077$: at $\lambda = 0.6$ ($X \approx 29$, $d = 624$), $C = 570.8/571.6$ (prime/zero), certified +0.018 against the asymptotic +0.071; at $\lambda = 0.8$ ($X \approx 88$, $d = 832$), $C = 710.6/711.9$, certified +0.279 against +0.319.

8.2 Final verdict and hand-off

For anyone resuming this line I leave a pointer recording the outcome in my own framing: the negative-index route is empty for upper bounds ($\kappa(X, T) = 0$ from primes in all ranges; visibility needs $\dim V \geq \#\text{zeros}$); the dual gives the claimed $N_0^{\text{dist}} \geq (\frac{1}{2} - o(1))N$, which would beat the 5/12 record if correct and carries “too short to be new” as its top risk. The advice: start from the referee proof and the ranked risk list, and do not re-derive the $\kappa(X, T) = 0$ / warped-sinc diagnosis.

My closing message to the orchestrator, verdict first:

Verdict: PARTIAL — negative-index route EMPTY as an upper-bound machine;
its DUAL (positive-index inertia) yields a claimed unconditional theorem: $\geq 1/2$
of the zeros are on the line (as distinct zeros). Referees requested.

followed by the four findings.

- (1) Theorem 1, proven by my own account: the negative index of the Weil Hermitian form on C_c^∞ equals the number of *distinct* zeros with $\operatorname{Re} \rho > \frac{1}{2}$, an m -fold off-line zero being one negative square, multiplicity intrinsically invisible; $\kappa(L)$ finite, monotone, $\rightarrow \kappa$; RH $\iff \kappa(L) = 0$ for all L — the brief’s item (1), sharpened.
- (2) $\kappa(X, T)$ done honestly is identically 0 — a prime-side computation for $X = 10^2, \dots, 10^6$, $T \leq 2400$, with $d - \#\text{zeros}$ exact nulls reproduced from primes to 10^{-6} ; R4’s 33–153 negatives belong to a warped-sinc operator, not to a compression of the Weil form; nothing to fit, $\kappa/N \rightarrow 0$.
- (3) Upper bounds via κ are structurally empty: $\kappa(V) \leq N_{\text{off}}^{\text{dist}}$ is the wrong direction; visibility of all zeros below T needs $\dim V \gtrsim N(T)$ (measured $L_{\text{ann}}^* \approx 1.12 \log(t_0/2\pi)^{0.89} \delta^{-0.07}$), where $\kappa \leq \dim V$ is trivial; and Theorem 2 (an isolated zero visible at $L \geq C\sqrt{\log t/\delta}$) cannot be made configuration-free.
- (4) Theorem 4 (claimed): $n_+(A^*QA) \leq n_+(Q)$, together with the off-line pair $\{\rho, 1 - \bar{\rho}\}$ as a signature-(1, 1) block counted twice by Riemann–von Mangoldt, gives $N_0^{\text{dist}}(I') \geq 2n_+^\theta(W|_V) - N(I')$; Cauchy–Schwarz gives $n_+ \geq (\operatorname{tr} R)^2 / \operatorname{tr} R^2$, prime-computable; the Montgomery–Vaughan-range evaluation gives $\lambda N / (1 + \lambda^2/3) \rightarrow 3N/4$ at $X = T/2\pi$ (under RH exactly the classical “3/4 distinct”); hence $\liminf N_0^{\text{dist}}/N \geq \frac{1}{2}$ unconditionally, against the record 5/12 that I quote from memory and never checked in this run.

Numerics reported with it: prime-only $C/N = 0.766/0.762/0.757$ at $T = 10^4/5 \cdot 10^4/10^6$ against the law’s 0.750; certified +0.50; inertia checked on synthetic configurations with 0%, 25%, 100% of zeros on the line. Top risk: “too short to be new”.

That is where I stop: the brief’s proposed route answered in the negative with the mechanism identified, and its dual left as a claimed theorem that I have tested numerically in every way available to me here but have neither located in, nor excluded from, the literature.

Part II

The two-thirds run

The rank–trace lemma and two thirds

Summary

Earlier in this campaign a run (E2) compressed Weil’s explicit-formula Hermitian form $W(f, g) = \sum_{\rho} m_{\rho} h_f(\gamma_{\rho}) \overline{h_g(\bar{\gamma}_{\rho})}$ to the Gabor test space $f_k(u) = \varphi(u) e^{-i\tau_k u}$, $\tau_k = T + 2\pi k/L$, $L = \lambda \log(T/2\pi)$, obtaining a $d \times d$ Gram matrix G whose zero side is a sum of rank-one positive pieces from on-line zeros and signature- $(1, 1)$ blocks from hypothetical off-line pairs $\{\rho, 1 - \bar{\rho}\}$, and whose traces $\text{tr } G$, $\text{tr } G^2$ are computable from primes (PNT-strength sums and Montgomery–Vaughan, no RH). Counting positive eigenvalues by Cauchy–Schwarz and comparing with Riemann–von Mangoldt, it claimed $n_{\text{on}}^{\text{dist}} \geq 2n_+ - N$ with $n_+ \geq \frac{3}{4}N$ at $\lambda \rightarrow 1^-$, hence at least one half of the zeros in $[T, 2T]$ on the line as distinct points (“Theorem 4”); two referees checked the block structure and supplied coefficient coordinates and an exact Poisson identity $\sum_k \hat{\varphi}(z - \tau_k) \hat{\varphi}(z' - \tau_k) = L \Phi(z - z')$. The orchestrator’s brief asked this run to verify its own quick algebra of a single pair block at depth δ (claimed $v^T v = L^2 a$ real and δ -independent, eigenvalues $mL^2 a(1 \pm A(\delta))$ with $A(\delta) \geq 1$), to decide whether the negative spectrum alone can raise $1/2$ (forecast: no, since tight pairs $\delta \rightarrow 0$ have vanishing negative eigenvalue), to analyse a spectral “integrality” analogue of Montgomery’s RH argument (which gives $2/3$ simple, $5/6$ distinct by the brief’s elementary arithmetic), and to pursue higher prime-side moments $\text{tr } R^3$, $\text{tr } R^4$ as the likely unconditional route to the target $2/3$.

The block facts were confirmed numerically to 10^{-14} , including the Poisson identity at complex arguments; cross terms between blocks were found sign-indefinite, so termwise diagonal dominance of $\text{tr } R^2$ is false. Extremal experiments over vector configurations showed that with normalised pair vectors ($v^T v = 1$) the minimum of $\|M\|_F^2$ is exactly $n_{\text{on}} + 4n_p$, strictly above the Cauchy–Schwarz floor, while every per-pair quadratic pricing with free mass or free phase fails. This led the run to a claimed and proved inequality, linear in the masses: if $M = M_{\text{on}} + M_p$ with $M_{\text{on}} \succeq 0$ of rank $\leq r$ and M_p having at most b positive eigenvalues, then $r \geq 2 \text{tr } M_{\text{on}} + 4 \text{tr } M_p - 4b - \|M\|_F^2$, by von Neumann’s trace inequality (recalled from memory) and Cauchy–Schwarz. Applied to the windowed zero side in units $\tilde{R} = G/(L^2 a)$, using only $|u_i|^2 \leq 1$, inertia of blocks and the refereed tail bound, it yields the certificate $n_{\text{on}}^{\text{dist}} \geq 4 \text{tr } A - 2N - \|A\|_F^2 \geq (2 - 1/\lambda - \lambda/3 - o(1))N \rightarrow \frac{2}{3}N$, containing no pair data. Verdict, by the run’s account: unconditionally at least $2/3$ of zeros are distinct and on the line, sharp for this information (extremal: $2N/3$ lattice simples plus $N/6$ tight pairs); on true zeros at $T = 1000$ the certificate gives $0.61\text{--}0.63N$ against Theorem 4’s $0.38\text{--}0.40N$.

The negative spectrum alone was confirmed useless against tight pairs, as forecast; the brief’s forecast that higher moments are the real lever was contradicted: $\text{tr } R^3$ is admissible for $\lambda < 1$ but an odd moment cannot improve a count on $[0, \infty)$ without a Lindelöf-strength bound on λ_{max} , and $\text{tr } R^4$ needs $\lambda < 1/2$ where $d < N/2$. The $(m-1)(m-2) \geq 0$ integrality level is destroyed by interactions; simple on-line zeros come out only at $1/3$; the coordinator’s requested

per-pair estimate $s_j \geq 1 - o(1)$ was found unnecessary. The RH comparison 5/6 distinct and the Rudnick–Sarnak unconditionality were stated from memory without citation check. The run’s top caveat, inherited from the referees: everything rests on the prime-side upper bound for $\|\tilde{R}\|_F^2$.

1 The problem as posed: off-line pair blocks and the negative spectrum of the compressed Weil form

The setting is a claimed unconditional theorem produced earlier in this campaign, called Theorem 4: by compressing Weil’s explicit-formula Hermitian form to a finite-dimensional Gabor test space attached to the window of ordinates $[T, 2T]$, counting the positive eigenvalues of the resulting matrix by Cauchy–Schwarz from two prime-side traces, and comparing that count with the Riemann–von Mangoldt zero count, it asserts that at least one half of the nontrivial zeros in the window lie on the critical line as distinct points. Under the Riemann Hypothesis, Montgomery’s pair-correlation method gives two thirds of the zeros simple; the orchestrator wants the unconditional one half pushed to that two thirds. Two runs are launched in parallel on this: a sibling optimises the test-space weight, and I am told to exploit instead the two pieces of structure that Theorem 4 throws away — the 2×2 blocks contributed by hypothetical off-line zero pairs, and the negative eigenvalues of the compressed form. The orchestrator’s brief asks me to verify its quickly derived algebra of a single pair block, to decide whether the negative spectrum by itself can improve the constant (it forecasts not), to analyse an “integrality” mechanism modelled on Montgomery’s argument, and to pursue higher prime-side moments as the unconditional route.

The brief, from the orchestrator (the campaign’s main thread, which wrote every brief and relayed all reports; later messages in this run call it “the coordinator”), set out here in full and rewritten for readability:

Role and reading. Research mathematician, maximal effort, a precise target. Read first: (1) the write-up of Theorem 4 by the earlier run that produced it (labelled E2 in the campaign: briefed to bound the *negative* index of inertia of a prime-computable compression of the Weil form, it found that index identically zero and instead lower-bounded the *positive* index from prime-side traces by Cauchy–Schwarz, double-counting off-line pairs against the Riemann–von Mangoldt count, and so claimed unconditionally that at least one half of the zeros are on the line as distinct zeros); (2) the verdict of the blind hostile referee who checked Theorem 4’s counting identity (referee C: it verified the signature blocks of the zero side and the factor 2) — read it for the block structure; (3) the verdict of the referee who closed Theorem 4’s technical write-up gaps (referee D: it made rigorous the repair “never orthonormalise, work in coefficient coordinates”, proved an exact Poisson identity for the Gabor system, and gave explicit tail exponents) — read it for coefficient coordinates and the Poisson identity; (4) the blind from-scratch re-derivation of the prime-side trace asymptotics by a separate run (E2-rederive). A sibling run (E2-kernel) is optimising the *kernel* — the weight placed on the test space — in order to raise the constant 3/4; do not duplicate that. Your lever is the *structure of the off-line pair blocks* and the *negative spectrum*, both of which Theorem 4 discards.

Context. Theorem 4 is the conjunction of two inequalities. The first is

$$n_{\text{on}}^{\text{dist}} \geq 2n_+ - N,$$

the number of distinct zeros on the line in the window being at least twice the positive index of inertia of the compressed form minus the zero count. The second is the Cauchy–Schwarz count

$$n_+ \geq \frac{(\operatorname{tr} R)^2}{\operatorname{tr} R^2} \longrightarrow \frac{3}{4}N$$

as the bandwidth parameter tends to 1, the two traces being evaluated from the prime side. Together: $2 \cdot \frac{3}{4} - 1 = \frac{1}{2}$. The loss from $3/4$ to $1/2$ has one cause: an off-line pair $\{\rho, 1 - \bar{\rho}\}$ of multiplicity m is counted $2m$ times in N , but contributes only *one* positive square to the form.

Facts to verify first. I derived the following quickly; check them rigorously via the Poisson identity, which schematically reads

$$\sum_k \hat{\varphi}(x - \tau_k - iy) \hat{\varphi}(x - \tau_k - iy') = L \int \varphi^2(u) e^{(y+y')u} (\dots) du + \text{aliasing terms}$$

(the oscillatory factor in the integrand is left unwritten). Take the Gabor system with a real even taper φ , normalised so that $\int \varphi^2 = L(1 + O(\eta))$, and grid points τ_k . An off-line pair at depth δ — that is, $\rho = \frac{1}{2} + \delta + it$ together with $1 - \bar{\rho}$, whose spectral parameters are $\gamma = t - i\delta$ and $\bar{\gamma}$ — has two evaluation functionals, with coordinates

$$v_k = \hat{\varphi}(t - \tau_k - i\delta), \quad w_k = \hat{\varphi}(t - \tau_k + i\delta) = \overline{v_k}$$

(careful here: $\hat{\varphi}(\bar{z}) = \overline{\hat{\varphi}(z)}$ because φ is real and even). I claim

$$\|v\|^2 = \|w\|^2 = L^2 A(\delta), \quad A(\delta) := \frac{1}{\int \varphi^2} \int \varphi^2(u) e^{2\delta u} du \approx \frac{\sinh(\delta L)}{\delta L} \geq 1,$$

and, writing $z_k = t - \tau_k - i\delta$,

$$\langle v, w \rangle := \sum_k v_k \overline{w_k} = \sum_k \hat{\varphi}(z_k)^2 = L^2,$$

which is *real, independent of δ* , and equal to the value for a zero on the line. Hence, on the span of v and w , the block contributed by the pair — the rank-two Hermitian form $m(|v\rangle\langle w| + |w\rangle\langle v|)$ — has nonzero eigenvalues

$$m(\operatorname{Re}\langle w, v \rangle \pm \|v\| \|w\|) = mL^2(1 \pm A),$$

that is, one positive eigenvalue $p = mL^2(A + 1)$ and one negative eigenvalue $-q$ with $q = mL^2(A - 1)$. These are in the units of the raw Gram matrix G ; divide by L for the normalised matrix R . *Check the signs and the normalisation numerically with a synthetic single pair.*

Consequences of these facts. (i) The pair contributes $p - q = 2mL^2$ to the trace, which is exactly what $2m$ simple on-line zeros would contribute: consistent with the trace $\operatorname{tr} R = LN$ carrying no information about whether zeros are on the line. (ii) An *isolated* pair contributes $p^2 + q^2 = 2m^2L^4(A^2 + 1) \geq 4m^2L^4$ to $\operatorname{tr} R^2$, against m^2L^4 for an on-line zero of multiplicity m . (iii) As $\delta \rightarrow 0$ one has $A \rightarrow 1$, so $q \rightarrow 0$ and $p \rightarrow 2mL^2$: the pair becomes spectrally indistinguishable from an on-line zero of multiplicity $2m$. This is the true enemy, and it cannot be beaten by any method that is also unable to count multiplicity. So the honest target is: distinct zeros on the line $\geq cN$ with c matching what RH plus Montgomery’s

pair-correlation argument gives for *simple* zeros, namely $c = 2/3$ — because under RH a “double on-line zero” and unconditionally a “tight off-line pair” are one and the same obstruction.

Program. Theorem 4 uses, of the compressed form R , only the two moments $(\sum_i \lambda_i, \sum_i \lambda_i^2)$ of its spectrum and only the positive index n_+ . Use more.

(a) The negative spectrum. Split $R = R_+ - R_-$ into positive and negative parts, with positive eigenvalues p_i and negative eigenvalues $-q_j$. The prime side gives, exactly,

$$\sum_i p_i - \sum_j q_j = \operatorname{tr} R = LN(1 + o(1)), \quad \sum_i p_i^2 + \sum_j q_j^2 = \operatorname{tr} R^2 = \frac{4}{3}L^2N(\cdots).$$

Inertia: $n_+ \leq n_{\text{on}}^{\text{dist}} + n_{\text{pair}}^{\text{dist}}$ and $n_- \leq n_{\text{pair}}^{\text{dist}}$, since each pair block gives at most one negative square and on-line zeros give none. (Check: is $n_-(A) = n_{\text{pair}}^{\text{dist}}$ *exactly* for the windowed form? Yes if the functionals are linearly independent; in any case $n_-^\theta(R) \leq n_{\text{pair}}^{\text{dist}}$.) That is an *upper* bound on n_- , whereas we want *lower* bounds on n_{on} . Combine with the count $N \geq n_{\text{on}} + 2n_{\text{pair}}$ (every multiplicity is at least 1): we want to show either that n_{pair} is small or that n_{on} is large. Cauchy–Schwarz applied to the positive part alone gives

$$n_+ \geq \frac{(\sum p)^2}{\sum p^2} = \frac{(LN + \sum q)^2}{\frac{4}{3}L^2N - \sum q^2},$$

which is *increasing* in the negative mass — so negative mass helps n_+ . Moreover pairs with $\delta \gg 1/L$ have $q \approx p \approx mL^2A$ large, so they consume the $\operatorname{tr} R^2$ budget quickly, so there are few of them. This would be a density estimate of the shape $\#\{\text{pairs with } A(\delta) \geq A_0\} \leq \frac{4}{3}N/(2A_0^2)$, roughly — can it be made rigorous? It needs “ $\operatorname{tr} R^2 \geq \sum_{\text{pairs}} (p^2 + q^2)$ ”, which is a *diagonal-dominance* statement, false in general for indefinite summands, because the cross terms $\operatorname{tr}(R_j R_k)$ between different blocks can be negative. Perhaps something survives for the negative part alone, $\sum_j q_j^2 = \operatorname{tr} R_-^2$ with R_- dominated by something — unclear. Think about what is rigorously available for $R = P + \sum_j B_j$ with $P \succeq 0$ the on-line part and B_j the pair blocks: Cauchy interlacing gives $n_-(R) \leq \sum_j \operatorname{rank}_-(B_j) = n_{\text{pair}}$; there are $\lambda_{\min}$ -type bounds; Weyl gives $\lambda_i(P + B) \leq \lambda_i(P) + \lambda_{\max}(B)$; Ky Fan says the sum of the k smallest eigenvalues of R is at most $\sum \langle x, Rx \rangle$ over any k orthonormal vectors x — so choose the x ’s in the span of the w -directions of far-off pairs to force large negative Rayleigh quotients, whence a lower bound on $\operatorname{tr} R_-$; combined with $\sum p - \sum q = LN$, $\sum p^2 + \sum q^2$ fixed, $n_+ \leq n_{\text{on}} + n_{\text{pair}}$ and $n_- \leq n_{\text{pair}}$, *optimise the worst case*. Concretely, set up the finite-dimensional extremal problem: nonnegative reals $p_1, \dots, p_a$ and $q_1, \dots, q_b$ with $a \leq n_{\text{on}} + n_{\text{pair}}$, $b \leq n_{\text{pair}}$, $\sum p - \sum q = LN$, $\sum p^2 + \sum q^2 \leq \frac{4}{3}L^2N$, together with any additional *provable* constraints linking the p and q of the same pair. For a spectrally isolated pair these would be $p_j - q_j = 2m_jL^2$ and $p_jq_j = m_j^2L^4(A^2 - 1) \geq 0$, but interaction with other zeros perturbs them. Can one prove a robust version — for each pair, nearly orthogonal directions $x_j^\pm$ with $\langle x_j^+, Rx_j^+ \rangle$ bounded below and $\langle x_j^-, Rx_j^- \rangle \leq -(\text{something} \geq 0)$? The Rayleigh quotient in the normalised w -direction, $\langle \hat{w}, R\hat{w} \rangle$, equals the contribution of pair j itself (something like $2mL^2 \cdot (1/A) \cdot (\cdots)$; compute it) plus the contributions of all the other zeros, which are positive semidefinite from on-line zeros (hence ≥ 0) and indefinite from other pairs — so single-direction quotients do not obviously produce negativity. Then perhaps the cleanest extra information really is n_- and $\operatorname{tr} R_-$ via Ky Fan with a cleverly chosen subspace. Solve the resulting

LP/QP: minimise n_{on} subject to the constraints and see whether the minimum exceeds $N/2$. Even without pair-linking constraints, does adding “ $n_- \leq n_{\text{pair}}$ ” and using $(\sum p)^2 / \sum p^2$ with $\sum q \geq 0$ free change anything? *Probably not by itself*: the worst case is $\sum q = 0$, i.e. every pair has $\delta \rightarrow 0$. Indeed — the $\delta \rightarrow 0$ pairs are the enemy and they have $q \rightarrow 0$; so the negative spectrum helps only against *far* pairs, which are already handled. Confirm this reasoning. If it is confirmed, lever (a) alone cannot beat $1/2$, and the write-up should say so crisply: “the extremal configuration for Theorem 4 is $N/4$ tight off-line pairs (depth $\delta = o(1/\log T)$) plus $N/2$ simple on-line zeros, which is spectrally identical to $N/4$ on-line double zeros plus $N/2$ simple zeros — Montgomery’s extremal configuration for $2/3$ ”. (On reflection that configuration has $N_{\text{simple}} = N/2$, not $2N/3$; let me check Montgomery’s extremal configuration for $\sum m^2 \leq \frac{4}{3}N$: one third of the zeros in doubles, i.e. $N/6$ double points, plus $2N/3$ simple zeros, giving $\sum m^2 = \frac{N}{6} \cdot 4 + \frac{2N}{3} = \frac{4}{3}N$ and $N_{\text{dist}} = \frac{N}{6} + \frac{2N}{3} = \frac{5}{6}N$ — whereas Cauchy–Schwarz gave only $3/4$. The discrepancy is because $(\sum m)^2 / \sum m^2$ is not tight for *integer* m . Montgomery’s bound uses integrality: $m^2 \geq 2m - 1$ summed over distinct zeros gives $\sum m^2 \geq 2N - N_{\text{dist}}$, so $N_{\text{dist}} \geq 2N - \sum m^2 \geq \frac{2}{3}N$; and for simple zeros, $\sum m^2 \geq N_{\text{simple}} + \sum_{m \geq 2} m^2 \geq N_{\text{simple}} + 2 \sum_{m \geq 2} m = N_{\text{simple}} + 2(N - N_{\text{simple}}) = 2N - N_{\text{simple}}$, so $N_{\text{simple}} \geq 2N - \sum m^2 \geq \frac{2}{3}N$. And $N_{\text{dist}} \geq N_{\text{simple}} \geq \frac{2}{3}N$, or better $\geq \frac{3}{4}N$ by Cauchy–Schwarz.)

(b) The integrality lever — likely the real one. Theorem 4 used the Cauchy–Schwarz count $n_+ \geq (\sum \lambda)^2 / \sum \lambda^2$, the spectral analogue of $N_{\text{dist}} \geq N^2 / \sum m^2$. Montgomery’s simple-zero bound instead uses that multiplicities are integers ≥ 1 : $\sum m^2 \geq 2N - N_{\text{simple}}$. The spectral analogue: the positive eigenvalues of R are not arbitrary — an isolated on-line zero of multiplicity m contributes an eigenvalue mL^2 , an isolated tight pair $2mL^2$, and interactions smear these values. If one could prove, in a robust sense, that “every positive eigen-direction carries integer mass ≥ 1 in units of L^2 ” — say an inequality like $\sum p_i^2 \geq L^2(2 \sum p_i - L^2 n_+)$, the analogue of $\sum m^2 \geq 2 \sum m - \#\{\text{points}\}$ — then $n_+ L^4 \geq 2L^2 \sum p - \sum p^2$ and, with $\sum p \geq L^2 N$ and $\sum p^2 \leq \frac{4}{3}L^4 N$ in G -units, $n_+ \geq 2N - \frac{4}{3}N = \frac{2}{3}N$: the same $2/3$ as Montgomery’s simple-zero constant. Fed into “on-line $\geq 2n_+ - N$ ” that would give only $N/3$, worse than the Cauchy–Schwarz route; but of course one takes the maximum of the two lower bounds for n_+ . The point is this: the final bound $2n_+ - N$ with $n_+ = \frac{3}{4}N$ gives $1/2$; to get $2/3$ on the line one needs $n_+ \geq \frac{5}{6}N$. Under RH, is $n_+ = N_{\text{dist}} \geq \frac{5}{6}N$ provable from $\sum m^2 \leq \frac{4}{3}N$? By integrality, yes: $N_{\text{dist}} = N - \sum (m - 1)$, and to minimise the number of points given $\sum m = N$ and $\sum m^2 \leq \frac{4}{3}N$ one uses doubles as much as possible; k doubles plus $N - 2k$ simples have $\sum m^2 = 4k + N - 2k = N + 2k \leq \frac{4}{3}N$, so $k \leq N/6$ and $N_{\text{dist}} \geq N - \frac{N}{6} = \frac{5}{6}N$. So under RH integrality gives $\frac{5}{6}$ distinct, better than the Cauchy–Schwarz $\frac{3}{4}$. (Is “ $5/6$ distinct under RH” known? Surely — Montgomery’s paper or folklore; *check the literature*.) And then, Theorem-4-style, distinct on the line $\geq 2 \cdot \frac{5}{6}N - N = \frac{2}{3}N$: *exactly the target*. So the *entire* problem of reaching $2/3$ is to prove the spectral integrality inequality unconditionally — something of the shape “ $n_+^\theta(R) \geq 2 \text{tr } R/u - \text{tr } R^2/u^2$ ” in appropriate units, i.e. that the positive eigenvalues of the windowed Weil form come in integer quanta ≥ 1 , robustly. Let me be precise about what is needed. With quantum $u = L^2$ (in G -units), the inequality $\sum_{\lambda_i > \theta} \lambda_i(2u - \lambda_i) \leq u^2 n_+^\theta$ would hold if every positive eigenvalue were $\geq u$ — but in fact $\lambda(2u - \lambda) \leq u^2$ holds for *every* real λ , being $(\lambda - u)^2 \geq 0$. So that inequality is always true, and gives $n_+^\theta \geq (2u \sum' \lambda - \sum' \lambda^2) / u^2$, the sums over $\lambda > \theta$; with $\sum' \lambda \geq \text{tr } R_+ - \theta d$, $\text{tr } G = L^2 N$ and $u = L^2$, this is

$n_+ \geq 2N - \text{tr } G^2/L^4 = 2N - \frac{4}{3}N = \frac{2}{3}N$ — only $2/3$, below the Cauchy–Schwarz $3/4$, precisely because no integrality has been used: $(\lambda - u)^2 \geq 0$ is free. Genuine integrality would say $\lambda \in \{u, 2u, 3u, \dots\}$, hence $(\lambda - u)(\lambda - 2u) \geq 0$, hence $\lambda^2 \geq 3u\lambda - 2u^2$, hence $\sum \lambda^2 \geq 3u \sum \lambda - 2u^2 n_+$, hence

$$n_+ \geq \frac{3u \sum \lambda - \sum \lambda^2}{2u^2} = \frac{3N - \frac{4}{3}N}{2} = \frac{5}{6}N.$$

That is the inequality wanted; it requires “no eigenvalue of the zero-side form strictly between u and $2u$, or below u ” — and this is robustly false because of interactions: the eigenvalues of $\sum_\rho m_\rho u_\rho u_\rho^*$ are the multiplicities only if the vectors u_ρ are orthogonal, whereas $\langle u_\rho, u_{\rho'} \rangle = D_L(\gamma - \gamma') \neq 0$ for $|\gamma - \gamma'| \lesssim 1/L$, so zeros closer than the mean spacing times l/L interact. Under RH Montgomery needs no eigenvalues — he reads $\sum m^2$ directly off the diagonal of the pair correlation. Unconditionally we replaced $\sum m^2$ by $\sum \lambda^2$ and lost integrality. Can a robust integrality be recovered? Options: (i) work at slightly smaller bandwidth and use that $D_L(\gamma - \gamma')$ is small unless $|\gamma - \gamma'| < 2\pi/L \sim \text{mean spacing}/\lambda$ — but interactions affect a positive fraction of zeros; no. (ii) Use the determinant, or higher traces: $\text{tr } R^3$, $\text{tr } R^4$ from the prime side? $\text{tr } R^3$ involves triple correlations of primes, $\sum \Lambda(n_1)\Lambda(n_2)\Lambda(n_3)$ with conditions of the type $n_1 n_2 = n_3$; the diagonal is computable, the off-diagonal needs GUE triple correlation, i.e. Hardy–Littlewood — not unconditional. But perhaps with support restrictions: are Rudnick–Sarnak’s n -level correlations with restricted Fourier support unconditional? Do they assume RH? They prove GUE statistics for test functions with $\sum_j |\xi_j| < 2$ (or < 1 ?); I believe Rudnick–Sarnak 1996 is unconditional in the smoothed form for restricted support — *check*. If $\text{tr } R^3$ is prime-computable for λ small enough (a condition like $3\lambda/2 < 1$?), then with three moments ($\sum \lambda, \sum \lambda^2, \sum \lambda^3$) one gets sharper counting inequalities — something like $n_+ \geq (\sum \lambda^2)^2 / \sum \lambda^4$, or the optimal three-moment Markov–Krein bound — so compute what the RH-predicted value of $\text{tr } R^3$ gives for the on-line bound; this could beat $1/2$ even without integrality. (iii) Accept Cauchy–Schwarz and attack the two-for-one cost through lever (a) for non-tight pairs together with a separate argument that tight pairs ($\delta < c/L$) are rare — no; there is no unconditional tool for that, it is RH-hard. So (ii), higher prime-side moments with restricted support, is the promising unconditional lever.

Do. Verify all the block and eigenvalue facts numerically. Settle whether lever (a) is *provably* useless against tight pairs (I expect yes). Then pursue (b)(ii): what is the largest k such that $\text{tr } R^k$ is unconditionally computable from primes at bandwidth λ — a condition like $k\lambda/2 < 1$, or $(k-1)\lambda < 2$? Compute the RH/GUE-predicted main terms of $\text{tr } R^3$, and of $\text{tr } R^4$ if admissible, for the sinc system, as functions of λ . Then solve the moment problem: given $(m_1, m_2, m_3)(\lambda) = (\sum \lambda_i, \sum \lambda_i^2, \sum \lambda_i^3)$ over the positive spectrum — careful, odd moments mix in the signs of negative eigenvalues; handle that via $n_- \leq n_{\text{pair}}$ and worst-casing, or use even moments only — what is the best lower bound on n_+ , hence on $2n_+ - N$, optimised over admissible λ ? Does it exceed $1/2$? Does it reach $2/3$? Report the numbers together with the exact admissibility condition and its literature status (Rudnick–Sarnak, Hughes–Rudnick restricted-support n -level results: which of them are unconditional).

Deliverable. A report, with a final summary covering: are the block facts confirmed?; the verdict on lever (a); the integrality analysis ($5/6$ under RH via integers — known?); the higher-moment route: admissible $k(\lambda)$, predicted moments, the resulting unconditional on-line constant; is $2/3$ reached, and if not, exactly what input would reach it.

1.1 The brief's objects, notation and sources, spelled out

For a reader meeting this corner cold, here is what every symbol and coined term in the brief denotes, in the sense the campaign uses them and consistently with the definitions fixed by the documents I read below; and, where the brief indicates it, who supplied what.

The form and its two sides. The zeros are the nontrivial zeros $\rho = \beta + it$ of ζ , each with multiplicity m_ρ ; the *spectral parameter* of ρ is $\gamma_\rho := (\rho - \frac{1}{2})/i = t - i(\beta - \frac{1}{2})$, real exactly when ρ is on the critical line. For a test function f supported in $[-L/2, L/2]$ put $h_f(z) = \hat{f}(z) = \int f(u)e^{izu} du$, an entire function. The *Weil form* is the Hermitian form $W(f, g) = \sum_\rho m_\rho h_f(\gamma_\rho) \overline{h_g(\bar{\gamma}_\rho)}$; this expression is its *zero side*. Weil's explicit formula rewrites $W(f, g)$ as an integral of $h_f \bar{h}_g$ against an explicit density built from the digamma function, the pole of ζ , and the prime sum $\sum_{n \leq X} \Lambda(n) n^{-1/2} \cos(\tau \log n)$ with $X = e^L$; that expression is the *prime side*, and “prime-computable” means evaluable from it without knowledge of zeros.

Window, bandwidth, counts. The *window* is the ordinate range $I = [T, 2T]$, with $l := \log(T/2\pi)$; $N = N(I) \sim Tl/2\pi$ is the number of zeros with ordinate in I , counted with multiplicity (Riemann–von Mangoldt). The *bandwidth* is $\lambda \in (0, 1)$ and $L := \lambda l$ is the length of the support interval of the test functions, so $X = e^L \leq (T/2\pi)^\lambda$; “ $\lambda \rightarrow 1^-$ ” is the limit in which the constants $3/4$ and $1/2$ appear. N_{dist} , N_{simple} are the numbers of distinct, respectively simple, zeros in the window; $n_{\text{on}}^{\text{dist}}$ (also written N_0^{dist}) is the number of distinct zeros on the line $\beta = \frac{1}{2}$; an *off-line pair* is $\{\rho, 1 - \bar{\rho}\}$ with $\beta \neq \frac{1}{2}$ (the two members share the ordinate t and the multiplicity m , by the functional equation), and $n_{\text{pair}}^{\text{dist}}$ counts distinct such pairs. The *depth* of a pair is $\delta = \beta - \frac{1}{2}$; a *tight* pair has $\delta = o(1/L)$, a *far* pair $\delta \gg 1/L$. Since each distinct on-line point has multiplicity ≥ 1 and each pair contributes ≥ 2 to the count, $N \geq n_{\text{on}}^{\text{dist}} + 2n_{\text{pair}}^{\text{dist}}$; this is the “Riemann–von Mangoldt bookkeeping”.

The Gabor system and the matrices G and R . The test space is spanned by the *Gabor system* $f_k(u) = \varphi(u)e^{-i\tau_k u}$, $0 \leq k < d$, where $\tau_k = T + 2\pi k/L$ is a grid of frequencies filling the window, $d = \lfloor LT/2\pi \rfloor \approx \lambda N$, and the *taper* φ is a real even C^2 function on $[-L/2, L/2]$ equal to 1 on the middle $(1 - \eta)$ fraction of the interval, η being a small smoothing parameter eventually sent to 0; the *sinc system* is the untapered case $\varphi = \mathbf{1}_{[-L/2, L/2]}$, whose $\hat{\varphi}$ is the sinc function. The *compressed Weil form* is the $d \times d$ Gram matrix $G_{kl} = W(f_k, f_l)$; “ G -units” refers to this raw matrix. The brief's R is G divided by L (in Theorem 4's write-up R was G orthonormalised by the mass matrix $\int f_k \bar{f}_l$, which is approximately $L \cdot \text{Id}$; referee D's repair is to use G/L outright). In the brief's units $\text{tr } R = LN(1 + o(1))$ and $\text{tr } R^2 \approx \frac{4}{3}L^2N$ at $\lambda \rightarrow 1$ (more precisely $\lambda l^2 N(1 + \lambda^2/3)$); these are the prime-side evaluations the campaign calls *Step 2*. Later in this document I renormalise once more, to $\tilde{R} = G/(L^2 a)$, so that a simple isolated on-line zero has eigenvalue 1.

Inertia and counting. n_+ and n_- are the positive and negative indices of inertia of R (numbers of positive and negative eigenvalues); n_+^θ , n_-^θ count eigenvalues above a threshold θ (respectively below $-\theta$), the threshold absorbing a small tail error. A “positive square” is a rank-one positive semidefinite summand $m|l\rangle\langle l|$ of the form; the *Cauchy–Schwarz count* is the elementary fact that a Hermitian matrix with positive trace has at least $(\text{tr } R)^2 / \text{tr } R^2$ positive eigenvalues. The *windowed form* A is the part of the zero side coming from zeros with ordinate in a slightly enlarged window I' ; the remainder E is the *tail*. Cauchy interlacing, Weyl's inequalities and Ky Fan's principle are the standard eigenvalue inequalities for sums and compressions of Hermitian matrices, as the brief states them.

The pair block. For an off-line pair, v and $w = \bar{v}$ are the coordinate vectors of the two evaluation functionals $f \mapsto h_f(\gamma)$, $f \mapsto h_f(\bar{\gamma})$ on the Gabor basis; $A(\delta) \geq 1$ is the φ^2 -weighted exponential moment defined in the brief, and $p, -q$ are the two nonzero eigenvalues of the pair's rank-two block. The inner-product kernel the brief writes as $D_L(\gamma - \gamma') = \langle u_\rho, u_{\rho'} \rangle$ is the kernel governing overlaps of evaluation functionals at nearby spectral parameters — for the sinc system a Dirichlet/sinc-type kernel of width $\sim 1/L$; the brief does not define it further, and in what follows its role is played by $\Phi = \text{FT}(\varphi^2)$ from referee D's Poisson identity. The *quantum* $u = L^2$ is the eigenvalue of an isolated simple on-line zero in G -units.

External results invoked. “RH plus Montgomery” is Montgomery's 1973 theorem: under RH, the pair correlation of zeros with Fejér weight gives $\sum m_\rho^2 \leq (\frac{4}{3} + o(1))N$, whence at least $2/3$ of zeros are simple; the brief's “ $5/6$ distinct under RH” is its own elementary deduction from the same inequality, flagged as surely known but unchecked. Rudnick–Sarnak (1996) and Hughes–Rudnick are results identifying n -level correlations of zeros with GUE predictions for test functions of restricted Fourier support; whether they are unconditional is left by the brief as a hedge to be checked. “Hardy–Littlewood” is the prime k -tuple conjecture. The “three-moment Markov–Krein bound” is the classical extremal problem of bounding the mass or number of atoms of a measure on $[0, \infty)$ from finitely many moments.

Provenance. Theorem 4 and its Steps 1–3 are E2's; the block description of the zero side is referee C's; coefficient coordinates and the exact Poisson identity are referee D's; the prime-side traces were re-proved independently by the re-derivation run. The single-pair algebra ($\|v\|^2$, $\langle v, w \rangle$, p, q , consequences (i)–(iii)), the inertia inequalities of lever (a), the extremal-configuration picture, the target $2/3$, the integrality arithmetic and the higher-moment proposal are the orchestrator's own, offered as quick derivations for me to verify. The optimisation of φ belongs to the sibling kernel run.

So the target $(2/3)$, the pair-block algebra, the inertia facts, the extremal picture and the integrality arithmetic are all handed to me; what the brief wants from me is verification, a crisp verdict on lever (a), and the higher-moment computation.

1.2 Theorem 4 as E2 wrote it

I begin with the reading list, in the order given.

First E2's write-up of Theorem 4 (claimed): “at least one half of the nontrivial zeros of zeta are on the critical line (as distinct zeros)”. The objects I will be living with are fixed here. Zeros are $\rho = \beta + it$ with multiplicity m_ρ , and $\gamma_\rho := (\rho - \frac{1}{2})/i = t - i(\beta - \frac{1}{2})$; the Weil form is

$$W(f, g) := \sum_{\rho} m_{\rho} h_f(\gamma_{\rho}) \overline{h_g(\bar{\gamma}_{\rho})}, \quad h_f(z) = \int f(u) e^{izu} du,$$

Hermitian by the functional equation, and equal by the explicit formula to $\int h_f \bar{h}_g \nu_X$ with an explicit density ν_X — a digamma term, a pole term, and the prime sum $-\frac{1}{\pi} \sum_{n \leq X} \Lambda(n) n^{-1/2} \cos(\tau \log n)$ — whenever $\text{supp } f, g \subset [-L/2, L/2]$ with $X = e^L$. Three linear-algebra lemmas are recorded: *inertia does not increase under pullback*, $n_+(A^*QA) \leq n_+(Q)$, i.e. a Hermitian form transported to another space through any linear map acquires no new positive squares; the Cauchy–Schwarz count $n_+^\theta(R) \geq (\text{tr } R - \theta d)^2 / \text{tr } R^2$ for the number n_+^θ of eigenvalues above a threshold θ ; and Weyl's perturbation inequality. The set-up: $l = \log(T/2\pi)$, $L = \lambda l$, and the Gabor system

$$f_k(u) = \varphi(u) e^{-i\tau_k u}, \quad \tau_k = T + \frac{2\pi k}{L}, \quad 0 \leq k < d = \left\lfloor \frac{LT}{2\pi} \right\rfloor,$$

with a C^2 taper φ equal to 1 on the middle $(1 - \eta)$ fraction of $[-L/2, L/2]$.

Step 1 is the part the brief told me to exploit. The zero-side matrix is split into A , the contribution of zeros with ordinate in the slightly enlarged window $I' = [T - D_0, 2T + D_0]$, $D_0 = T^{1/2}$, and a tail E . Grouping A by distinct points, an on-line zero of multiplicity m at height t gives the rank-one square $m|l_t|^2$ of its evaluation functional, while an off-line $\rho = \frac{1}{2} + \delta + it$ together with $\rho^* = 1 - \bar{\rho}$ gives the Hermitian block $\begin{bmatrix} 0 & m \\ m & 0 \end{bmatrix}$, of signature $(1, 1)$, in the two functionals $l_\gamma, l_{\bar{\gamma}}$. Hence the abstract block form Q , of which A is a pullback, has $n_+(Q) = n_{\text{on}}^{\text{dist}} + n_{\text{pair}}^{\text{dist}}$, while Riemann–von Mangoldt counting gives $N(I') \geq n_{\text{on}}^{\text{dist}} + 2n_{\text{pair}}^{\text{dist}}$; with the tail bound $\|E\| \ll T^{-1+o(1)} =: \theta_0$ this yields

$$n_{\text{on}}^{\text{dist}}(I') \geq 2n_+^\theta(R) - N(I').$$

Step 2 is the prime-side claim

$$\begin{aligned} \text{tr } R &= LT \langle \mu \rangle_I (1 + o(1)), \\ \text{tr } R^2 &= 2\pi LT \left[\langle \mu \rangle_I^2 + \frac{L^2}{12\pi^2} \right] (1 + o(1)), \end{aligned}$$

with $\langle \mu \rangle_I$ the mean over $I = [T, 2T]$ of the digamma density, proved by Poisson/Shannon for the sinc kernel, the diagonal sum $\sum_{n \leq X} \Lambda(n)^2 n^{-1} (1 - \log n/L)$, and Montgomery–Vaughan for the off-diagonal; it gives $n_+^\theta(R) \geq [\lambda/(1 + \lambda^2/3)] N(I)(1 + o(1))$, consistent under RH with Montgomery’s pair-correlation evaluation. Step 3 concludes

$$\liminf_T \frac{N_0^{\text{dist}}([T, 2T])}{N([T, 2T])} \geq \frac{2\lambda}{1 + \lambda^2/3} - 1 \longrightarrow \frac{1}{2} \quad (\lambda \rightarrow 1^-),$$

nontrivial for $\lambda > 3 - \sqrt{6}$. E2’s comparison list at the end — Levinson 1/3, Conrey 0.4088, Bui–Conrey–Young 0.4105, Feng 0.4128, Pratt–Robles–Zaharescu–Zeindler 5/12, simple on the line $\geq 0.407N$; under RH $\geq 2/3$ simple (Montgomery), $\geq 3/4$ distinct, “later ~ 0.8467 distinct?? (from memory)” — is E2’s own recollection, flagged as such in the file.

1.3 The two referee verdicts on Theorem 4

Next referee C’s verdict — the blind hostile referee on the counting identity — headed “SURVIVES” for the linear-algebra and counting joints. What matters for me is the exact form of the zero-side matrix in E2’s one-sided complex Gabor space. Since $\hat{\varphi}$ has real Taylor coefficients,

$$\begin{aligned} G_{kl} &= W(f_k, f_l) = \sum_{\rho} m_\rho \hat{\varphi}(\gamma_\rho - \tau_k) \hat{\varphi}(\gamma_\rho - \tau_l) \\ &= \sum_{\text{distinct } \rho} m_\rho (a_\rho a_\rho^T)_{kl}, \end{aligned}$$

each distinct zero contributing a complex-symmetric rank-one piece aa^T , “NOT aa^* ”. An on-line zero has a real, giving a positive semidefinite rank-one piece; an off-line ρ and its partner ρ^* have $a_{\rho^*} = \bar{a}_\rho$, and $m(aa^T + \bar{a}a^*)$ is the pullback of $\begin{bmatrix} 0 & m \\ m & 0 \end{bmatrix}$ under $f \mapsto (l_\gamma f, l_{\bar{\gamma}} f)$: at most one positive square in all cases, linear dependence of the functionals included. The conjugate zeros at height $-t$ sit in the tail, so there is no double-counting of quadruples. Also $\text{rank}(G_A) \leq n_{\text{on}}^{\text{dist}} + 2n_{\text{pair}}^{\text{dist}}$, the inertia statement $n_+ \leq n_{\text{on}} + n_{\text{pair}}$ being the sharper one. The Riemann–von Mangoldt bookkeeping and the factor 2 are redone (“a pair costs two R–vM zeros but buys at most one positive square”).

The verdict also carries a table from referee C’s synthetic-configuration test — true zeros on $[600, 1200]$, all doubled, all turned into pairs at various depths, half the ordinates paired, mixed

worlds, coincident heights. Writing $C = (\text{tr } R)^2 / \text{tr } R^2$ for the Cauchy–Schwarz count, so that $2C - N(I')$ is the *certificate* — the lower bound for $n_{\text{on}}^{\text{dist}}$ that Theorem 4 actually certifies from the data — the table shows in every case $n_+ \leq n_{\text{on}}^{\text{dist}} + n_{\text{pair}}^{\text{dist}}$ and $2C - N(I') \leq n_{\text{on}}^{\text{dist}}$, and that pairing or multiplicity always lowers C at fixed count. On true zeros the explicit formula is confirmed to $1.1 \cdot 10^{-8}$, $n_+ = d = 436$, and $C/N = 0.744$ against the asymptotic law 0.750. The referee’s “rank corollary” — replacing inertia by rank gives $N_{\text{dist}} \geq \frac{3}{4}N$ and $N_{\text{simple}} \geq \frac{1}{2}N$ unconditionally from the same Step 2 — is recorded as the red flag. Finally the literature anchor I will lean on later: referee C cites Baluyot–Goldston–Suriajaya–Turnage–Butterbaugh, arXiv 2306.04799 = Acta Arith. 214 (2024), Thm 1, as an unconditional pair-correlation asymptotic

$$F(\alpha) = T^{-2\alpha}(\log T + O(1)) + \alpha + O(1/\sqrt{\log T})$$

with complex γ allowed, which integrates to E2’s $\text{tr } R^2$ main term at $\lambda = 1$, and notes that BGSTB extracted only “61.7% simple ASSUMING all zeros in $|\beta - 1/2| < 1/(2 \log T)$ ”. The burden, says C, lies entirely on Step 2’s upper bound for $\text{tr } R^2$.

Then referee D’s verdict on the technical gaps. The structural change I will adopt wholesale: never orthonormalise; work with the real symmetric matrix G/L in coefficient coordinates, where inertia, Weyl and Cauchy–Schwarz need no inner product, so the ill-conditioned mass matrix of the Gabor system disappears from the argument. D’s concrete taper is $\varphi = \mathbf{1}_{[-c,c]} * \text{Hann}_w$ with ramp width $w = \eta L/2$ and $c = (L - w)/2$, with closed-form transform $\hat{\varphi}(r) = (2 \sin(rc)/r) \widehat{\text{Hann}_w}(r)$, and taper functionals

$$a := \frac{1}{L} \int \varphi^2, \quad b := \frac{1}{L} \int \varphi^4, \quad J, \quad \Phi := \text{FT}(\varphi^2), \quad \Phi(0) = La,$$

J being a third functional of the taper that carries the λ^2 term below. The tail bound is made explicit using only the r^{-2} decay of $\hat{\varphi}$: $\theta_0 = 2A_0 C_1^2 e^{L/2} \log(4T) D_0^{-2} = T^{\lambda/2-1+o(1)}$, with constants A_0, C_1 from D’s notes, and only $\theta_0 = o(1)$ is needed.

The item the brief singled out is D’s exact Poisson lemma: for any $\varphi \in L^2$ supported in $[-L/2, L/2]$ and the full grid $\tau_k = T + 2\pi k/L$, $k \in \mathbb{Z}$,

$$\sum_{k \in \mathbb{Z}} \hat{\varphi}(\tau - \tau_k) \hat{\varphi}(\tau' - \tau_k) = L \Phi(\tau - \tau'), \quad \sum_{k \in \mathbb{Z}} \hat{\varphi}(\tau - \tau_k)^2 = L^2 a,$$

proved by Poisson summation (the Fourier transform in s of $s \mapsto \hat{\varphi}(\tau - s) \hat{\varphi}(\tau' - s)$ vanishes for frequencies $|\xi| \geq L$, so only the $\xi = 0$ term survives), stated for real arguments and checked numerically to $8 \cdot 10^{-11}$. This is the rigorous form of the identity the brief wrote schematically, without aliasing terms: on the full grid there are none. Its consequence is the limiting ratio

$$\frac{\lambda a^2}{b + \lambda^2 J} = \frac{\lambda}{1 + \lambda^2/3} (1 - \kappa \lambda \eta + O(\eta^2)), \quad \kappa \approx 0.27,$$

and the taper constants are tabulated: at $\eta = 0.2$, $a = 0.87933$, $b = 0.86245$, $J = 0.22702$; at $\eta = 0.1$, $a = 0.93967$; at $\eta = 0.05$, $a = 0.96983$. D’s prime-side prediction matches the true-zero traces to $3 \cdot 10^{-4}$ at $T = 2000$, and the numerical table at $T = 2000$ shows certificates $(2C_G - N(I'))/N$ of $+ .402$ and $+ .424$ at $\lambda = 1$, $\eta = 0.1$ and 0.05 . None of the repairs moves the constant; D too places the whole weight on the Step-2 main term.

1.4 The re-derived prime-side traces and a declared gap

Last on the list is the re-derivation run’s proof of the prime- and gamma-side traces — the blind, from-scratch re-derivation of Step 2 alone. It fixes $l = \log(T/2\pi)$, $0 < \lambda < 1$, $L = \lambda l$,

$X = e^L$, $N = Tl/2\pi$, splits the explicit-formula density as $\nu_X = \mu + \Pi + P$ (digamma, pole and prime parts) and works entirely on the u -side. Two points in it matter for me.

First, a structural warning. The mass matrix of the Gabor system,

$$M_{kl} = \int \varphi^2(u) e^{-2\pi i(k-l)u/L} du = L T_d[s],$$

is a Toeplitz matrix whose symbol $s = \varphi^2(L\theta/2\pi)$ takes every value in $[0, 1]$, so M is *not* $L(\text{Id} + O(\eta))$: by Szegő about ηd of its eigenvalues are spread over $[0, L/2]$, the smallest exponentially small, and the orthonormalised “ $R = M^{-1/2}GM^{-1/2}$ ” is genuinely ill-conditioned. The author therefore treats two systems: (A) an exactly orthogonal system with spacing $2\pi/L_0$, $L_0 = (1 - \eta)L$, and a Princen–Bradley window satisfying $\sum_m \varphi^2(u + mL_0) \equiv 1$, so that $M = L_0 \text{Id}$ exactly and $R = G/L_0$; (B) the brief’s spacing $2\pi/L$, with R_S the compression of the form to the spectral subspace $S = \{M \geq L/2\}$ of the mass matrix, $\dim S \geq (1 - 2\eta)d$.

Second, the results. Theorem A (system A) and Theorem B (system B) both give

$$\begin{aligned} \text{tr } R &= \lambda l N (1 + o(1)), \\ \text{tr } R^2 &= \lambda l^2 N \left(1 + \frac{\lambda^2}{3}\right) (1 + o(1)), \\ \frac{(\text{tr } R)^2}{\text{tr } R^2} &= \frac{\lambda}{1 + \lambda^2/3} N (1 + o(1)), \end{aligned}$$

with $o(1) = O(\eta + 1/l)$; the off-diagonal prime terms are $O(\lambda l(T^{\lambda-1} + T^{-\lambda/4}))$ relative to the main term, the $\lambda^2/3$ is $2 \cdot \frac{1}{6}$ from the Fejér autocorrelation of the u -window integrated against $d(\log^2 n/2)$, and the $\mu\mu'$ constant is exactly 1. The inputs are Stirling, Mertens/Chebyshev, the prime number theorem with classical error (only for a constant), and Montgomery–Vaughan’s Hilbert inequality and mean-value theorem — no zeros, no RH, no density theorems. The numerical section confirms the finite main terms to between 10^{-4} and $7 \cdot 10^{-4}$.

And a declared gap, Remark 7.4: for the literal unrestricted $M^{-1/2}GM^{-1/2}$ of system (B) the method proves nothing, because on the $\sim \eta d$ bad eigendirections of M the prime part of the Rayleigh quotient is bounded unconditionally only by $\sup |P| \sim X^{1/2}$ — “any method not controlling $\sup_{\tau \in I} |P|$ on short windows, i.e. Lindelöf-type input” does not decide it. That obstruction I will meet again.

2 Verifying the off-line pair block numerically

2.1 What the sibling owns, and my normalised matrix

I look at what the sibling kernel agent — the one the brief assigned the optimisation of the taper — has produced in the meantime, reading the opening of its report so as not to duplicate it. Its structural fact is this. For any finite family of test functions h_j the zero side of the form is governed by the kernel

$$k(z, z') = \sum_j h_j(z) \overline{h_j(z')},$$

which on $\mathbb{R} \times \mathbb{R}$ is a finite-rank Gram kernel; hence in any translation-invariant limit its Fourier transform $w := \hat{k}$ is nonnegative (Bochner) with $\text{supp } w \subset [-L/2, L/2]$. The pair-correlation kernel seen by the Cauchy–Schwarz count is $K = |k|^2$ with $K(0) = (\int w)^2$, the same k entering both $\text{tr } G$ and $\text{tr } G^2$; conversely every admissible w is realised exactly by the Gabor system with $\varphi = \sqrt{w}$, via referee D’s Poisson identity. So, in the sibling’s words, “within inertia +

(tr, tr²) Cauchy–Schwarz + explicit formula with $X \leq T$ the whole design freedom is one function $w \geq 0$ on $[-L/2, L/2]$ — $w = \varphi^2$, Theorem 4 as written being $w = 1$ with K the Fejér kernel — and a signed w is unreachable. Redoing Step 2 of Theorem 4 for general w , the sibling obtains the certificate constant

$$c_\lambda(v) = \frac{\lambda(\int v)^2}{\int v^2 + \lambda^2 \iint |s - s'| v(s)v(s') ds ds'} \quad (v \geq 0 \text{ on } [-\tfrac{1}{2}, \tfrac{1}{2}]),$$

which it identifies with Montgomery’s pair-correlation evaluation and with the very functional appearing in Montgomery’s RH bound; $v = 1$ recovers $\lambda/(1 + \lambda^2/3)$. Its extremal problem leads to $v'' + 2\lambda^2 v = 0$, $v = A \cos(\sqrt{2} \lambda s)$ — and there my excerpt stops. The taper is the sibling’s business; mine is the blocks.

I now fix the notation I will use for everything that follows. As before $l = \log(T/2\pi)$, $L = \lambda l$, $N = N([T, 2T]) \sim Tl/2\pi$, and the test space has dimension $d = \lfloor LT/2\pi \rfloor = \lambda N(1 + O(1/l))$; the Gabor system is $f_k(u) = \varphi(u)e^{-i\tau_k u}$ with $\tau_k = T + 2\pi k/L$, φ a real even C^2 taper on $[-L/2, L/2]$, and $a := \frac{1}{L} \int \varphi^2$. I adopt referee D’s coefficient coordinates (no orthonormalisation, so no mass matrix) and referee C’s form of the zero-side matrix:

$$G = \sum_{\text{distinct } \rho} m_\rho u_\rho u_\rho^T \quad (\text{transpose, not adjoint}),$$

$$u_\rho := (\hat{\varphi}(\gamma_\rho - \tau_k))_k, \quad \gamma_\rho = t - i\delta \text{ for } \rho = \tfrac{1}{2} + \delta + it.$$

The one choice of my own is the normalisation: I work with

$$\tilde{R} := G/(L^2 a),$$

so that an isolated simple on-line zero should contribute eigenvalue exactly 1 (to be checked below). In these units the prime-side asymptotics of Theorem 4, Step 2 — equivalently the re-derivation agent’s Theorems A/B — read

$$\text{tr } \tilde{R} = N(1 + o(1)), \quad \text{tr } \tilde{R}^2 = N\left(\frac{1}{\lambda} + \frac{\lambda}{3}\right)(1 + o(1)),$$

and Theorem 4 itself reads $n_{\text{on}}^{\text{dist}} \geq 2n_+ - N$ with

$$n_+ \geq \frac{(\text{tr } \tilde{R})^2}{\text{tr } \tilde{R}^2} = \frac{\lambda}{1 + \lambda^2/3} N.$$

Finally, with $\Phi := \text{FT}(\varphi^2)$ the kernel of referee D’s Poisson identity, I set $\tilde{\Phi} := \Phi/(La)$, so that $\tilde{\Phi}(0) = 1$, and

$$A(\delta) := \tilde{\Phi}(-2i\delta) = \frac{\int \varphi^2(u) e^{2\delta u} du}{\int \varphi^2} \geq 1,$$

which is exactly the brief’s $A(\delta)$ (its $\sinh(\delta L)/(\delta L)$ being the sharp-cutoff approximation), here expressed through referee D’s Poisson kernel.

2.2 The synthetic single-pair test

Now the brief’s first task item: verify the block facts “with a synthetic single pair”. I set up referee D’s taper concretely — $\varphi = \mathbf{1}_{[-c,c]} * \text{Hann}_w$ with ramp width $w = \eta L/2$ and $c = (L - w)/2$, so that $\varphi \equiv 1$ on $|u| \leq (1 - \eta)L/2$ and on the ramp, in the normalised ramp coordinate $t \in (0, 1)$, φ has the closed form $t + \sin(2\pi(t - \frac{1}{2}))/ (2\pi)$ — together with a trapezoidal

evaluation of $\hat{f}(z) = \int f(u) e^{izu} du$ valid for *complex* z , which is what an off-line zero requires. The test takes $L = 6$, $\eta = 0.2$, the grid $\tau_k = 2\pi k/L$ for $|k| \leq 4000$, and synthetic zeros

$$(t, \delta) \in \{(0.3, 0), (0.3, 0.05), (1.234, 0.2), (0.77, 0.5), (2, 1)\},$$

forms the functional $v_k = \hat{\varphi}(t - i\delta - \tau_k)$ of the off-line point, and compares

$$\begin{aligned} v^T v & \text{ with } L^2 a, \\ \|v\|^2 & \text{ with } L^2 a A(\delta), \\ \text{the extreme eigenvalues of } vv^T + \bar{v}\bar{v}^T & \text{ with } -L^2 a (A - 1), +L^2 a (A + 1), \end{aligned}$$

these being the brief's "facts to verify first" rewritten in my taper-general normalisation.

I add two checks to the brief's list. First, referee D stated and tested the Poisson identity for real arguments only, whereas the brief's sketch of it already carried complex shifts; I test

$$\sum_k \hat{\varphi}(z - \tau_k) \hat{\varphi}(z' - \tau_k) = L \Phi(z - z')$$

at complex z, z' , since that is the form that will govern every cross term involving an off-line zero. Second, a first look at the interactions the brief had warned could be negative: the cross term between an on-line zero at height t' , with real functional u , and a pair at (t, δ) contributes

$$\text{tr}(uu^T(vv^T + \bar{v}\bar{v}^T)) = 2 \text{Re}(u \cdot v)^2$$

to $\text{tr} G^2$, and by Poisson $u \cdot v = L \Phi(t - t' - i\delta)$; I evaluate this for a pair at $t = 0$, $\delta = 0.3$ and $t' \in \{0, 0.4, 0.8, 1.047, 1.3\}$, in units of $L^4 a^2$.

The dense eigendecomposition of the truncated block is needlessly slow; the obvious rank-two reduction — with $W = [v, \bar{v}]$ the block is $B = W \sigma_x W^*$, $\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and its nonzero eigenvalues are those of the 2×2 matrix $\sigma_x W^* W$ — is what I use from here on, though the numbers below still come from the original dense computation on the $|k| \leq 4000$ grid.

Here is what comes back. The taper constant is $a = 0.879332$ (referee D's 0.87933 at $\eta = 0.2$), so $L^2 a = 31.655945$. For every one of the five zeros $v^T v = 31.655945$ with imaginary part at most $5.7 \cdot 10^{-14}$ — real, positive and δ -independent, exactly as the brief asserted — and $\|v\|^2$ matches $L^2 a A(\delta)$ to all printed digits. The block spectra:

t	δ	$A(\delta)$	$\lambda_{\min}$	$\lambda_{\max}$	rank
0.3	0	1.000000	-0.000000	63.311891	1
0.3	0.05	1.011679	-0.369707	63.681598	2
1.234	0.2	1.196943	-6.234416	69.546307	2
0.77	0.5	2.647667	-52.158467	115.470358	2
2.0	1.0	18.833064	-564.522489	627.834380	2

The predicted $-L^2 a (A - 1)$ and $+L^2 a (A + 1)$ agree with $\lambda_{\min}$ and $\lambda_{\max}$ to every digit shown. At $\delta = 0$ the block has rank one with the single eigenvalue $63.311891 = 2L^2 a$: this is the brief's "true enemy" made concrete — a pair collapsed onto the line is spectrally an on-line zero of multiplicity 2, eigenvalue 2 in $\tilde{R}$ units.

The Poisson identity holds at complex arguments as cleanly as at real ones: for

$$(z, z') \in \{(0.4 - 0.1i, 1.7 + 0.3i), (2 - 0.6i, -1.1 - 0.2i), (0.5, 0.5 + 0.9i)\}$$

the two sides agree to within $1.4 \cdot 10^{-14}$, the last case giving the real value 71.22414602, as it must, $z - z'$ being purely imaginary and φ^2 even.

And the cross term equals its Poisson prediction $2 \operatorname{Re} [L\Phi(t - t' - i\delta)]^2$ exactly, its ratio to $L^4 a^2 = 1002.10$ running

$$+2.4558, \quad +1.4719, \quad +0.0574, \quad -0.1883, \quad -0.0357 \quad \text{at } t' = 0, 0.4, 0.8, 1.047, 1.3.$$

So the interaction between an on-line zero and a nearby pair is not sign-definite: large and positive when they nearly coincide, negative about one grid cell ($2\pi/L \approx 1.047$) away. Whatever per-pair lower bound on $\operatorname{tr} \tilde{R}^2$ I hope to use cannot come from termwise positivity of cross terms.

3 Integrality needs quantised, phase-locked mass

3.1 Integrality survives interactions under fixed normalisation

The brief’s integrality lever rests on its consequence (ii), which in $\tilde{R}$ units says that an isolated off-line pair costs $2(A^2 + 1) \geq 4$ in $\operatorname{tr} \tilde{R}^2$ while contributing only 2 to $\operatorname{tr} \tilde{R}$; and the form of attack the brief suggested was an extremal LP/QP over eigenvalue configurations. The sign-indefinite cross terms just computed are the threat to any such per-pair accounting. I recast the question as a finite-dimensional extremal problem for the Frobenius norm at the pure linear-algebra level — this formulation is mine, not the brief’s or any earlier report’s. Take

$$M = \sum_{i=1}^{n_{\text{on}}} u_i u_i^T + \sum_{j=1}^{n_p} (v_j v_j^T + \bar{v}_j \bar{v}_j^T),$$

$$u_i \in \mathbb{R}^d, \quad |u_i| = 1, \quad v_j \in \mathbb{C}^d, \quad v_j^T v_j = 1,$$

where the constraint $v_j^T v_j = 1$ is precisely the δ -independent normalisation verified in the single-pair test, and no other restriction is placed on the vectors, so every kind of interaction is allowed. Here n_{on} plays the role of the number of distinct on-line zeros, n_p the number of distinct off-line pairs, and $N := n_{\text{on}} + 2n_p$ the zero count with multiplicity. Two questions:

- (Q1) Is $\min \|M\|_F^2 \geq n_{\text{on}} + 4n_p$? That is “integrality”, or diagonal dominance, and would give $2/3$.
- (Q2) How does the true minimum compare with the Cauchy–Schwarz floor $N^2/(n_{\text{on}} + n_p)$, which is all Theorem 4 uses?

I minimise $\|M\|_F^2$ by BFGS with 40 random restarts over thirteen cases from $(d, n_{\text{on}}, n_p) = (4, 0, 1)$ to $(8, 1, 3)$, recording the spectrum of the minimiser and the pair norms $A_j := \|v_j\|^2$ (the analogue of $A(\delta)$: $A_j = 1$ is a tight pair, $A_j > 1$ a deep one).

The answer is unambiguous: in every case the numerical minimum equals $n_{\text{on}} + 4n_p$ to five decimals, strictly above the Cauchy–Schwarz floor whenever $n_{\text{on}} > 0$. A selection:

(d, n_{on}, n_p)	N	$\min \ M\ _F^2$	$n_{\text{on}} + 4n_p$	C–S floor	spectrum; A_j
(4, 1, 1)	3	5.00000	5	4.5000	$\{1, 2\}; 1$
(4, 2, 1)	4	6.00000	6	5.3333	$\{1, 1, 2\}; 1$
(6, 2, 2)	6	10.00000	10	9.0000	$\{1, 1, 2, 2\}; 1.139, 1.139$
(6, 4, 1)	6	8.00000	8	7.2000	$\{1^4, 2\}; 1$
(8, 2, 3)	8	14.00000	14	12.8000	$\{1, 1, 2, 2, 2\}; 1.044, 1.039, 1.018$
(8, 4, 2)	8	12.00000	12	10.6667	$\{1^4, 2, 2\}; 1.55, 1.55$
(8, 0, 4)	8	16.00000	16	16.0000	$\{2^4\}; 1.075, \dots, 1.674$

The minimising spectrum is always $\{1 (\times n_{\text{on}}), 2 (\times n_p)\}$ — on-line zeros at eigenvalue 1, pairs looking like double zeros — and the pair norms at the minimum are ≥ 1 but need not equal 1 (they reach 1.674), so the optimiser is genuinely using non-tight, interacting pairs and

still cannot get below $n_{\text{on}} + 4n_p$. Put differently, the Cauchy–Schwarz extremal spectrum, all eigenvalues equal to $N/(n_{\text{on}} + n_p)$, is not realisable by on-line points plus normalised pairs.

So at the linear-algebra level, with $v^T v$ pinned to 1, the per-pair cost 4 appears to survive the sign-indefinite interactions — numerical evidence only, and somewhat against the spirit of the brief’s warning that spectral integrality is robustly false because of interactions. The natural next question is which of my constraints is doing the work: can the normalisation $v_j^T v_j = 1$ be dropped in favour of a free inequality in the quantities $|u_i|$ and $v_j^T v_j$?

3.2 Unnormalised and phase-locked variants all fail

I now strip the normalisation and look for a free quartic inequality in the raw vectors. With

$$M = \sum_i u_i u_i^T + \sum_j 2 \operatorname{Re}(v_j v_j^T)$$

for *arbitrary* real u_i and complex v_j — no normalisation, no phase condition — and writing $s_j := v_j^T v_j$, I test three candidate inequalities of my own devising:

$$(G2) \quad \|M\|_F^2 \geq \sum_i |u_i|^4 + 4 \sum_j |s_j|^2,$$

$$(G2\text{re}) \quad \|M\|_F^2 \geq \sum_i |u_i|^4 + 4 \sum_j (\operatorname{Re} s_j)^2,$$

$$(G1) \quad \|M\|_F^2 \geq \sum_i |u_i|^4 + 2 \sum_j |s_j|^2 + 2 \sum_j \|v_j\|^4.$$

Everything is homogeneous of degree four, so I fix the total Euclidean norm of all the vectors to 1 and minimise lhs – rhs by BFGS with 60 restarts over the seven cases

$$(d, n_{\text{on}}, n_p) \in \{(3, 1, 1), (4, 2, 1), (4, 0, 2), (5, 2, 2), (6, 3, 2), (6, 1, 3), (8, 4, 3)\}.$$

All three fail, and fail cleanly: G2 and G1 return $\min(\text{lhs} - \text{rhs}) = -4.000000$ in every case, and G2re returns -1.000000 with one pair and -2.000000 with two or more.

A gap of exactly -4 at total norm 1 means the left side is being driven to zero while $|s| = 1$. I extract the G2 minimiser at $(d, n_{\text{on}}, n_p) = (3, 1, 1)$: $u \approx 0$ (entries of size 10^{-8}), $\|v\|^2 = 1$, and

$$v^T v = 2.3 \cdot 10^{-7} - 1.0i,$$

the printed v having componentwise equal and opposite real and imaginary parts, i.e. $v = e^{-i\pi/4} r$ with r a real unit vector. Then $vv^T = -i r r^T$ is purely imaginary, $2 \operatorname{Re}(vv^T)$ is the zero matrix, M has eigenvalues between $-4.9 \cdot 10^{-9}$ and $4.7 \cdot 10^{-7}$, and $\text{lhs} = 2.2 \cdot 10^{-13}$ against $\text{rhs} = 4$. So it is the free *phase* of $v^T v$ that kills G2: a “pair” whose $v^T v$ has the wrong argument contributes a vanishing block while being charged full price.

Next attempt: charge only for the positive real part of s_j ,

$$(G3) \quad \|M\|_F^2 \geq \sum_i |u_i|^4 + 4 \sum_j (\operatorname{Re} s_j)_+^2 \quad ?$$

for arbitrary real u_i and complex v_j , now with 80 restarts over twelve cases from $(2, 1, 1)$ to $(8, 2, 4)$. Still false. With a single pair the gap is $-5.000 \cdot 10^{-1}$ every time, the minimiser having $\operatorname{Re} s = -0.25$ and one u with $|u|^2 = 0.75$ (any further u_i at 0); with two or more pairs the gap is -1.333 every time, the minimiser having one pair with $\operatorname{Re} s \approx 0.667$, the remaining pairs sharing $\operatorname{Re} s \approx -0.333$ between them (for instance $\{0.667, -0.032, -0.302\}$ at $(8, 4, 3)$), and all

$|u_i|^2 \rightarrow 0$. Pairs with negative $v^T v$ are priced at zero on the right-hand side but still subtract on the left.

So the relevant version is the phase-locked one. The functional-equation pairing $\rho \leftrightarrow 1 - \bar{\rho}$ together with the Poisson identity forces $v^T v = L^2 a > 0$ for a genuine off-line pair — this is exactly the positivity verified in the single-pair test — so I now take the real u_i arbitrary and the complex v_j with $v_j^T v_j$ *real and positive* but of free magnitude (implemented by rotating each v_j through $e^{-\frac{i}{2} \arg s_j}$), and test

$$(G4) \quad \left\| \sum_i u_i u_i^T + \sum_j 2 \operatorname{Re}(v_j v_j^T) \right\|_F^2 \geq \sum_i |u_i|^4 + 4 \sum_j (v_j^T v_j)^2.$$

My first attempt, 100 restarts over seventeen cases up to $(10, 5, 5)$, is cut off at the time limit before returning anything, so I rerun it with 25 restarts, dropping the two largest cases $(8, 4, 3)$ and $(10, 5, 5)$, and leave it running.

4 A trace inequality coupling rank and negative spectrum

4.1 Phase-locking fails; a rank-plus-inertia claim

I read the results of the phase-locked test (G4), in which the on-line vectors u_i are arbitrary real, the pair vectors v_j are complex with $s_j := v_j^T v_j$ forced real and positive but of free magnitude, and the candidate inequality charges $4s_j^2$ per pair. Phase-locking does not rescue per-pair quadratic pricing either:

(d, n_{on}, n_p)	min gap	$s_j = v_j^T v_j$	$A_j = \ v_j\ ^2 / s_j$	$ u_i ^2$
(2, 1, 1)	$-2.500 \cdot 10^{-1}$	0	30168	0.75
(2, 0, 2)	$-6.666 \cdot 10^{-1}$	0.667, 0	1, $4.8 \cdot 10^4$	—
(4, 2, 1)	$-2.499 \cdot 10^{-1}$	0	5730	0, 0.751
(5, 2, 2)	$-6.666 \cdot 10^{-1}$	0, 0.667	$1.0 \cdot 10^4$, 1	0, 0
(4, 0, 3)	$-7.251 \cdot 10^{-1}$	0, 0.637, 0	905, 1, $2.0 \cdot 10^4$	—
(6, 2, 3)	$-7.250 \cdot 10^{-1}$	0.633, 0, 0	1, 2526, 9978	0, 0
(8, 2, 4)	$-7.276 \cdot 10^{-1}$	0, 0.633, 0, 0	893, 1, $1.2 \cdot 10^5$, 660	0, 0

The gap $\min(\text{lhs} - \text{rhs})$ is -0.25 with one pair, -0.667 whenever $n_p = 2$, and between -0.724 and -0.728 for $n_p = 3, 4$. The minimisers always consist of one honest pair ($s \approx 2/3$ or 0.63 , $A = 1$, i.e. tight) together with degenerate “pairs” having $v^T v \rightarrow 0$ and $A \sim 10^3$ – 10^5 , the on-line vectors either vanishing or, in the one-pair cases, sitting at $|u|^2 = 0.75$. The mechanism is plain: writing $v = x + iy$ with $|x| = |y|$, a pair with vanishing $v^T v$ costs nothing on the right-hand side of G4, while its block

$$2 \operatorname{Re}(v v^T) = 2(x x^T - y y^T)$$

remains free to move mass around on the left. So the normalisation $v^T v = 1$ in the one experiment that succeeded was doing two jobs at once — fixing the phase *and* quantising the mass — and the magnitude cannot be set free: a per-pair price of the form $4(v^T v)^2$ is simply false without quantised mass.

I therefore abandon block-by-block pricing and write down a claim of a structurally different kind, one that uses no block structure at all — only the two global features of the windowed zero-side matrix that survive everything above: the on-line part is positive semidefinite of small rank, and the pair part has few positive eigenvalues but large trace. Precisely: let $M_{\text{on}} \succeq 0$ with $\operatorname{rank} M_{\text{on}} \leq r$ and $\operatorname{tr} M_{\text{on}} = t_{\text{on}}$; let $M_p = P - Q$ with $P \succeq 0$ of rank at most b and $Q \succeq 0$; put $t' := \operatorname{tr} M_p > 0$ and $s := t'/b$.

Claim (abstract inequality).

$$\|M_{\text{on}} + M_p\|_F^2 \geq r g_s\left(\frac{t_{\text{on}}}{r}\right) + \frac{t'^2}{b}, \quad g_s(\mu) = \begin{cases} \mu^2, & \mu \leq s, \\ 2s\mu - s^2, & \mu > s. \end{cases}$$

Corollary (as I state it, for $s \geq 2 - o(1)$ and $t_{\text{on}} + t' = N$, writing $M = M_{\text{on}} + M_p$):

$$r \geq 2N - \|M\|_F^2.$$

For provenance: the question — whether the pair-block and negative-spectrum structure can lift Theorem 4’s constant towards $2/3$ — is the brief’s, and so are both ingredients of the claim. The bound $n_+(M_p) \leq b$, one positive eigenvalue per pair block, is the inertia fact of the brief’s lever (a); and $s \geq 2 - o(1)$ is the brief’s consequence (i), $p - q = 2mL^2$, i.e. trace $2m \geq 2$ per block in $\tilde{R}$ units — the very fact the brief set aside as “consistent with $\text{tr } R = LN$ carrying no information”. What is mine is coupling the rank of the on-line part with the inertia and trace of the pair part in a single inequality that is linear in the masses rather than quadratic. (Editorially: with $\|\tilde{R}\|_F^2 \rightarrow N(1/\lambda + \lambda/3) \rightarrow \frac{4}{3}N$ from the earlier agents’ prime-side computation, the corollary would read $r \geq \frac{2}{3}N - o(N)$ — an inference; nothing to this effect is written down at this point. Note too that $2N - \frac{4}{3}N = \frac{2}{3}N$ is the very arithmetic the brief had already done at its

Before trying to prove the claim I try to break it. I draw 400 random instances with $d \in \{2, \dots, 6\}$ and with r, b and the rank of Q each uniform in $\{1, \dots, d\}$; I parametrise $M_{\text{on}} = AA^T$, $P = BB^T$, $Q = CC^T$ by unconstrained real factors of those ranks and minimise lhs – rhs by BFGS, penalising $t' \leq 0$. The claim is homogeneous of degree four in the factors — μ^2 , $s\mu$ and s^2 are all of degree four — so I also track the gap normalised by the fourth power of the factor norm. Printing each new record low, every value that appears is nonnegative,

$$+1.783 \cdot 10^{-9}, \quad +3.539 \cdot 10^{-13}, \quad +1.186 \cdot 10^{-13}, \quad +1.776 \cdot 10^{-15}, \quad +2.220 \cdot 10^{-16}, \quad +0.000,$$

and the run ends with “overall most negative gap found: 0.0”. The optimiser reaches equality and never crosses it. Numerical support, not a proof — but this claim, unlike G1–G4, has resisted a serious attempt at falsification.

4.2 Locating cached zeta zeros for true-zero tests

I will want to see the corollary’s right-hand side on actual zeta matrices, so I go looking for cached zeros in the campaign tree. E2’s tables hold only its experiment logs; referee C’s material consists of its verdict, the BGSTB paper, and its small test computations (the linear-algebra lemmas, the explicit-formula check, the synthetic-configuration test, a Deuring–Heilbronn test). The zero caches turn up elsewhere: an array of ordinates cached by referee B together with its outputs for the windows $[600, 1200]$, $[1000, 2000]$ at $\lambda = 0.85, 1.0$ and $[9000, 10000]$; a file of zeros produced by the D2 agent; and a large list of zeros at the root of the campaign tree.

Inspecting them: referee B’s array holds 3700 ordinates, from 14.13472514 to 4218.23078863; the large list has 100,000 ordinates to nine decimals (14.134725142, 21.022039639, 25.010857580, ...); the D2 agent’s file has 700 zeros to about twenty digits. I also reread the set-up of referee C’s synthetic-configuration test, since it is the template for what I want to do. It takes the window $[600, 1200]$ with a reach of 25 on either side (zeros within 25 of the window are included in the windowed matrix), builds the Gram matrix exactly on the zero side, and checks the certificate $2C - N$ — with $C = (\text{tr } R)^2 / \text{tr } R^2$ the Cauchy–Schwarz count, so that $2C - N$ is the lower bound for $n_{\text{on}}^{\text{dist}}$ that Theorem 4 certifies — against $n_{\text{on}}^{\text{dist}}$, and n_+ against $n_{\text{on}} + n_{\text{pair}}$, at $\lambda \in \{1.0, 0.85\}$, over the scenarios: (a) true zeros; (b) all doubled;

(b') all tripled; (c) every ordinate turned into a pair at $\delta \in \{10^{-4}, 0.05, 0.2, 0.45\}$, so that N doubles; (c2) every other ordinate paired and the rest deleted, preserving the Riemann–von Mangoldt count; (d) ordinates with index $\equiv 0 \pmod 4$ paired, $\equiv 1, 2$ on-line, $\equiv 3$ deleted (N preserved), at $\delta \in \{10^{-3}, 0.1\}$; (e) an on-line zero and a pair at the same height; (f) two pairs at the same height at depths 0.05 and 0.3.

4.3 The pair block, the extremal problem, the theorem

I set down in my report — the document the referees and coordinator will read — everything established so far: the verified block facts, the extremal experiments, and, now in a cleaner form than the g_s claim and with a proof, the abstract theorem.

The pair block. First referee D's Poisson identity in the generality I actually used, for arbitrary complex arguments: for $\phi \in L^2$ supported in $[-L/2, L/2]$ and the full grid $\tau_k = T + 2\pi k/L$, $k \in \mathbb{Z}$,

$$(P) \quad \sum_{k \in \mathbb{Z}} \hat{\phi}(z - \tau_k) \hat{\phi}(z' - \tau_k) = L \Phi(z - z'), \quad \Phi = \text{FT}(\phi^2),$$

with Φ entire, real on $\mathbb{R}$, and $\Phi(\bar{z}) = \overline{\Phi(z)}$. The proof is one line: the Fourier transform in s of $s \mapsto \hat{\phi}(z - s) \hat{\phi}(z' - s)$ at frequency ξ is

$$2\pi \int \phi(u) \phi(\xi - u) e^{izu + iz'(\xi - u)} du,$$

which vanishes for $|\xi| \geq L$, so only the $\xi = 0$ Poisson term survives; complex z changes nothing (and numerically the two sides agreed to 10^{-14}). Then, for an off-line pair $\rho = \frac{1}{2} + \delta + it$, $\rho^* = 1 - \bar{\rho}$, with spectral parameters $\gamma = t - i\delta$, $\gamma^* = \bar{\gamma}$ and functionals $v := u_\rho$, $u_{\rho^*} = \bar{v}$, the brief's facts as verified in my units are

$$\begin{aligned} v^T v &= L \Phi(0) = L^2 a \quad (\text{real, positive, } \delta\text{-independent; to } 10^{-14}), \\ \|v\|^2 &= L \Phi(-2i\delta) = L^2 a A(\delta), \quad A(\delta) = \frac{\int \phi^2 \cosh(2\delta u)}{\int \phi^2} \geq 1, \end{aligned}$$

the last equal to $\sinh(\delta L)/(\delta L)$ for the sharp window. The pair contributes the real symmetric block

$$B = m(vv^T + \bar{v}\bar{v}^T) = 2m \text{Re}(vv^T) = m W \sigma_x W^*, \quad W = [v, \bar{v}],$$

whose nonzero eigenvalues are those of

$$m \sigma_x W^* W = m \begin{bmatrix} s & A' \\ A' & \bar{s} \end{bmatrix}, \quad s = v^T v = L^2 a, \quad A' = \|v\|^2,$$

namely $mL^2 a (1 \pm A)$ — verified at five depths. In $\tilde{R}$ units the block has eigenvalues $m(1 + A)$ and $-m(A - 1)$, trace $2m$ (δ -independent, the value for $2m$ on-line zeros), and $\text{tr} B^2 = 2m^2(1 + A^2) \geq 4m^2$. Writing $v/\sqrt{L^2 a} = r + iq$ with r, q real, the normalisation $v^T v = 1$ says $r \cdot q = 0$ and $|r|^2 - |q|^2 = 1$, and

$$\frac{B}{mL^2 a} = 2(rr^T - qq^T), \quad |r|^2 = \frac{A+1}{2}, \quad |q|^2 = \frac{A-1}{2}.$$

As $\delta \rightarrow 0$, $A - 1 \sim (2\delta)^2 \langle u^2 \rangle_{\phi^2} \sim \delta^2 L^2 / 3$, and the pair becomes spectrally an on-line zero of multiplicity $2m$ — the true enemy, as the brief said. An on-line zero of multiplicity m contributes $m uu^T$, positive semidefinite of rank one, eigenvalue m .

Then the part the brief had only flagged in passing (By (P), for an on-line zero at height t_u with functional u and a pair at (t_v, δ) ,

$$\text{tr}(uu^T \cdot 2 \text{Re}(vv^T)) = 2 \text{Re}(u \cdot v)^2 = 2 \text{Re}[L \Phi(t_v - t_u - i\delta)]^2,$$

and $\text{Re} \Phi(x - iy)^2 = C^2 - S^2$ with

$$C = \int \phi^2(u) \cos(xu) \cosh(yu) du, \quad S = \int \phi^2(u) \sin(xu) \sinh(yu) du,$$

which is negative near the zeros of C ; the computed example had offsets $x = 0, 0.4, 0.8, 2\pi/L, 1.3$ giving ratios $+2.46, +1.47, +0.06, -0.19, -0.04$. Worse, the pair-pair cross terms

$$4 \text{Re} \left[\tilde{\Phi}(x - i(\delta_1 - \delta_2))^2 + \tilde{\Phi}(x - i(\delta_1 + \delta_2))^2 \right]$$

can be of size $\sim -4A^2$: two deep pairs ($\delta L \gg 1$) offset by half a grid cell have blocks that are nearly opposite “quadrupoles” $2(rr^T - qq^T)$ and nearly cancel. So termwise diagonal dominance, $\text{tr} \tilde{R}^2 \geq \sum_{\text{sites}} (\text{self terms})$, is false, and whatever survives must be global. This is the same phenomenon that forced BGSTB (2024) — as reported in referee C’s verdict — to assume $|\beta - \frac{1}{2}| < 1/(2 \log T)$ in order to extract simple zeros from their unconditional pair correlation. Finally, truncation to the actual grid $0 \leq k < d$: for an on-line zero $|u_{\text{trunc}}|^2 \leq L^2 a$ exactly, by dropping nonnegative terms from (P) — and this, I flag, is the only normalisation fact the final argument will need.

The extremal problem. The linear-algebra model of the windowed zero side, in $\tilde{R}$ units, is

$$M = \sum_i m_i u_i u_i^T + \sum_j m_j 2 \text{Re}(v_j v_j^T), \quad u_i \text{ real unit}, \quad v_j^T v_j = 1, \quad m \in \{1, 2, \dots\},$$

where integer masses are coincident unit atoms, so without loss all masses are 1 with coincidences allowed, and the prime side supplies $\text{tr} M = N$ and $\|M\|_F^2 = T_2 := N(1/\lambda + \lambda/3)$. The findings of the experiments are three.

1. In every case $\min \|M\|_F^2 = n_{\text{on}} + 4n_p$ exactly, with spectrum $\{1^{(n_{\text{on}})}, 2^{(n_p)}\}$ at the minimiser, strictly above the Cauchy–Schwarz/inertia floor $N^2/(n_{\text{on}} + n_p)$ that Theorem 4 uses whenever $n_{\text{on}} n_p > 0$; the Cauchy–Schwarz extremal spectrum “all eigenvalues = $4/3$ ” is not realisable by on-line points plus pairs. This is the integrality phenomenon, and it survives interactions.
2. What is false: with free masses $s_j = v_j^T v_j > 0$ the analogue fails (gap $\rightarrow -0.73$, degenerate pairs with $s \rightarrow 0$, $A \rightarrow \infty$ acting as free mass movers); with free phases everything fails ($v = e^{-i\pi/4} r$ gives the zero block). Quantised mass and phase-locking — the latter forced by the functional equation together with Poisson — are both essential, exactly as, I remark from memory, integrality $m \in \{1, 2, \dots\}$ is essential in Montgomery’s $N_{\text{simple}} \geq 2N - \sum m^2$.
3. The counterexamples show that the correct inequality is *linear* in the masses, like $\sum m^2 \geq \sum (2m - 1)$, not quadratic.

This is what led to the following.

Theorem (abstract; proved; this is the lever). Let $M = M_{\text{on}} + M_p$ be Hermitian $d \times d$ with $M_{\text{on}} \succeq 0$, $\text{rank } M_{\text{on}} \leq r$, $\text{tr } M_{\text{on}} = t_{\text{on}}$, and M_p Hermitian with at most b (≥ 1) strictly positive eigenvalues and $\text{tr } M_p = t'$. Then for every $c > 0$

$$\|M\|_F^2 \geq c t_{\text{on}} - \frac{c^2 r}{4} + 2c t' - c^2 b, \quad (3.1)$$

and in particular, at $c = 2$,

$$r \geq 2t_{\text{on}} + 4t' - 4b - \|M\|_F^2. \quad (3.2)$$

The proof is five lines, and von Neumann's trace inequality (which I invoke from memory) is the only tool. Expand

$$\|M\|^2 = \|M_{\text{on}}\|^2 + \|M_p\|^2 + 2 \operatorname{tr}(M_{\text{on}} M_p).$$

Let $\mu_1 \geq \dots \geq \mu_r \geq 0$ be the eigenvalues of M_{on} ; let $M_p = M_p^+ - M_p^-$ be the Jordan decomposition, $p_1, \dots, p_{b'} > 0$ ($b' \leq b$) the positive eigenvalues of M_p , $\nu_1 \geq \nu_2 \geq \dots \geq 0$ the eigenvalues of M_p^- , and $y := \sum \nu$.

(i) $\operatorname{tr}(M_{\text{on}} M_p^+) \geq 0$ (two positive semidefinite matrices), and $\operatorname{tr}(M_{\text{on}} M_p^-) \leq \sum_k \mu_k \nu_k$ by von Neumann, both sequences sorted decreasing. Pointwise, for $\mu, \nu \geq 0$,

$$2\mu\nu \leq 2c\nu + \nu^2 + (\mu - c)_+^2,$$

trivially for $\mu \leq c$, and for $\mu > c$ the difference of the two sides is $(\nu - \mu + c)^2$. Hence

$$2 \operatorname{tr}(M_{\text{on}} M_p) \geq -2cy - \sum \nu^2 - \sum_k (\mu_k - c)_+^2.$$

(ii) By Cauchy–Schwarz on at most b positive eigenvalues, with $\sum p = t' + y$,

$$\|M_p\|^2 = \sum p^2 + \sum \nu^2 \geq \frac{(\sum p)^2}{b} + \sum \nu^2 = \frac{(t' + y)^2}{b} + \sum \nu^2.$$

(iii) $\|M_{\text{on}}\|^2 - \sum_k (\mu_k - c)_+^2 = \sum_k g_c(\mu_k)$, where $g_c(\mu) = \mu^2$ for $\mu \leq c$ and $2c\mu - c^2$ for $\mu \geq c$; since $g_c(\mu) \geq c\mu - c^2/4$ for all $\mu \geq 0$,

$$\sum_{k \leq r} g_c(\mu_k) \geq ct_{\text{on}} - \frac{rc^2}{4}.$$

Summing,

$$\begin{aligned} \|M\|^2 &\geq ct_{\text{on}} - \frac{rc^2}{4} + \frac{(t' + y)^2}{b} - 2cy \\ &\geq ct_{\text{on}} - \frac{rc^2}{4} + \min_y \left[\frac{(t' + y)^2}{b} - 2cy \right] \\ &= ct_{\text{on}} - \frac{rc^2}{4} + 2ct' - c^2b. \end{aligned}$$

Three checks. (a) The minima $n_{\text{on}} + 4n_p$ from the normalised experiment are exactly (3.2) with equality ($t_{\text{on}} = r = n_{\text{on}}$, $t' = 2n_p$, $b = n_p$). (b) The 400 random instances of the sharper intermediate form

$$\|M\|^2 \geq r g_s(t_{\text{on}}/r) + \frac{t'^2}{b}, \quad s = t'/b,$$

gave most negative gap 0.0. (c) With $b = t' = 0$, optimising over c returns $\|M_{\text{on}}\|^2 \geq t_{\text{on}}^2/r$, so the Cauchy–Schwarz count is a special case. The general optimisation over c I leave for the application; the case $c = 2$ is what matters at the extremal configuration.

5 Interruption, then the certificate tested on true zeros

5.1 The run is cut off; the coordinator writes out two thirds

At this point the run is cut off by a server error — immediately after the abstract theorem (3.1)/(3.2) and its proof reach my report, and before anything about its application to the Weil form has been written down.

The coordinator resumes me with a message which, notably, is the first place the consequence $(2 - 1/\lambda - \lambda/3)N \rightarrow \frac{2}{3}N$ of my (3.2) is written out — by the coordinator, not by me; my own “would give $2/3$ ” at the extremal-problem stage was a forecast for (Q1), not this derivation — together with a checklist for the bookkeeping. Condensed, with every item kept:

You were killed by a server error after writing the abstract theorem (3.1)/(3.2); the report survived through that point and the proof checks out line by line. Now write the application to the windowed Weil form, giving

$$\begin{aligned} n_{\text{on}}^{\text{dist}}(I') &\geq 2 \operatorname{tr} A + 2t' - 4b - \|A\|_F^2 \\ &\geq (2 - \frac{1}{\lambda} - \frac{\lambda}{3} - o(1))N \longrightarrow \frac{2}{3}N, \end{aligned}$$

with every bookkeeping point explicit:

- (a) A is the windowed zero-side matrix, $A = M_{\text{on}} + M_p$ in $\tilde{R}$ -units (divide G by L^2a); $M_{\text{on}} \succeq 0$ of rank $\leq n_{\text{on}}^{\text{dist}}$; M_p the sum of pair blocks, with $n_+(M_p) \leq b = n_{\text{pair}}^{\text{dist}}$ by subadditivity of inertia;
- (b) $\operatorname{tr} A$ and $\|A\|_F^2$ from the prime side (Theorem 4 Step 2, or the re-derivation) minus the tail E , bounded using $\theta_0 = lT^{\lambda/2-1}$ and shown to be $o(N)$;
- (c) $t' = \sum_j 2m_j s_j$ with $s_j = v_j^T v_j$ on the truncated grid — prove $s_j \geq 1 - o(1)$ for interior pairs, absorb the $O(D_0 l)$ edge pairs, hence $t' \geq 2b(1 - o(1)) - o(N)$;
- (d) t_{on} uses $|u_i|^2 \leq 1$ only through $t_{\text{on}} + t' = \operatorname{tr} A$ — confirm there is no other use;
- (e) conclude and optimise over c and λ : show $c = 2$, $\lambda \rightarrow 1^-$ optimal given $t' \geq 2b$; note $c = 8/3$ would give $3/4 - b/N$, which is circular;
- (f) state the final theorem, $\liminf N_0^{\text{dist}}(T, 2T)/N(T, 2T) \geq 2/3$ unconditionally; compare Montgomery 1973 (RH, $2/3$ simple); does the argument also give *simple-on-line* $\geq 2/3$? — try to sharpen (3.1) using that on-line masses are integers ≥ 1 ; if not, state distinct;
- (g) numerics: verify (3.2) on actual zeta matrices at $T \sim 1000$ – 2000 with true zeros ($t' = b = 0$ under observed RH) and compare with $n_{\text{on}}^{\text{dist}}$ and $(2 - 1/\lambda - \lambda/3)N$.

Then higher moments (brief, defer to the HL agent’s report) and a final verdict of at most 250 words, verdict first.

5.2 The new certificate on true zeros beats Theorem 4

I take item (g) first. I build the windowed zero-side matrix A in $\tilde{R}$ units directly from zeros: $l = \log(T/2\pi)$, $L = \lambda l$, $d = \lfloor LT/2\pi \rfloor$, the grid $\tau_k = T + 2\pi k/L$ for $0 \leq k < d$, referee D’s closed-form $\hat{\phi}$ (the flat-top transform times the Hann-ramp transform, ramp width $w = \eta L/2$, $c = (L - w)/2$) evaluated at complex arguments, L^2a obtained by quadrature of $|\hat{\phi}|^2/2\pi$, and every zero with height in the enlarged window $I' = [T - 30, 2T + 30]$, i.e. a reach of 30 on either side. An on-line zero of multiplicity m at height t contributes $m u u^T$ with $u = (\hat{\phi}(t - \tau_k))_k$ real; an off-line pair at (t, δ) contributes $m \cdot 2 \operatorname{Re}(v v^T)$ with $v = (\hat{\phi}(t - i\delta - \tau_k))_k$. I accumulate t_{on} and t' as the traces of the on-line and pair parts, count b as the number of distinct pairs, write N' for the zero count of I' with multiplicity, and print two certificates — the lower bounds for

$n_{\text{on}}^{\text{dist}}(I')$ certified respectively by (3.2) and by Theorem 4’s Cauchy–Schwarz count:

$$\text{NEW} := 2 \operatorname{tr} A + 2t' - 4b - \|A\|_F^2 \quad (\text{from (3.2)}),$$

$$\text{OLD} := \frac{2(\operatorname{tr} A)^2}{\|A\|_F^2} - N' \quad (\text{Theorem 4}),$$

both as multiples of $N(I) = \#\{\gamma \in (T, 2T]\}$, together with $n_{\text{on}}^{\text{dist}}$ and whether $\text{NEW} \leq n_{\text{on}}^{\text{dist}}$ holds. The zeros are the 100,000 ordinates located earlier.

Besides the true zeros I feed in synthetic worlds symmetric under the functional equation, in the spirit of referee C’s synthetic-configuration test but chosen to probe the new inequality: all zeros doubled; every fourth ordinate turned into a pair at depth δ with the next ordinate deleted, so that the Riemann–von Mangoldt count is preserved; an adversarial lattice at the mean spacing $2\pi/l$ whose pattern, per six zeros, is four on-line sites plus one tight pair — on-line fraction $2/3$ and $T_2/N = (4 + 4)/6 = 4/3$, the configuration that should saturate the target; and “deep dephased pair-couples”, in which from each run of eight true zeros four are kept and the other four replaced by two pairs at depth δ offset from each other by π/l — the nearly cancelling quadrupoles found in the cross-term analysis.

At $T = 1000$, with $N(I) = 868$, $d = 806$ at $\lambda = 1$ and 726 at $\lambda = 0.9$, and $N' = 921$ true zeros in I' :

configuration	$\operatorname{tr} A$	$\ A\ _F^2$	$\ A\ ^2 / \operatorname{tr} A$ (law)	NEW	OLD	$n_{\text{on}}^{\text{dist}}$
true, $\lambda = 1$, $\eta = 0.1$	867.42	1203.26	1.387 (1.333)	+0.612N	+0.380N	921
true, $\lambda = 0.9$, $\eta = 0.1$	868.13	1288.83	1.485 (1.411)	+0.515N	+0.286N	921
true, $\lambda = 1$, $\eta = 0.05$	867.42	1184.95	1.366 (1.333)	+0.634N	+0.402N	921
all doubled ($N' = 1842$)	1734.84	4813.05	2.774	−1.548N	−0.681N	921
every 4th $\rightarrow$ pair, $\delta = 0.001$	867.42	1578.33	1.820	+0.116N	+0.036N	460
every 4th $\rightarrow$ pair, $\delta = 0.4$	867.95	2359.76	2.719	−0.782N	−0.327N	460
adversary lattice, $\delta = 0.001$	805.99	1116.56	1.385 (1.333)	+0.534N	+0.356N	571
adversary lattice, $\delta = 0.1$	805.99	1131.45	1.404	+0.517N	+0.338N	571

Here “law” is the asymptotic value $1/\lambda + \lambda/3$ of $\|A\|_F^2 / \operatorname{tr} A$ predicted by the prime side. The rows not shown: the every-fourth-pair world at $\delta = 0.05$ gives NEW +0.107N against OLD +0.031N, and at $\delta = 0.2$ both go negative (−0.034N and −0.048N), all four depths having $t_{\text{on}} = 433.52$, $t' \approx 434$, $b = 231$; the dephased deep couples at $\delta = 0.3$ and 0.6 push $\|A\|^2 / \operatorname{tr} A$ up to 2.463 and 4.471 and the certificates down to −0.527N/−0.248N and −2.534N/−0.613N respectively — vacuous, as they should be for worlds carrying this much Frobenius mass.

In the adversary lattice $t_{\text{on}} = 537.95$, $t' = 268.04$, $b = 142$, $N' = 855$, and $n_{\text{on}}^{\text{dist}} = 571$, so the new certificate, 463.5, sits just under the truth, while the old one gives 308.6. In every row the check $\text{NEW} \leq n_{\text{on}}^{\text{dist}}$ comes back true, as it must for a theorem about A . And on the true zeros the new certificate certifies 0.61–0.63N distinct zeros on the line where Theorem 4’s Cauchy–Schwarz certificate gives 0.38–0.40N — this with the observed ratio $\|A\|^2 / \operatorname{tr} A$ still somewhat above its asymptotic law $1/\lambda + \lambda/3$ at this height.

6 Applying the inequality: two thirds distinct, unconditionally

6.1 Bookkeeping on the windowed form; no pair data needed

Now the application of (3.2) to the windowed Weil form, following the coordinator’s checklist but simplifying it where I can.

Objects. Everything is in referee D’s coefficient coordinates and my normalisation: $\tilde{R} := G/(L^2a)$ with $G_{kl} = W(f_k, f_l)$, $0 \leq k, l < d$. On the zero side $\tilde{R} = A + E$, where A collects the

zeros with height in the enlarged window $I' = [T - D_0, 2T + D_0]$, $D_0 = T^{1/2}$, and E is the tail. Grouping A by distinct points, the on-line part is

$$M_{\text{on}} := \sum_{\text{on-line } t_i \in I'} m_i u_i u_i^T, \quad u_i = \frac{(\hat{\phi}(t_i - \tau_k))_k}{\sqrt{L^2 a}} \text{ real},$$

so $M_{\text{on}} \succeq 0$ and $\text{rank } M_{\text{on}} \leq r := n_{\text{on}}^{\text{dist}}(I')$; the pair part is

$$M_p := \sum_j m_j 2 \text{Re}(v_j v_j^T)$$

over the distinct off-line pairs in I' . Each pair block is a pullback of $\begin{bmatrix} 0 & m \\ m & 0 \end{bmatrix}$, hence has at most one positive eigenvalue by Lemma 1 of Theorem 4 (inertia does not increase under pullback, dependent functionals included), and the positive index of inertia is subadditive, so $n_+(M_p) \leq b := n_{\text{pair}}^{\text{dist}}(I')$. Thus $A = M_{\text{on}} + M_p$, and referee C's Riemann–von Mangoldt bookkeeping gives

$$N(I') \geq N_{\text{on}}(I') + 2b, \quad N_{\text{on}}(I') := \sum_i m_i,$$

the on-line zeros counted with multiplicity. This is the coordinator's item (a) as stated.

The one normalisation fact. The only property of the vectors I use is

$$|u_i|^2 = \frac{1}{L^2 a} \sum_{0 \leq k < d} \hat{\phi}(t_i - \tau_k)^2 \leq \frac{1}{L^2 a} \sum_{k \in \mathbb{Z}} \hat{\phi}(t_i - \tau_k)^2 = 1,$$

by dropping nonnegative terms and applying the Poisson identity (P). Hence

$$t_{\text{on}} := \text{tr } M_{\text{on}} = \sum_i m_i |u_i|^2 \leq N_{\text{on}}(I').$$

Nothing whatever is assumed about the individual pair traces $t'_j = 2m_j \text{Re}(v_j^T v_j)_{\text{trunc}}$ on the truncated grid — they may be anything near the window edges or for deep zeros; only the total $t' := \text{tr } M_p = \text{tr } A - t_{\text{on}}$ enters. Here I depart from the coordinator's checklist explicitly: its item (c), “prove $s_j \geq 1 - o(1)$ for interior pairs, absorb edge pairs”, is not needed, and edge pairs require no separate treatment. Item (d), the request to confirm that $|u_i|^2 \leq 1$ is used nowhere else, is settled by the same observation.

The certificate. Applying (3.2) to $A = M_{\text{on}} + M_p$,

$$r \geq 2t_{\text{on}} + 4t' - 4b - \|A\|_F^2 = 2 \text{tr } A + 2(t' - 2b) - \|A\|_F^2,$$

and

$$\begin{aligned} t' - 2b &= \text{tr } A - t_{\text{on}} - 2b \\ &\geq \text{tr } A - N_{\text{on}}(I') - (N(I') - N_{\text{on}}(I')) \\ &= \text{tr } A - N(I'). \end{aligned}$$

Therefore

$$n_{\text{on}}^{\text{dist}}(I') \geq 4 \text{tr } A - 2N(I') - \|A\|_F^2. \quad (4.1)$$

This is cleaner than the coordinator's $2 \text{tr } A + 2t' - 4b - \|A\|_F^2$: it contains no pair data at all. It is the *certificate* in what follows — the lower bound for distinct on-line zeros that the argument actually certifies from the data of A .

The tail. From referee D's tail estimate,

$$\|E\|_{\text{op}} \leq \sum_{\rho \notin I'} m \|u_\rho\|^2 =: \theta_0 \leq 2A_0 L^2 \log(4T) T^{\lambda/2-1}$$

(in $\tilde{R}$ units, up to a factor $L/(L^2 a) < 1$), and the same sum bounds $|\text{tr } E|$ and the trace norm $\|E\|_1$, since E is a sum of blocks of trace norm at most $2m \|u_\rho\|^2$. So $\text{tr } A = \text{tr } \tilde{R} + O(\theta_0)$, and

$$\begin{aligned} \|A\|_F^2 &= \|\tilde{R} - E\|_F^2 \leq \|\tilde{R}\|_F^2 + 2\|E\|_{\text{op}} \|\tilde{R}\|_1 + \|E\|_1 \|E\|_{\text{op}}, \\ \|\tilde{R}\|_1 &\leq \|A\|_1 + \|E\|_1, \end{aligned}$$

with

$$\begin{aligned} \|A\|_1 &\leq \sum_{\text{on}} m_i |u_i|^2 + \sum_{\text{pairs}} 2m_j \|v_j\|_{\text{trunc}}^2 \\ &\leq N(I') \max(1, A(\delta)) \leq N(I') e^{L/2} = N(I') \left(\frac{T}{2\pi}\right)^{\lambda/2}, \end{aligned}$$

because $A(\delta) = \int \phi^2 e^{2\delta u} / \int \phi^2 \leq e^{\delta L} \leq e^{L/2}$ for $|\delta| \leq \frac{1}{2}$ and truncation only lowers $\|v\|^2$. Hence

$$\begin{aligned} \|A\|_F^2 &\leq \|\tilde{R}\|_F^2 + O(N T^{\lambda/2} \cdot L^2 l T^{\lambda/2-1}) \\ &= \|\tilde{R}\|_F^2 + O(N l^3 T^{\lambda-1}) = \|\tilde{R}\|_F^2 + o(N) \end{aligned}$$

for fixed $\lambda < 1$. I stress what is *not* used: no zero-density theorem, no bound on individual eigenvalues, no eigenvalue threshold θ ; (4.1) uses the rank and inertia of A exactly.

6.2 The constant two thirds and its sharpness

Prime side and the constant. The unconditional prime-side input is the one already refereed for Theorem 4 — its Step 2, equivalently the re-derivation agent's Theorems A/B and referee D's evaluation:

$$\text{tr } \tilde{R} = N(I) (1 + o(1)), \quad \|\tilde{R}\|_F^2 = N(I) t_\lambda (1 + o(1)),$$

where

$$t_\lambda := \frac{b_w + \lambda^2 J_w}{\lambda a_w^2} \longrightarrow \frac{1}{\lambda} + \frac{\lambda}{3}$$

as the taper parameter $\eta = 1/\log l \rightarrow 0$ (a_w, b_w, J_w being referee D's taper functionals), together with $N(I') = N(I)(1 + O(T^{-1/2}))$. Inserting these into (4.1),

$$n_{\text{on}}^{\text{dist}}(I') \geq (4 - 2 - t_\lambda - o(1))N(I) = \left(2 - \frac{1}{\lambda} - \frac{\lambda}{3} - o(1)\right)N(I),$$

and $N_0^{\text{dist}}(T - D_0, 2T + D_0) \geq n_{\text{on}}^{\text{dist}}(I')$ — the formula the coordinator wrote out from my (3.2). As $\lambda \rightarrow 1^-$ this is $2 - \frac{4}{3} = \frac{2}{3}$. It is positive iff $\lambda > 3 - \sqrt{6} = 0.5505$, the same threshold as Theorem 4; and for every $\lambda \in (0.55, 1]$ the new constant $2 - t$ beats Theorem 4's $2/t - 1$, because $(t-1)(t-2) \leq 0$ on $1 \leq t \leq 2$: at $\lambda = 1$, 0.667 against 0.500; at $\lambda = 0.9$, 0.589 against 0.417; at $\lambda = 0.8$, 0.483 against 0.324.

On the coordinator's item (e): optimising c in (3.1) rather than fixing $c = 2$ gives, with $x = t_{\text{on}}$ and $t = t_\lambda$,

$$r \geq \frac{(2N - x)^2}{tN} - 2(N - x),$$

which the adversary minimises at $x = (2 - t)N$ to the same value $(2 - t)N$, while at $x = N$ (no off-line zeros) it equals N/t , the Cauchy–Schwarz $3N/4$. So $c = 2$ is optimal at the extremal configuration and nothing is lost globally; and $c = 8/3$ would formally give $3N/4$ minus terms involving b — circular, as the coordinator noted.

On composition with the sibling’s kernel optimisation: the bound is $2 - t^*(\lambda)$ for whatever t^* the optimal taper achieves. It is the exact analogue of Montgomery’s $N_{\text{simple}} \geq (2 - F\text{-functional})N$, so — from memory, not pursued and not checked here — Montgomery–Taylor/Cheer–Goldston-type kernels (RH-simple 0.6725...) would transfer verbatim if their prime-side evaluation were available for the Gabor realisation. The sibling’s report says the admissible pair-correlation kernels are exactly $|\text{FT } w|^2$ with $w \geq 0$, a restriction relative to Montgomery–Taylor, so on that route $2/3$ improves to roughly 0.67 at best.

Theorem 4’ (claimed). Unconditionally, with the same analytic input as Theorem 4 — Weil’s explicit formula, Riemann–von Mangoldt, PNT-strength $\sum \Lambda^2/n$, Montgomery–Vaughan — plus von Neumann’s trace inequality:

$$\liminf_T \frac{N_0^{\text{dist}}([T, 2T])}{N([T, 2T])} \geq \frac{2}{3}, \quad \text{and for support parameter } \lambda, \quad \geq 2 - \frac{1}{\lambda} - \frac{\lambda}{3}.$$

The constant $2/3$ is sharp for the information $\{\text{tr } \tilde{R}, \|\tilde{R}\|^2, \text{block structure}\}$: take $2N/3$ simple on-line zeros on the lattice $(2\pi/l)\mathbb{Z}$ (mutually orthogonal for the sinc kernel at $\lambda = 1$) plus $N/6$ tight off-line pairs ($\delta \rightarrow 0$) on the remaining sites; then

$$\text{tr } \tilde{R} = N, \quad \|\tilde{R}\|^2 = \frac{2N}{3} + \frac{4N}{6} = \frac{4N}{3}, \quad n_{\text{on}}^{\text{dist}} = \frac{2N}{3}$$

— the brief’s extremal picture, spectrally identical to Montgomery’s RH-extremal “ $2N/3$ simples plus $N/6$ doubles”.

The comparisons I record are all from memory, with no lookup: Montgomery 1973 under RH gives $\geq 2/3$ simple; RH plus the integer inequality $(m - 1)(m - 2) \geq 0$ gives $\geq 5/6$ distinct (elementary — the brief’s own arithmetic; I believe it folklore or known but could not verify a citation offline; Bui–Heath-Brown-type papers quote 0.8466 distinct under RH via better kernels, consistent with $5/6 = 0.8333$ being the Fejér-kernel value); Cauchy–Schwarz gives $3/4$ distinct. Theorem 4’ is the $(m - 1)^2 \geq 0$ level of integrality, “ $\sum m^2 \geq 2N - N_{\text{dist}}$ ”, transplanted off the line: $2/3$ distinct on-line, no RH — the level the brief had itself written down for n_+ and set aside.

Simple zeros: no. On the coordinator’s item (f), sharpening to *simple* on-line zeros, the answer is negative. Rank counts distinct points, so directly one gets only

$$N_0^{\text{simple}} \geq 2n_{\text{on}}^{\text{dist}} - N_{\text{on}} \geq (3 - 2t)N = \frac{N}{3} \quad (\lambda = 1),$$

below Bui–Conrey–Young’s 0.407 (again from memory). I tried to sharpen (3.1) using that on-line masses are integers: $\|M_{\text{on}}\|^2 \geq \sum m_i^2 \geq 2N_{\text{on}} - N_{\text{on}}^{\text{simple}}$ is true, but step (i) of the proof consumes the eigenvalue slack $\sum_k (\mu_k - c)_+^2$ of M_{on} , which integer masses do not control — coincident or nearly coincident distinct simple zeros and a double zero are indistinguishable to the spectrum of M_{on} . So: distinct, not simple. Counting on-line zeros with multiplicity, $N_{\text{on}}(I') \geq n_{\text{on}}^{\text{dist}} \geq 2N/3$ as well (against Pratt et al.’s $5/12$, from memory).

Numerics. For the record I collect the certificate computation on true zeros just run (zero side exact from the 100,000 cached ordinates, $T = 1000$, $I' = [970, 2030]$, $N(I) = 868$, $N(I') = 921$,

everything in units of $N(I)$, NEW the certificate from (3.2), OLD Theorem 4's): true zeros NEW +0.612 / OLD +0.380 with $\|A\|^2/\text{tr } A = 1.387$ against the law 1.333; at $\lambda = 0.9$, +0.515/ +0.286; at $\eta = 0.05$, +0.634/ +0.402; all doubled -1.548 (vacuous); every fourth ordinate a pair at $\delta = 0.001$: +0.116/ +0.036 with $n_{\text{on}}^{\text{dist}}/N = 0.530$, and NEW +0.107, -0.034 , -0.782 at $\delta = 0.05, 0.2, 0.4$; the adversary lattice +0.534/ +0.356 with $n_{\text{on}}^{\text{dist}}/N = 0.657$ — near-extremal, the gap being the finite- l excess 0.385 versus 0.333 — and +0.517 at pair depth 0.1; dephased deep couples -0.53 , -2.53 . On true zeros the new certificate is $0.61\text{--}0.63N$ (law: $2 - 1.387 = 0.613$, tending to $2/3$ as $l \rightarrow \infty$) versus $0.38\text{--}0.40N$ for Theorem 4, and (3.2) holds in every synthetic world, as it must, deep or dephased pairs only making $\|A\|^2$ larger.

6.3 The brief's other levers and higher moments

Lever (a) and lever (b). Lever (a), the negative spectrum *alone* — adding $n_- \leq n_{\text{pair}}$ and $\text{tr } R_-$ to the Cauchy–Schwarz count: as the brief anticipated, it is useless against tight pairs, since the negative eigenvalue $q = m(A - 1) \rightarrow 0$ as $\delta \rightarrow 0$, and the Cauchy–Schwarz-plus-inertia extremal is not moved by it. What works is negative spectrum and rank *jointly* — that is the theorem: step (i) charges the negative part of M_p against the large eigenvalues of M_{on} via von Neumann, step (ii) charges the positive part via $n_+ \leq b$. The tight pair is not beaten (it cannot be) but is correctly priced at the equivalent of $4m^2$, at least 2 per zero against 1 per simple on-line zero — integrality.

On lever (b): under RH, $\sum m^2 \leq 4N/3$ with integer m gives $N_{\text{dist}} \geq 5N/6$ (k doubles plus $N - 2k$ simples: $N + 2k \leq 4N/3$) and $N_{\text{simple}} \geq 2N/3$, against $3/4$ from Cauchy–Schwarz. Unconditionally the $(m - 1)^2 \geq 0$ level is recoverable robustly — that is (3.1) — but the $(m - 1)(m - 2) \geq 0$ level is not: eigenvalues of interacting simple zeros fill $(0, 2)$, and “no eigenvalue in $(1, 2)$ ” is false already under RH+GUE, where the Gram spectrum has mean 1 and variance $1/3$ at $\lambda = 1$. (Editorially: this confirms the brief's forecast that the $(\lambda - u)(\lambda - 2u) \geq 0$ level is robustly false; what the brief did not foresee is that the lower level, which it had found useless when aimed at n_+ , carries the whole result when aimed at rank.)

Higher moments, (b)(ii). On the zero side

$$\text{tr } \tilde{R}^k = \sum_{\gamma_1, \dots, \gamma_k} \prod_j \tilde{\Phi}(\gamma_j - \gamma_{j+1}),$$

exact via (P) with complex γ allowed — a Rudnick–Sarnak-type smooth k -correlation statistic whose Fourier support, in Montgomery's α -units, reaches $\sum_j |\xi_j| = k\lambda$ for k even and $(k - 1)\lambda$ for k odd. From memory — I did not fetch it — Rudnick–Sarnak 1996 (Duke 81) prove the n -level correlations are GUE for $\sum |\xi_j| < 2$ and state explicitly that RH is not assumed; on the prime side this is “only diagonal solutions contribute”, the off-diagonal being killed by Montgomery–Vaughan iff $X^{k/2} \ll T$. So: $k = 2$ needs $\lambda < 1$ (used); $k = 3$ needs $(k - 1)\lambda < 2$, i.e. $\lambda < 1$, admissible on the whole useful range (the *ppp* diagonal $n_3 = n_1 n_2$ forces prime powers of a single prime, lower order; main terms from μ^3 and $\mu \cdot PP$); $k = 4$ needs $4\lambda < 2$, i.e. $\lambda < 1/2$, inadmissible where useful, and for $\lambda < 1/2$ the dimension $d = \lambda N < N/2$ kills any on-line certificate anyway, since $n_+ \leq d$.

But an odd moment cannot improve a lower bound for a count on $[0, \infty)$: given (m_1, m_2) , the infimum m_1^2/m_2 of the number of positive eigenvalues is approached for any prescribed $m_3 \geq m_2^2/m_1$, by putting mass ε at height $h \rightarrow \infty$ (which feeds m_3 and nothing else); dually, an LP certificate $p(\mu) = c_1\mu + c_2\mu^2 + c_3\mu^3 \leq 1$ on $(0, \infty)$ needs $c_3 \leq 0$ and is then dominated by its quadratic part, and with d known a constant term c_0 must be ≤ 0 because $p \leq 0$ on $(-\infty, 0]$. The third moment would help only together with an a-priori bound $\lambda_{\text{max}}(\tilde{R}) = O(1)$, i.e. a pointwise bound for $|\sum_{n \leq X} \Lambda(n)n^{-1/2-i\tau}|$ on windows of length $1/L$ —

Lindelöf-strength, unavailable; the re-derivation agent’s Remark 7.4 is the same obstruction. So $\text{tr } \tilde{R}^3$ is unconditionally computable and useless, while $\text{tr } \tilde{R}^4$ would be useful and is not computable for $\lambda > 1/2$. Departing from the brief’s DO-list, I therefore do *not* compute the GUE constants $m_3(\lambda)$, $m_4(\lambda)$ — moot — and defer to the HL agent’s report for higher-level material. The moment route cannot reach $2/3$; the structural route does, with $k = 2$ only.

My verdict. The block facts are confirmed exactly. The new ingredient is Theorem (3.1) — elementary, proved above, numerically unfalsified — plus the bookkeeping (4.1), which uses only $\text{rank } M_{\text{on}} \leq n_{\text{on}}^{\text{dist}}$, $n_+(M_p) \leq n_{\text{pair}}^{\text{dist}}$, $|u_i|^2 \leq 1$, $N(I') \geq N_{\text{on}} + 2n_{\text{pair}}$, and the Theorem-4/referee-D tail bound. Given the refereed prime-side asymptotics ($\text{tr } \tilde{R} \sim N$, $\|\tilde{R}\|^2 \sim (1/\lambda + \lambda/3)N$, $\lambda < 1$), the constant $1/2$ of Theorem 4 becomes $2/3$ for distinct zeros on the critical line, unconditionally, and sharp for this information; simple-on-line is only $1/3$ by this route. Referees C’s and D’s extraordinary-strength caveat applies verbatim, now even more so: everything rests on Step 2’s upper bound for $\|\tilde{R}\|_F^2$. Nothing in the theorem or the bookkeeping is asymptotic or delicate; the one place $\lambda < 1$ enters the application is $\|E\|_{\text{op}}\|A\|_1 = O(Nl^3T^{\lambda-1})$. With that the report goes to the referees as final.

My closing message to the orchestrator puts the verdict first: “the pair-block/negative-spectrum lever reaches exactly $2/3$ (distinct on-line), via a new elementary inequality; higher moments cannot.” The rest recapitulates, with nothing new computed or looked up: the block facts confirmed exactly (the Poisson identity for complex arguments, $v^T v = L^2 a$ real and depth-independent, $\|v\|^2 = L^2 a A(\delta)$, pair eigenvalues $mL^2 a(1 \pm A)$), cross terms sign-indefinite so that termwise diagonal dominance is false; lever (a) alone useless against tight pairs, as expected, but negative spectrum plus rank jointly give (3.1), verified numerically with the linear-algebra minima matching it with equality; with $c = 2$ and the Riemann–von Mangoldt bookkeeping,

$$n_{\text{on}}^{\text{dist}}(I') \geq 4 \text{tr } A - 2N(I') - \|A\|_F^2 \geq \left(2 - \frac{1}{\lambda} - \frac{\lambda}{3} - o(1)\right)N \longrightarrow \frac{2N}{3},$$

using only $|u_i|^2 \leq 1$, inertia and rank of blocks, the tail bound (needing $\lambda < 1$) and the already-refereed prime-side traces; numerics at $T = 1000$ on true zeros giving $0.61\text{--}0.63N$ against Theorem 4’s $0.38\text{--}0.40N$, every synthetic and adversarial world satisfying the inequality and the lattice-plus- $N/6$ -tight-pairs configuration extremal; under RH integers give $5/6$ distinct (likely folklore; citation unverified offline) while unconditionally only the $(m-1)^2$ level transfers, and simple-on-line is only $1/3$; $\text{tr } R^3$ admissible for $\lambda < 1$ but provably unable to improve the count without a λ_{max} bound, $\text{tr } R^4$ needing $\lambda < 1/2$ where $d < N/2$ kills everything. What could go beyond $2/3$: better kernels (the sibling’s route, about 0.67) or the $(m-1)(m-2)$ integrality level, which interactions destroy.

Appendix

Literature along the way

A Literature along the way

This appendix ties the ideas of Parts I and II to the published literature: which works were in play at each step, whether they were recalled from the model’s prior knowledge or actually retrieved during the campaign, by which agent and when, and what role each played. It is reconstructed from the campaign’s complete logs, in which every web search, every download, every file read and every message between the coordinator (Part I calls it the orchestrator) and an agent is recorded. An influence that came only through the model’s training is invisible to such logs, and we say so where it matters. Unless stated otherwise, times are given relative to the moment the one-half run of Part I returned its result; the two-thirds run of Part II was launched about seven and a half hours after that. The appendix traces provenance; it certifies no mathematics.

A.1 How literature could enter

Every agent in the campaign could search the open web and download papers — arXiv abstracts, PDFs and L^AT_EX sources, open-access journals, EuDML, zbMATH and public abstract pages — and many did: across the three days roughly a hundred arXiv papers and some ninety other scholarly pages were retrieved, all from the open web. The two runs narrated in Parts I and II retrieved nothing: neither made a network request of any kind. Nor did the coordinator, at any point in the campaign; and it read almost nothing from disk — over the whole campaign a handful of files, all written by agents that had themselves used no network. Consequently the only route by which downloaded material could travel from one agent to another was: the downloading agent’s files, if a later agent was told to read them; or the short final message with which each agent reported back to the coordinator, and whatever of it the coordinator then put into a later brief. Both routes are visible in the logs, and both are followed below. The inputs of the one-half run were the coordinator’s brief, the report, notes and code of one earlier run (R4), the reports of two others (N6, frontier-1) — none of which had used the network — and two tables of zeta zeros used for validation. The inputs of the two-thirds run were its brief and one later message, the one-half run’s proof file, the verdict files of referees C and D, the re-derivation agent’s proof, and, once, the first six thousand bytes of its sibling’s report; of these only referee C’s verdict carried anything derived from a download (a paragraph identifying the one-half run’s prime-side trace with a published theorem as read from the PDF, two sentences of published proportions of distinct and simple zeros, and an arXiv number), and referee D’s verdict names the same paper once — none of it bearing on the lemma or the bookkeeping of Part II.

A.2 Before the one-half run: the frame

The proposal to treat the condition space as a Pontryagin space Π_κ and to use the Pontryagin (1944)/Kreĭn–Langer bound on non-real spectrum to bound off-line zeros is the coordinator’s,

stated in the brief that launched the one-half run (Part I, §1). The words “Pontryagin”, “Kreĭn–Langer” and “ Π_κ ” occur in essentially no earlier record of any agent in the campaign (one unrelated use of “Pontryagin” aside) and in no document any agent had downloaded; R4’s report, which prompted the idea by finding its operator indefinite (“Kreĭn, not Hilbert”), cites nothing and was produced without network access, as were the reports R4 itself had read. The frame was not read from any paper. One qualification the logs impose: notes from an earlier session of the same project, which the coordinator had skimmed at the very start of the campaign as a list of one-line summaries, already described Weil’s 1952 positivity criterion “recast as a Gram/signature statement” on a finite, prime-computable matrix; whether that exposure, some twenty hours earlier, cued the reframing cannot be decided from the logs. The brief’s pointers for the conjectured identity (negative index of W = number of off-line zeros) — Bombieri’s 2000 note on Weil’s quadratic functional, Yoshida’s work of the 1990s, Burnol — are recalled and explicitly hedged (“Yoshida? Bombieri?”); none had been retrieved by anyone at that point. Weil’s criterion and the explicit formula are classical and recalled; the normalisation is the textbook one that N6’s earlier run had validated numerically. The Turán/Nazarov-type lemmas and the Selberg–Levinson–Conrey benchmark in the brief (“can the negative-index route REPRODUCE a positive-proportion result?”) are likewise recalled; the run showed the former bound the wrong operator and used the latter only as the target it then met.

A.3 The one-half run (Part I): memory only

Every paper the one-half run names it names from memory, and it says so in its own notes (“I do not know a reference; the closest are . . .”, “Literature (from memory, unverified offline): Yoshida 1992 . . ., Bombieri 2000 . . ., Burnol”, “KNOWN RESULTS FOR COMPARISON: . . . (from memory)”). When the dual argument of Part I, §5 produced the prime-side lower bound $(\text{tr } R)^2 / \text{tr } R^2 \rightarrow \frac{3}{4}N$, the run recognised the number as the classical consequence, by Cauchy–Schwarz, of Montgomery’s 1973 RH-conditional theorem (“ $\geq \frac{3}{4}$ distinct”) and the evaluation as Montgomery’s prime-side computation via the Montgomery–Vaughan mean value theorem — both recalled; by its own account the inequality was applied from memory, with no way to check its constant during the run. Nothing the run had read mentions Montgomery, pair correlation or $\frac{3}{4}$; the only agents downloading anything during those hours were working on Epstein, Dedekind and Beurling zeta functions for unrelated branches, into files this run never opened. In its red-team notes (Part I, §7), about a quarter of an hour before returning, the run listed the closest literature it could recall — Montgomery 1973, Gallagher–Mueller, and “Baluyot–Goldston–Suriajaya–Turnage–Butterbaugh’s unconditional pair-correlation $F(\alpha)$ (2023/24), where — as far as I recall — no unconditional proportion *on the line* is drawn” — and ranked “too short to be new” as its top risk. This is a pointer, not a citation: no title, identifier or theorem statement was available to it; its role was to tell the referees where to look. The records it quotes (5/12, Pratt–Robles–Zaharescu–Zeindler; 0.407 simple on the line, attributed to Bui–Conrey–Young) and the line “Under RH: $\geq 2/3$ simple (Montgomery), $\geq 3/4$ distinct (C-S)” are marked “from memory”.

A.4 The referee round and the literature agent

Three blind referees (from about four minutes after the run returned). Each was given the run’s proof file and notes and one joint of the argument to break; they were told not to read each other’s work, and did not. Referees A and C turned to the web about 80 minutes after the run returned, referee B about two and a half hours after; all three searched for the four author names the run had recalled, located *An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function* (Baluyot, Goldston, Suriajaya,

Turnage-Butterbaugh; arXiv 2306.04799, Acta Arith. 2024), and read it (abstract page or PDF). What this established: the prime-side asymptotic the run had evaluated is, in smoothed form, Theorem 1 of that paper — published and unconditional — so the entire burden of the one-half claim sits on that single evaluation; and the same paper’s introduction gave referee C the standard list of records (Pratt–Robles–Zaharescu–Zeindler 41.7%; Montgomery, Montgomery–Taylor and Chirre–Gonçalves–de Laat under RH). Referees A and B also opened the abstract of the sequel on proportions of simple and critical zeros (arXiv 2501.14545), and referee A the abstract of the pair-correlation-conjecture paper (arXiv 2503.15449); a search by referee C returned Farmer’s 1995 *Counting distinct zeros of the Riemann zeta-function*. All of this is *verification and positioning*. Two things the referees added were their own: referee C’s observation that rank in place of inertia gives $\geq \frac{3}{4}$ distinct, hence $\geq \frac{1}{2}$ simple (its code computed the rank statistic about twenty minutes before its first search, and it had read the run’s own line on $\frac{2}{3}$ simple and $\frac{3}{4}$ distinct under RH an hour earlier; the integer step it uses appears in no text it downloaded, though whether the downloaded abstract’s discussion of proportions of simple zeros prompted the *phrasing* cannot be decided from the logs); and referee A’s discovery of the mass-matrix gap and its “never orthonormalise” repair, which concern Gram matrices and owe nothing to the pair-correlation papers it had opened. Referee B derived the exact reduction to the Montgomery–Vaughan mean value theorem about twenty minutes before its first search and used the download only to identify the published theorem. Referee D, later, worked without network access; the exact Poisson identity it proved had been suggested by the coordinator in its brief. This is the model case for the whole chain: a paper named from memory inside a run, located on the web by the next agents, and used to verify and to position.

The blind re-derivation (launched about two and three-quarter hours after). Forbidden to read the run’s proof, it re-derived the prime-side traces with error terms in two repaired normalisations, with no network access, citing nothing it did not recall.

The literature agent (launched about two and three-quarter hours after; reported about four hours after). This is where papers were actually read. It downloaded and read the L^AT_EX sources of arXiv 2306.04799, 2501.14545 (Baluyot–Goldston–Suriajaya–Turnage-Butterbaugh, *Pair correlation of zeros of the Riemann zeta function I*), 2503.15449 (Goldston–Lee–Schettler–Suriajaya, *Pair correlation conjecture for the zeros of the Riemann zeta-function I: simple and critical zeros*), 2511.20059 (Goldston–Suriajaya, *Zeta zeros on the critical line*) and 2603.28104 (Goldston–Suriajaya, *Zeta zeros in a narrow vertical box*); then Farmer (1995), the abstract of Wu’s *Distinct zeros of the Riemann zeta-function* (Q. J. Math. 2015; more than 66% distinct), the abstract of Bui–Heath-Brown (arXiv 1302.5018), Conrey–Ghosh–Gonek’s *Simple zeros of the Riemann zeta-function* (1998), Bombieri’s 2000 note via EuDML, and two of Suzuki’s screw-function papers (arXiv 2606.09096, 2206.03682). Its findings, by role:

- *Verification.* The run’s $\text{tr } R^2$ is literally the Fejér-kernel sum of the 2024 paper’s Theorem 1 (equivalently “Aryan’s lemma” in Goldston–Suriajaya, $(\frac{4}{3} + o(1))N$ unconditionally); the published works lose unconditionality only when they pass to diagonal terms by termwise kernel positivity (the thin-box hypothesis), which the inertia argument does not do.
- *Positioning.* Goldston–Suriajaya state the removal of RH from Montgomery’s argument — which would give $\frac{2}{3}$ simple and on the line — as an open problem; no published work combines the *positive* index of the compressed Weil form with a pair-correlation second moment to count zeros on the line; Bombieri (2000) already has the negative-index half of the run’s Theorem 1 for large truncations, so that half is prior art; and the records against which the claim would be measured (Wu 66.04% distinct — a number obtained from the download, the agent’s own search query having guessed a different one; 5/12 on the line). The two Suzuki papers were checked for a precedent linking signature to off-line zeros and

yielded none.

- *Formulation.* Reading Goldston–Lee–Schettler–Suriajaya’s and Goldston–Suriajaya’s “horizontal multiplicity” $H(\gamma)$ (the number of zeros, with multiplicity, on the horizontal line $t = \gamma$; $H(\gamma) = 1$ forces a zero to be simple *and* critical), the literature agent observed that the zeros certified by the run’s certificate are $H(\gamma) = 1$ zeros — “the Cauchy–Schwarz-weakened form” of Goldston–Suriajaya’s Theorem 2. The coordinator put this into the brief for the write-up (nearly five and a half hours after the run) as something to verify from the inertia bookkeeping and to state only if proven; the paper-writing agent verified it by its own one-line count and made “simple and on the critical line” the statement of the one-half theorem in the draft. The proof is the run’s; the *form of the statement* is the one place on the whole chain where a downloaded paper contributed wording that survives.

The paper-writing agent, launched nearly five and a half hours after the run returned, read the Goldston–Suriajaya source and the first pages of the other downloads directly from the literature agent’s files, for its bibliography and related-work section. No agent anywhere in the campaign made a network request during the three and three-quarter hours that followed the literature agent’s last download — the stretch in which the referee for the mass-matrix gap reported, the paper-writing agent drafted, and the two-thirds push was decided.

A.5 The two-thirds push (Part II and its siblings)

The target. Two thirds as Montgomery’s RH-conditional constant had been in the coordinator’s own words since shortly after the one-half run returned (its brief to referee B, six minutes after, asks whether the classical RH result “is Montgomery’s $\geq 2/3$ simple (RH) ... — where does $3/4$ come from?”), and the run’s proof file records “under RH: $\geq 2/3$ simple (Montgomery)”. The Goldston–Suriajaya sentence — that an RH-free version of Montgomery’s computation would give two thirds simple and on the line, and that this is open — reached the coordinator in the literature agent’s report about four hours after the run returned and was plainly live when the target was set: about seven and a quarter hours after the run returned, answering the question of what would come next, the coordinator wrote that one half to two thirds “is a real, well-posed target (it’s exactly the Cauchy–Schwarz loss versus Montgomery’s RH-conditional $2/3$)” and, in the same message, suggested sending the draft to the authors of the open problem, “since it’s their theorem being used and their open problem being claimed”. Three minutes later the human typed “Push it to $2/3$ ”; the two-thirds run’s brief followed fifteen minutes after that. The logs cannot say what weight the published open-problem statement carried against the recalled constant and that instruction; they do show that it supplied no technique — the route of Goldston and Suriajaya (termwise positivity under a box hypothesis) is precisely the obstruction the literature agent had diagnosed, and Part II goes the other way. The draft’s later framing sentence, “matching Montgomery’s RH-conditional 1973 constant unconditionally ... Goldston–Suriajaya’s open problem”, was dictated by the coordinator with the instruction to cite their statement of the problem precisely.

The integer template and $5/6$. The arithmetic in the two-thirds brief — Montgomery’s use of integrality, $\sum m^2 \geq \sum (2m - 1)$, giving $2/3$ simple under RH; the observation that integrality also gives $5/6$ distinct; and “ $2 \cdot (5/6)N - N = 2N/3$ ✓✓ EXACTLY THE TARGET” — was derived by the coordinator while writing the brief (the text carries its own running check marks and the aside “is $5/6$ -distinct-under-RH known? surely — Montgomery’s paper or folklore; CHECK literature”), seven minutes after it had set out the two levers. No text upstream of the coordinator contains it. One coincidence is worth recording: the literature agent’s findings file had, for about four hours by then, a line noting that integer combinatorics gives $\geq 5/6$ distinct under RH, attributed there to Conrey–Ghosh–Gonek; the coordinator never read that

file, and the agent’s report to it did not carry the line. (The optimal-window sibling read the file two minutes after the two-thirds run was launched, its own brief having already set “target $c^* \geq 5/6$ ”. When the draft briefly attributed the $5/6$ to Conrey–Ghosh–Gonek, a cold-read referee of the draft (the cold referee below), working from the downloaded texts, re-attributed it to Montgomery plus integrality.)

Pair blocks and the Poisson identity. The block facts the brief asked the run to verify — $v^T v = L^2 a$ real and depth-independent, $\|v\|^2 = L^2 a A(\delta)$, eigenvalues $mL^2 a(1 \pm A)$ — were the coordinator’s own quick algebra, done on top of referee C’s block-structure lemma and referee D’s exact Poisson identity $\sum_k \hat{\varphi}(z - \tau_k) \hat{\varphi}(z' - \tau_k) = L \Phi(z - z')$, both obtained without network access. Nothing here has a published source among the run’s inputs.

The trace inequality. The rank–trace lemma of Part II and its five-line proof are the run’s. The one classical tool it invokes, von Neumann’s trace inequality, was recalled: the phrase occurs nowhere in the campaign’s logs before the run’s report, and the brief had offered Weyl’s inequalities, Cauchy interlacing and Ky Fan’s principle instead. The bookkeeping of the application (Part II, §6) uses only quantities already established in the campaign (the refereed prime-side traces, referee D’s tail bound) and Riemann–von Mangoldt.

Higher moments, and why they were closed. The brief’s alternative lever — prime-side $\text{tr } R^3$, $\text{tr } R^4$ in a restricted-support regime, with “Rudnick–Sarnak 1996 / Hughes–Rudnick: which are unconditional? — CHECK” — was recalled by the coordinator. The run settled its own case itself ($\text{tr } R^3$ admissible for $\lambda < 1$ but unable to improve the count; $\text{tr } R^4$ needs $\lambda < 1/2$). The higher-moments sibling (“the HL agent” of Part II), launched forty-one minutes after the two-thirds run, proved the general no-go (“Theorem U”) without downloading anything; its only network use was two web searches, about ten hours after the one-half run returned, for the abstracts of Goldston–Lee–Schettler–Suriajaya (arXiv 2503.15449: the pair correlation conjecture for all α implies 100% simple and critical) and Goldston–Suriajaya, used to position its conditional theorem as complementary to theirs; the “PCC $\Rightarrow$ 100%” fact had already been relayed to it in its brief, from the literature agent’s report. The generalisations agent and its two literature sub-agents — launched about eight and a quarter hours after the one-half run returned, with their downloads falling between eight and two-thirds and nine and three-quarter hours after — fetched the Dirichlet- L and record-proportion papers listed in the table, chiefly to audit where GRH enters Chandee–Lee–Liu–Radziwiłł and what the unconditional support range of Hughes–Rudnick is; the outcome was a hedged remark and one line of records in the draft’s section on variants.

The optimal window (the sibling). The optimal-window sibling’s brief, written five minutes before the two-thirds run’s, carried from the coordinator’s memory Montgomery’s functional $c(w) = (\int w)^2 / (\int w^2 + \iint |a - b| w(a)w(b))$, the Euler–Lagrange equation $w'' + 2w = 0$ with the $\cos \sqrt{2}$ ansatz, the sentence “Montgomery–Taylor (1974?) improved $2/3$ -simple to 0.6725 via optimized kernel”, and the pointer to Carneiro–Chandee–Littmann–Milinovich’s Crelle paper with the instruction “(recall precisely; fetch arXiv 1406.5462 if network works)”. That arXiv number occurs nowhere earlier in any log of the campaign: it was typed from memory, correctly. (The number 0.6725 itself had first been written by the coordinator, from memory and with a question mark, in the literature agent’s brief about two and three-quarter hours after the run returned; the literature agent’s report later confirmed it from arXiv 2501.14545.) The sibling read its inputs — including the literature agent’s findings file, which lists “Montgomery–Taylor: .6725; Cheer–Goldston .672753; Chirre–Gonçalves–de Laat: .679” as names and numbers only — and then solved the variational problem

itself: its solver, written nineteen minutes after its launch, already contains the closed form $c_\lambda^* = \sqrt{2} \tan(\lambda/\sqrt{2}) / (1 + (\lambda/\sqrt{2}) \tan(\lambda/\sqrt{2}))$ with optimiser $\cos(\sqrt{2}\lambda s)$; four minutes later it printed $c^* = 0.7532961$, $2c^* - 1 = 0.50659$, and three minutes after that its report identified $2 - 1/c^* = 0.6725007$ as “exactly the Montgomery–Taylor constant”. Only then, thirty-one minutes after its launch (about eight hours after the one-half run returned), did it make its first network request: one arXiv API call for 1406.5462 and 1810.08843 — the second number, Chirre–Gonçalves–de Laat’s semidefinite-programming paper, supplied from its own memory — followed by both e-prints. From the Carneiro–Chandee–Littmann–Milinovich source it read Corollary 15 and the remark giving $N^*(T) \leq (2^{-1/2} \cot 2^{-1/2} + \frac{1}{2} + o(1))N(T)$, checked that this equals its $1/c^*$, and recorded that their extremal kernel lies inside its admissible class, so that the restriction $\hat{k} \geq 0$ costs nothing; from Chirre–Gonçalves–de Laat it took only the bound 1.3208, to estimate what signed weights could buy, and classified that as beyond the method. The downloads thus verified an identification already made and supplied the optimality remark and the cotangent closed form that later appear in the draft’s Theorem D; none of it bears on the two-thirds argument, and none of it reached the two-thirds run, whose one look at the sibling’s report took in only its first six thousand bytes, ending inside the statement of the extremal problem and before the literature section the sibling appended later.

The interruption. Three and a quarter hours after its launch, immediately after the lemma and its proof had been written to its report, the two-thirds run was cut off by an infrastructure error; the coordinator, having read the partial report and checked the proof line by line, resumed it seven minutes later with the message reproduced in Part II, §5, which is where the consequence $(2 - 1/\lambda - \lambda/3)N \rightarrow \frac{2}{3}N$ was first written out. Nothing was fetched or read from outside the campaign in between. The two referees of the lemma that followed worked without network access.

A.6 What reached the paper draft from downloaded sources

All of it is positioning or bibliography, apart from the one formulation already described, and all of it entered through the paper-writing agent and the cold referee rather than through either run: the bibliography’s journal data, mined from the reference lists inside the downloaded e-prints and completed by five bibliographic web searches the paper-writing agent made about eleven hours after the one-half run returned (journal data for Wu, Bui–Conrey–Young, Feng and Farmer, Montgomery–Vaughan’s 1974 Hilbert-inequality paper, Yoshida 1992); the records paragraph of the introduction, including Wu’s 66.04% distinct and Farmer’s bound; the remark relating $\text{tr } G^2$ at $\lambda = 1$ to “Aryan’s lemma” as it appears in Goldston–Suriajaya’s narrow-box paper and to Theorem 1 of Baluyot–Goldston–Suriajaya–Turnage–Butterbaugh, and the courtesy that the prime-side input is used “in the form given to it” in those papers; the related-work subsections on the work of Goldston and his coauthors and on Goldston–Lee–Schettler–Suriajaya; Bombieri’s 2000 negative-index observation cited as prior art; Theorem D’s optimality sentence; two of the cold referee’s fourteen corrections that were sourced from downloaded text (quote Goldston–Suriajaya’s Theorem 2 exactly; attribute 5/6 to Montgomery plus integrality); and, when the main theorem became two thirds, the framing sentence about Montgomery’s constant and Goldston–Suriajaya’s open problem. On the whole chain from the brief of Part I to the two-thirds revision of the draft, no downloaded paper supplied a lemma, an inequality, a constant or a proof step: not the inertia identity, the double count, the Cauchy–Schwarz count, the coefficient-coordinate repair, the Poisson identity, the Montgomery–Vaughan reduction, the variational c^* , the rank–trace lemma or its bookkeeping. What the model knew from training — evidently including Weil’s criterion, Pontryagin–Kreĭn theory, Montgomery’s, Montgomery–Vaughan’s and Montgomery–Taylor’s results, the Carneiro–Chandee–Littmann–Milinovich paper down to its arXiv number, and the existence of the 2023/24 unconditional

pair-correlation paper — is outside what logs can trace.

A.7 The works, in one table

The table lists every work that was downloaded or opened by an agent on the chain of Parts I and II and its write-up (the three blind referees and referee D; the literature, re-derivation and paper-writing agents; the two-thirds run, its optimal-window and higher-moments siblings and the two referees of its lemma; the generalisations agent with its two literature sub-agents; the cold referee), together with the principal works that were recalled and used but never fetched. “When” is in hours after the one-half run returned; the two-thirds run was launched at +7.6 h. Roles: *recalled only*; *recalled, then verified at source*; *verification* (used to check a claim already made in the campaign); *positioning* (related work, records, bibliography); *formulation* (wording of a statement entered); *frame* (named the target as an open problem); *dead end*.

authors	short title	arXiv / venue	fetches by; when	role
<i>Recalled and used, never fetched on this chain</i>				
Weil (1952); Pontryagin (1944), Kreĭn–Langer	explicit formula and positivity criterion; operators on Π_κ spaces	—	recalled by the coordinator in the one-half run’s brief; the terms occur in essentially no earlier record, and in no download	recalled only — the frame of Part I
Yoshida (1992); Burnol	Weil positivity on small support; Sonine spaces	—	recalled by the coordinator (“Yoshida? Bombieri?”) and by the run (“from memory and unverified offline”); Yoshida’s journal data searched by the paper-writing agent (+11 h)	recalled only — frame pointers (Bombieri below)
Turán; Nazarov	zeros and sign changes of exponential sums	—	recalled by the coordinator	recalled only (proposed tool; shown to bound the wrong operator) recalled only
Montgomery (1973)	The pair correlation of zeros of the zeta function	Proc. Symp. Pure Math. 24	recalled by the run on recognising $\frac{3}{4}$; by the coordinator ($\geq \frac{2}{3}$ simple under RH, +6 min; the integer template in the two-thirds brief, +7.6 h)	
Montgomery– Vaughan (1974)	Hilbert’s inequality	J. London Math. Soc.	recalled by the run; reduction re-derived by referee B ($2\frac{1}{4}$ h, before its first search) and by the re-derivation agent (no network); journal data searched by the paper-writing agent (+11 h)	recalled only (the analytic input)

authors	short title	arXiv / venue	fetched by; when	role
Gallagher– Mueller	pair correlation and simple zeros	—	recalled by the run among “the closest” works; by the coordinator in the higher-moments brief	recalled only
Montgomery– Taylor (in Montgomery 1975); Cheer–Goldston (1993)	the 0.6725 / 1.3275 simple-zero constant; optimised kernels	—	recalled by the coordinator (0.6725 with a “?” and Cheer–Goldston at $2\frac{3}{4}$ h; Montgomery–Taylor at $7\frac{1}{2}$ h); confirmed in arXiv 2501.14545 and 1406.5462	recalled, then verified at source
Rudnick–Sarnak (1996); Ky Fan, Weyl, Cauchy interlacing; von Neumann’s trace inequality; Markov–Kreĭn Selberg, Levinson, Conrey; Bui– Conrey–Young; Feng; Pratt–Robles– Zaharescu– Zeindler	restricted-support n -level correlations; eigenvalue inequalities	—	recalled (the two-thirds brief; von Neumann by the two-thirds run itself)	recalled only
	proportion-on-the-line theorems and records (5/12; “0.407 simple”)	—	recalled by the coordinator (benchmark, from the campaign’s first hour) and by the run (“for comparison ... from memory”); exact figures later checked by the paper-writing agent’s searches (+11 h) and the records sub-agent (arXiv 1802.10521, +8 $\frac{3}{4}$ h)	recalled; positioning
<i>Downloaded after the one-half run returned: the referees, the</i>			<i>literature agent, the paper-writing agent</i>	
Baluyot– Goldston– Suriajaya– Turnage– Butterbaugh (2024)	An unconditional Montgomery theorem for pair correlation of zeros of the Riemann zeta function	2306.04799	named from memory by the run (four author names, no title or number); referees C, A ($1\frac{1}{3}$ h), referee B ($2\frac{1}{2}$ h), literature agent e-print ($2\frac{5}{6}$ h)	recalled, then verified at source (the run’s asymptotic = Theorem 1); positioning (records paragraph)
Baluyot– Goldston– Suriajaya– Turnage– Butterbaugh (2025)	Pair correlation of zeros of the Riemann zeta function I: proportions of simple zeros and critical zeros	2501.14545	referee A abstract ($1\frac{1}{3}$ h), referee B abstract ($2\frac{1}{2}$ h), literature agent e-print ($2\frac{5}{6}$ h), cold referee (10 h)	verification; positioning (the literature agent confirmed the recalled .6725 from its Theorem 2)
Goldston–Lee– Schettler– Suriajaya (2025)	Pair correlation conjecture for the zeros of the Riemann zeta-function I: simple and critical zeros	2503.15449	referee A abstract ($1\frac{1}{2}$ h), literature agent e-print ($2\frac{5}{6}$ h; horizontal multiplicity read 3 h), higher-moments sibling search (10 h)	formulation (“simple and on the line”, one-half theorem); positioning (“PCC $\Rightarrow$ 100%”)

authors	short title	arXiv / venue	fetches by; when	role
Goldston– Suriajaya (2025)	Zeta zeros on the critical line	2511.20059	literature agent e-print (3 h); paper-writing agent read that copy (6 h); records sub-agent ($8\frac{5}{6}$ h); cold referee (10 h)	frame (removing RH from Montgomery’s proof stated as open); formulation; positioning
Goldston– Suriajaya (2026)	Zeta zeros in a narrow vertical box	2603.28104	literature agent e-print ($3\frac{1}{2}$ h)	frame (with the above); verifica- tion/positioning (“Aryan’s lemma” = the run’s $\text{tr } R^2$ at $\lambda = 1$)
Bombieri (2000)	Remarks on Weil’s quadratic functional	Rend. Lincei (EuDML)	literature agent ($3\frac{3}{4}$ h); cold referee search hit	recalled with a “?” in the one-half brief, then verified at source; positioning (prior art for the negative-index half of Theorem 1)
Farmer (1995)	Counting distinct zeros of the Riemann zeta-function	Electron. J. Combin. 2	search hits for referees C, A ($1\frac{1}{2}$ h); literature agent ($3\frac{1}{2}$ h; title recalled in its brief); paper-writing agent front matter (6 h)	positioning
Wu (2015)	Distinct zeros of the Riemann zeta-function	Quart. J. Math. 66 (abstract)	literature agent ($3\frac{1}{2}$ h)	positioning (66.04% distinct; the 0.66 of the draft’s introduction)
Bui–Heath- Brown (2013)	On simple zeros of the Riemann zeta-function	1302.5018 (abstract)	literature agent ($3\frac{2}{3}$ h)	positioning (19/27 already recalled in referee B’s brief)
Conrey–Ghosh– Gonek (1998)	Simple zeros of the Riemann zeta-function	Proc. LMS (author’s copy)	literature agent ($3\frac{2}{3}$ h); paper-writing agent p. 1 (6 h)	positioning ($\geq 5/6$ distinct by integrality under RH sat in the findings file; see text)
Suzuki (2022; 2026)	Aspects of the screw function corresponding to the Riemann zeta function; Weil’s quadratic form via the screw function	2206.03682; 2606.09096	literature agent ($3\frac{3}{4}$ h)	dead end here (prior-art search for a link between signature and off-line zeros found nothing); used on the Weil-positivity branch
<i>Downloaded by the optimal-window sibling</i>				
Carneiro– Chandee– Littmann– Milinovich (2017)	Hilbert spaces and the pair correlation of zeros of the Riemann zeta-function	1406.5462	sibling (8 h; arXiv number recalled by the coordinator in its brief at $7\frac{1}{2}$ h)	recalled, then verified at source (Cor. 15 optimality; remark and closed form in Theorem D)
Chirre– Gonçalves-de Laat (2020)	Pair correlation estimates for the zeros of the zeta function via semidefinite programming	1810.08843	sibling (8 h; number recalled by the sibling)	positioning; dead end (bounds the signed-weight frontier; not in the draft)

authors	short title	arXiv / venue	fetches by; when	role
<i>Downloaded by the generalisations agent and its two literature sub-agents ($8\frac{2}{3}$ to $9\frac{3}{4}$ h), and by the cold referee</i>				
Chandee–Lee– Liu–Radziwiłł (2014)	Simple zeros of primitive Dirichlet L -functions and the asymptotic large sieve	1211.6725	literature sub-agent; generalisations agent	verification (where GRH enters) → hedged remark only
Hughes–Rudnick (2003)	Linear statistics of low-lying zeros of L -functions	math/ 0208230	literature sub-agent	verification (unconditional support range; recalled in the two-thirds brief)
Drappeau– Pratt–Radziwiłł (2020); Chandee–Lee (2017)	One-level density with extended support; n -level density for Dirichlet L -functions	2002.11968; 1706.02848	literature sub-agent	positioning
Wu (2019)	The twisted mean square and critical zeros of Dirichlet L -functions	1802.09704	records sub-agent; generalisations agent	positioning (the Dirichlet records line)
Pratt–Robles– Zaharescu– Zeindler (2020)	More than five-twelfths of the zeros of ζ are on the critical line	1802.10521	records sub-agent	positioning (record recalled from the start)
Conrey–Iwaniec– Soundararajan (2011)	Critical zeros of Dirichlet L -functions; Asymptotic large sieve	1105.1177; 1105.1176	literature sub-agent; generalisations agent	positioning
Sono (2021)	Zeros of Dirichlet L -functions on the critical line	2105.07422	both sub-agents; generalisations agent	positioning
Andersen– Thorner (2020); de Faveri (2021); Rezvyakova (2025)	Zeros of GL_2 L -functions on the critical line; Simple zeros of $GL(2)$ L -functions; linear combinations of degree-two L -functions	2004.03581; 2109.15311; 2501.00551	records sub-agent; generalisations agent	positioning (mostly not in the draft)
Conrey–Farmer– Kwan–Lin– Turnage- Butterbaugh (2025)	Short mollifiers of the Riemann zeta-function	2508.11108	records sub-agent	positioning
—	Pair correlation of zeros of Dirichlet L -functions: a possible path towards the conjectures of Chowla, Elliott–Halberstam and Montgomery; Levinson’s theorem and its generalization for Dirichlet L -functions	2411.19762; 2511.06109	literature sub-agent; generalisations agent	dead end
Özlük (1996)	On the q -analogue of the pair correlation conjecture	J. Number Theory (abstract pages)	literature sub-agent	positioning
Odlyzko	table of the first 10^5 zeros	author’s page	cold referee (9 h)	verification data (entrywise check of the prime-side matrix)

A.8 Other branches of the campaign

The same three days saw several other lines of work, run by other agents from other briefs, and for two of them the relation to the literature was the opposite of the one just described.

Weil positivity. The branch that ended in a computer-assisted certification of Weil positivity for test functions of larger support began from a frame recalled by the coordinator — Connes and Consani’s 2021 theorem for support in $[2^{-1/2}, 2^{1/2}]$, stated from memory in the very first brief of the campaign and again in the brief that asked an agent to “draw” the positivity margin as a function of the support. That agent searched for Yoshida’s 1992 paper, downloaded three of Suzuki’s screw-function papers (arXiv 2209.04658, 2206.03682, 2606.09096) and a July-2026 numerical study of Suzuki’s operator (arXiv 2607.24830), and fetched the abstract of Connes–Consani’s *Weil positivity and trace formula, the archimedean place* (arXiv 2006.13771), about nine and a half hours after the one-half run returned, before its own computation, and used them as positioning: it extended the published numerics by many orders of magnitude and found the exponential law of the margin empirically. The agent then asked to explain that law, about sixteen hours after the one-half run returned, was given two arXiv numbers from that report and a third the coordinator recalled (Connes–Consani–Moscovici, *Zeta zeros and prolate wave operators*, arXiv 2310.18423), downloaded some twenty papers of Connes and Consani, read *Spectral triples and zeta-cycles* (arXiv 2106.01715) and Connes–Consani–Moscovici’s *Zeta spectral triples* (arXiv 2511.22755), and wrote in its report that the source of the idea was there: the extremely small eigenvalues are explained in those papers by prolate vectors, and the earlier agent had “rediscovered the curve numerically without this identification”. The certifying agent, about six hours later, read the downloaded text of the 2021 paper for the exact published statement and normalisation it certified against, and fetched Bombieri 2000 and looked up Yoshida 1992 (which it could not obtain) for a priority caveat. On this branch, then, the mechanism was *supplied* by downloaded papers, and the agents say so; the certification scheme itself was the coordinator’s brief plus the certifying agent’s own design. The branch shares no files or messages with the chain of Parts I and II beyond the Bombieri/Yoshida remark both cite.

Zero-density estimates. This branch was literature-driven by construction. Its brief, on the first day, named the genre from memory — Guth and Maynard’s *New large value estimates for Dirichlet polynomials* (arXiv 2405.20552, number recalled correctly) and the Tao–Trudgian–Yang database of exponents — and the agent’s first acts were to download that paper and Tao–Trudgian–Yang’s *New exponent pairs, zero density estimates, and zero additive energy estimates: a systematic approach* (arXiv 2501.16779), to clone the public repository of the associated exponent database, and to transcribe from them the pool of published ingredients and the table to be beaten; it later fetched Kerr’s 2019 preprint (arXiv 1909.12075) to check that its candidate improvement was not already known. What it found — a published large-values lemma present in the database’s blueprint but not loaded by its optimiser, whose inclusion repairs three jump discontinuities of the table — is a recombination of published lemmas it names; its two referees re-derived the comparison from the Tao–Trudgian–Yang source table, Gafni–Tao (arXiv 2505.24017) and a 2026 survey (arXiv 2607.04632), and one of them fetched an Ivić lecture. Provenance of every ingredient is the literature, by design; the finding is the ablation.

Everything else. Fifty-four further arXiv papers and a comparable number of journal, database and lecture-note pages were fetched by agents on branches that led nowhere near the results above: zero-free regions and decoupling (the campaign’s first hours), Eisenstein zero-curves and Epstein zeta functions, a multivariate-stability lift, Beurling generalized primes,

a rate law for zeros of linear combinations of L -functions (one literature sub-agent alone made sixty requests), periods and theta lifts, and, after everything else was finished, a workflow on zero-density estimates near the critical line that re-downloaded three of the pair-correlation papers among a dozen others. The logs settle their relation to Parts I and II mechanically: the set of directories from which any agent on that chain ever read a file contains none written by these agents, the coordinator read none of their files, and nothing from their short reports to the coordinator appears in any brief on the chain. One of them deserves a footnote: the literature sub-agent of the rate-law branch had the text of Goldston–Lee–Schettler–Suriajaya and Goldston–Suriajaya on disk during the very hours in which the one-half run was deriving its dual argument — the only place in the campaign where that text then existed — in temporary files that no other agent ever opened and in a report that went only to its own parent.

branch	principal works fetched (arXiv)	role
Weil positivity	Connes–Consani 2006.13771, 2106.01715; Connes–Consani–Moscovici 2310.18423 (recalled, then fetched), 2511.22755; Suzuki 2206.03682, 2209.04658, 2606.09096; numerics 2607.24830; Yoshida 1992 (journal page); Bombieri 2000 (EuDML); some fifteen further Connes–Consani papers downloaded and set aside	frame recalled (the 2021 theorem); mechanism of the margin law <i>supplied</i> by 2106.01715 / 2511.22755, declared by the agent; published statement used as the certification target; priority caveat
zero density	Guth–Maynard 2405.20552 (recalled, then fetched); Tao–Trudgian–Yang 2501.16779; the public exponent-database repository and blueprint; Kerr 1909.12075; Gafni–Tao 2505.24017; survey 2607.04632; an Ivić lecture	inputs by design (ingredient pool and baseline table); verification by the referees
all other branches	54 papers: e.g. Ford 1910.08205/1910.08209 and a Bourbaki exposé 1707.00119 (zero-free regions); 2112.10561, 2110.09368 (Epstein zeros); 0704.3448 (finite Euler products); 2309.01567, 2409.10051, math/0410270 (Beurling primes); two dozen papers on zeros of linear combinations and derivatives of L -functions; 2605.22475 (Eisenstein series); 1805.07741, 1910.08274 and others (near-line density, last)	no read path into Parts I–II; not discussed further